Overconfidence and the Political and Financial Behavior of a Representative Sample *

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October 31, 2026

Abstract

Overprecision—excessive confidence in the accuracy of one's beliefs—affects decision-making across multiple domains. In this paper, we examine how overprecision relates to financial and political behavior using a nationally representative sample from the German Socio-Economic Panel Innovation Sample (SOEP-IS). We introduce the Subjective Error Method (SEM), which measures overprecision by comparing respondents' realized errors with the errors they expected to make. In the baseline analysis, higher SEM-measured overprecision is associated with larger stock market forecasting errors, less diversified investment portfolios, more extreme political views, and a lower likelihood of voting. A two-step analysis shows that financial behavior predicted using the SEM score is associated with political behavior among the same respondents. The relationship also holds in the reverse direction. Two preregistered companion surveys show that SEM measures are positively related across diverse tasks and support interpreting subjective errors as stated uncertainty. Taken together, these findings suggest that the SEM offers a simple, scalable way to study overprecision and its relationship with behavior across tasks in large, heterogeneous samples.

Keywords Overconfidence, SOEP, Survey

JEL Classification C83 · D91 · G41

*Corresponding author: cirilbosch@gmail.com. We would like to thank Agustin Casas, Antoni-Italo de Moragas, Benjamin Enke, Tilman Fries, Thomas Gräber, Andreas Haufler, Frank Heinemann, Don Moore, Guillem Riambau, Uwe Sunde, the participants at the CESifo Area Conference in Behavioral Economics as well as the participants of seminars at CUNEF, EMLyon, and FU Berlin for their insightful comments. We acknowledge the financial support from the Deutsche Forschungsgemeinschaft (DFG) through CRC TRR 190 "Rationality and Competition" (project number 280092119). The Scientific Advisory Board of DIW Berlin and the Ethics Committee of the Faculty of Economics at Ludwig-Maximilians-Universität München provided ethical approval, and all participants included in the analyses provided informed consent. Two online surveys were preregistered at aspredicted.org (#118284 and #123146). The authors are not listed alphabetically, as the first two authors (Bosch-Rosa and Kassner) contributed more to the paper.

What would I eliminate if I had a magic wand? Overconfidence.

—Daniel Kahneman, *The Guardian*, 18 July 2015

1 Introduction

Overconfidence is a pervasive and potent bias in human judgment (Mannes and Moore, 2013; Kahneman, 2011). It leads to wars (Johnson, 2004), to excessive entry into markets (Camerer and Lovallo, 1999), or to 80% of the population thinking that they are above-average drivers (Svenson, 1981). However, overconfidence is a general term encompassing three distinct phenomena: overestimation, overplacement, and overprecision (Moore and Healy, 2008; Moore and Schatz, 2017). Overestimation concerns absolute values—individuals believe they perform better than they actually do. Overplacement has to do with relative values—individuals believe their performance is better than others' when it is not. Overprecision, the focus of this paper, concerns the degree of certainty in judgments and beliefs—individuals perceive their judgments as more accurate than they really are. In other words, overprecision relates to the second moment of the distribution, such that a person may hold accurate beliefs on average but underestimate the variance of the possible outcomes (Malmendier and Taylor, 2015).

Overprecision has wide-ranging consequences. From an economic standpoint, it has been linked with suboptimal decision-making, including insufficient insurance purchasing (Grubb, 2015) or distortions in corporate investment decisions (Ben-David et al., 2013; Moore et al., 2015). In financial markets, overprecision contributes to portfolio underdiversification (Goetzmann and Kumar, 2008), excessive trading (Barber and Odean, 2001), and systematic forecasting errors (Deaves et al., 2019). In politics, it fosters ideological extremism, entrenches partisan identification (Ortoleva and Snowberg, 2015a,b; Stone, 2019), and increases susceptibility to misinformation, such as "fake news" (Thaler, 2023). However, despite these well-documented consequences, most of the existing literature has examined overprecision within specific, isolated domains of behavior. As a result, it remains unclear whether overprecision is domain-specific, with different individuals exhibiting miscalibration in different settings, or whether the same individuals are consistently overprecise across domains.

In this paper, we address this gap by examining how overprecision relates to behavior across different domains. Specifically, we study its relationship with behavior in two key domains: financial decision-making and political behavior. To measure overprecision, we introduce the "Subjective Error Method" (SEM) in the German Socio-Economic Panel Innovation Sample (SOEP-IS), a nationally representative panel of the German population. The SEM is a simplified version of the MISS approach to measuring overprecision proposed by Soll et al. (2022), and it proceeds in two steps. First, respondents answer a numerical question, such as the year of Saddam Hussein's capture or the height of the Eiffel Tower. Second, they estimate the "distance" (in the relevant units) between their answer and the correct one. In other words, respondents report the *absolute error* they expect to make (henceforth, subjective error). By comparing respondents' realized errors with their subjective errors, we obtain a simple, direct measure of overprecision.

The richness of our data allows us to examine the relationship between respondents' overprecision, as measured by the SEM, and their sociodemographic characteristics. We find that overprecision is positively correlated with narcissism and negatively correlated with age, years of education, gross income, and financial literacy, but does not differ by gender. More importantly, we find several associations that align with theoretical predictions in the financial and political domains. Overprecision is associated with lower portfolio diversification (Odean, 1998; Barber and Odean, 2000) and more extreme political positions (Ortoleva and Snowberg, 2015b). Contrary to the prediction of Ortoleva and Snowberg (2015b), however, it is associated with a lower likelihood of voting. To examine whether these financial and political associations are connected, we use a two-step analysis. We first estimate the part of behavior associated with the SEM score in one domain, then relate it to behavior in the other domain for the same respondents. We find links in both directions, bringing together two literatures that have largely developed separately.

We also conduct two preregistered companion surveys to examine how SEM measures relate across tasks and how respondents interpret the subjective-error question. The first survey implements the SEM across five domains: contemporary history, general knowledge, economics, four-week-ahead stock price predictions, and a neutral counting task.[1]

[1]These domains differ in the ability to ascertain the true answer. In contemporary history, general knowledge, and economics, a correct answer exists at the time of the survey and may be known depending on the respondents' prior knowledge. The neutral counting task also has a correct answer, but respondents can only estimate it. In contrast, for stock price predictions, the correct answer remains unknown on

Using various statistical methods, we find that SEM measures are positively related across the five domains, suggesting that they capture a common component across tasks. The second survey compares respondents' subjective errors to their elicited probability distributions over their responses to the factual question. The reported subjective errors are positively associated with the variance of respondents' elicited probability distributions, supporting the interpretation of the subjective error as a measure of stated uncertainty. However, additional analyses suggest that differences in respondents' knowledge may contribute to some of these behavioral associations when the SEM is applied to factual questions.[2]

Our paper contributes to the literature in four ways. First, we implement the SEM in a large, nationally representative sample. Much of the existing evidence comes from university students (e.g., Alpert and Raiffa, 1982) or specialized groups, such as finance professionals (Glaser and Weber, 2007) and IT professionals (McKenzie et al., 2008). Our rich dataset from the German population broadens this evidence and allows us to examine the socioeconomic correlates of overprecision. Second, we find support for several theoretical predictions about financial and political behavior—two of the most consequential domains of decision-making. Third, we show that the financial and political behaviors associated with the SEM measure are related. To our knowledge, this is the first study to bridge the literatures on overprecision in finance and politics by documenting these cross-domain associations. Finally, our two companion surveys show that SEM measures are positively related across domains and that respondents' subjective-error reports are positively associated with the dispersion of their elicited probability distributions.

Taken together, our findings support the SEM as a simple and practical tool for studying overprecision in large surveys. Respondents report their uncertainty in familiar units, and the questions can be incorporated into surveys covering many aspects of their lives. These features make the SEM useful for studying how people assess the accuracy of their own judgments, an aspect of metacognition. Researchers can use it to compare

the survey date. Since these last two tasks do not require recalling a known factual answer, we can test whether overprecision also arises when prior factual knowledge has less direct influence.

[2]Here, factual questions draw on recall of existing information, such as the year of a historical event. Tasks that do not require factual recall include forecasting a stock's future price, predicting the temperature on a future date, or estimating the age of an unfamiliar person from a photograph. These tasks may still draw on knowledge and skill, but do not require recalling a previously known answer.

calibration across people and tasks, track how it changes over time, and examine its relationship with a wide range of outcomes.

The remainder of this paper is organized as follows: Section 2 discusses the notion of overprecision (2.1) and introduces the Subjective Error Method (2.2). Section 3 details the SOEP-IS data set and examines the relationship between overprecision and its sociodemographic correlates. Section 4 relates the financial and political behaviors associated with the SEM measure. Section 5 examines how SEM measures relate across task domains. Section 6 concludes.

2 Overprecision Measurement and the Subjective Error Method

2.1 Overprecision Measurements

Overprecision, also known as miscalibration, is a form of overconfidence resulting from excessive confidence in the accuracy of one's judgments (Moore et al., 2015). It relates to the second moment of the belief distribution and, therefore, directly affects information processing. For this reason, overprecision is widely used in finance and political science to model overconfident agents. For example, Odean (1998) finds that overconfident traders engage in excessive trading and hold underdiversified portfolios because they believe their private signals are more precise than they actually are. Scheinkman and Xiong (2003) combine a short-sale constraint with overprecise traders to explain the formation of asset market bubbles.[3] In the political science literature, Ortoleva and Snowberg (2015b) show that more overprecise individuals tend to vote more, hold more extreme political views, and exhibit stronger partisan identification. Consistent with this, Stone (2019) suggests that overprecision increases partisanship through excessively strong inferences drawn from biased information sources. More recently, the literature has begun to study the role of overprecision in fake news dissemination (Pennycook et al., 2021; Thaler, 2023).

However, precisely because overprecision concerns the second moment of the belief distribution, it is difficult to measure (Moore et al., 2015). The most common way to measure it is to elicit the respondents' 90% confidence intervals (CI) for a series of numerical questions such as "How long is the Nile River?" (e.g., Alpert and Raiffa, 1982). Using

[3]For a longer discussion on the different models of overprecision used in the finance literature, see Daniel and Hirshleifer (2015).

this paradigm, a perfectly calibrated respondent would fail to capture the correct answer within the CI in one out of every ten questions. However, the literature has shown that this method is problematic because respondents are unfamiliar with CIs and do not fully grasp what is being asked (Moore et al., 2015). This confusion is corroborated by the finding that the elicited intervals resulting from asking 90% CIs are practically identical to those resulting from asking for 50% CIs (Teigen and Jørgensen, 2005), and that often the purported 90% CIs contain the correct answer only 30% to 60% of the time (e.g., Russo and Schoemaker, 1992; Bazerman and Moore, 2013; Moore et al., 2015).

While alternatives to CIs for measuring overprecision exist, they tend to be time-consuming or limited in the information they provide. For example, the two-alternative forced-choice (2AFC) method asks respondents to choose between two possible answers to a question and indicate their confidence that the answer is correct. By comparing the number of correct answers with stated confidence, one can measure whether respondents are, on average, overconfident. However, this method has several drawbacks as it cannot distinguish between overprecision and overestimation of one's own judgment (Moore et al., 2015), and it cannot capture continuous distributions (see Griffin and Brenner (2004) or Moore et al. (2015) for a further discussion of the 2AFC method and its statistical limitations). Another approach to measuring overprecision is the Subjective Probability Interval Estimates (SPIES) method by Haran et al. (2010), which elicits full probability distributions from respondents. Although the SPIES method may measure overprecision more accurately than CIs (Moore et al., 2015), it is time-consuming and requires respondents to understand the concept of probability distributions. Additionally, because distributions can only be elicited by partitioning the support into discrete bins, researchers must make *ad hoc* decisions about how to implement and define the desired 90% boundaries.

More recently, Ortoleva and Snowberg (2015b) and Chapman et al. (2023) employ a parametric estimation method by regressing a six-point scale self-reported measure of confidence in the accuracy of the answer on a polynomial of the realized error. The drawback is that the measure of accuracy is quite coarse because of its limited six-point scale. Moreover, the measure is population-relative: because a respondent's estimated overprecision depends on the relationship between confidence and realized error in the sample as a whole, the same respondent may be classified differently in samples with different

compositions.[4] Soll et al. (2022) examine how expert advice of varying quality affects subjects' uncertainty and point estimates. Because they are interested in subjects' prediction confidence, Soll et al. (2022) introduce the MISS Method, a three-step procedure for measuring overprecision. The first step involves explaining and describing to subjects the concept of MISS—namely, the concept of absolute expected error.[5] In the second step, subjects make a point prediction for a numerical question (e.g., what year was John Lennon born?). Finally, in the third step, subjects are asked to estimate their MISS for the previous question.[6] By comparing the reported MISS to the realized true error, the authors develop a measure of confidence for each subject.

While the MISS measure of overconfidence is similar to the SEM (which we discuss more thoroughly in Section 2.2), the introduction of the abstract concept of MISS in the first step and its use in eliciting subjects' expected absolute error make the MISS method cognitively taxing and may confuse respondents (see Footnotes 5 and 6). Indeed, as the authors report, subjects struggled to understand the MISS method and instructions even after being provided with numerical examples. As an alternative, Soll et al. (2022) recommend following what they call the "Within C Method". In this method, subjects are asked to report the probability that the point estimate for a numerical question is within C units of the truth. This allows the authors to back out subjects' expected absolute error and degree of confidence. However, this assessment requires imposing a normal distribution on subjects' errors, which may not always be appropriate (Soll and Klayman, 2004). Moreover, in the "Within C Method", subjects' levels of overconfidence are highly sensitive to C, which can result in the same person being classified as overconfident or underconfident depending on the presented value of C.

[4]See the discussion of the *residual measure* (op_i^{resid}) in Appendix C.1 for an in-depth comparison of the methods, including the example in Figure C.3.

[5]For example, in a task where subjects had to guess the number of calories in a dish, in the first step of the MISS method, the instructions read: "In addition to making a best guess [of calories in the picture], you will also estimate the accuracy of your guess. Accuracy is simply the distance to the truth, which we call the MISS. The MISS represents the amount by which your guess missed the truth. For example, suppose that Robin guesses the calories in a slice of cake. The slice has 600 calories, but Robin does not know this. If Robin guesses 350, then Robin's MISS is 250. If Robin guesses 1100, then the MISS is 500. The lower the MISS, the more accurate the guess. Remember, MISS is the amount by which you missed the correct answer, either too high or too low. If you guess exactly right the MISS is zero."

[6]Specifically, subjects are asked: "What is your best estimate of the MISS for the guess you just gave?". Notice that this question requires subjects to recall the definition of MISS, which was given in the first step.

To address the caveats listed above, we propose the Subjective Error Method. This method is like the MISS method but omits the first step, so it does not necessarily rely on subjects' knowledge of statistical concepts, requires no parametric assumptions, is easy to understand, is quick to implement, and does not depend on the properties of the entire sample.

2.2 The Subjective Error Method

The Subjective Error Method asks respondents two consecutive questions. The first question (a) can be on any topic but needs to have a numerical answer.[7] The second question (b) asks respondents how far away they expect their answer to question (a) to be from the true answer. In other words, the second question asks respondents to report their absolute subjective error. An example would be:

(a) *How long (in kilometers) is the Nile River?*

(b) *How far away (in kilometers) do you think your answer to (a) is from the true answer?*

By comparing the *subjective error* reported in (b) with the absolute *realized error* from question (a), we obtain a measure of how over- or underprecise respondents are in their judgments.

More formally, assume a respondent's true error is normally distributed with mean 0 and variance σ^2, as shown by the solid curve in Figure 1. A perfectly calibrated individual would, on average, accurately assess the true error distribution when answering questions using the SEM. However, for most respondents, the perceived distribution may not align with the true distribution. If a respondent is overprecise, their perceived variance of the error $\hat{\sigma}^2$ is smaller than the true variance; that is, the precision $\rho = 1/\hat{\sigma}^2$ is larger, as illustrated by the dashed curve in Figure 1. In this case, the subjective error would, on average, systematically deviate from the true error in all questions.[8]

[7] Some examples of numerical questions are the result of multiplying 385 by 67, the length of the Nile River, or the year of Lady Diana's death. Examples of questions that do not work include the name of Lady Diana's oldest son, the color of the Popemobile, or the gender of the current prime minister of the United Kingdom.

[8] Under the zero-mean normal distributions in Figure 1, the difference between corresponding quantiles is proportional to the difference in their standard deviations.

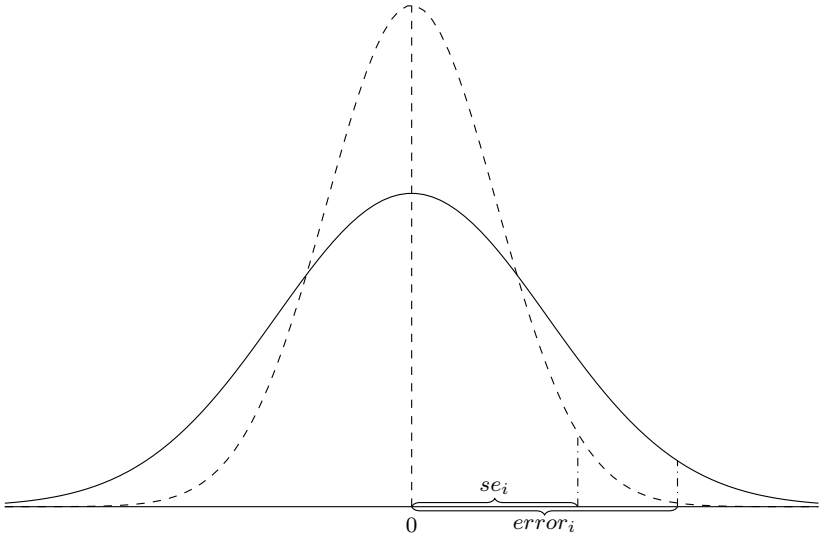


Figure 1: Two hypothetical normal distributions of the (subjective) error. The solid curve shows the true distribution of the error with a standard deviation of 2 (precision of .25). The dashed curve shows the perceived distribution by an overprecise respondent with a standard deviation of 1.25 (precision of .64). The dash-dotted vertical lines indicate the subjective error se_i and the absolute true error $error_i$ resulting with the same cumulative probability.

Let $a_{i,j}$ denote the answer of respondent i to question j, $se_{i,j}$ their subjective error, and ta_j the true answer. We measure respondent i's overprecision on question j as:

$$error_{i,j} = |a_{i,j} - ta_j|, \tag{1}$$

$$op_{i,j} = error_{i,j} - se_{i,j}, \tag{2}$$

where equation (1) measures the absolute realized error ($error_{i,j}$) of respondent i to question j. Importantly, we measure the *absolute error* because we care about the magnitude of the error but not its direction. In equation (2), we calculate the difference between realized error ($error_{i,j}$) and subjective error ($se_{i,j}$) for respondent i on question j. The sign of this difference determines whether the respondent is over- or underprecise. A respondent whose realized error is larger than their subjective error is considered *overprecise*, while a respondent whose realized error is smaller than their subjective error is considered *underprecise*. If the two errors are equal, the respondent is perfectly calibrated for that question.

2.3 Subjective Error Method – Discussion

The Subjective Error Method asks respondents for their best numerical estimate and how far they think it is from the true answer. Yet, giving a best estimate does not mean being certain that it is correct: respondents can choose their best answer while recognizing that it may be wrong. The subjective-error question asks them to assess the size of that possible error, regardless of whether their estimate is too high or too low. This is consistent with Bayesian reasoning, since even an optimal point estimate can have a positive expected absolute error.[9] Someone estimating the length of the Nile River can give their best answer while recognizing that it is unlikely to be exactly correct. The same applies to forecasts of future stock prices or electoral results, where the outcome has not yet been realized and respondents cannot be certain that their prediction will be correct. This metacognitive awareness need not involve formal probability calculations. Respondents may draw on rules of thumb, past experience, and contextual cues, and the SEM lets them express uncertainty without requiring statistical expertise.[10]

Metacognitive awareness also connects the SEM to the literature on cognitive uncertainty. Enke and Graeber (2023) define cognitive uncertainty as people's "subjective uncertainty about what the optimal action is." This captures doubts about whether one has correctly processed the available information or identified the best decision. Overprecision concerns whether people are more confident in their judgments than their accuracy warrants. The two concepts can coexist: people can recognize that they might be wrong and still underestimate how wrong they could be. For example, an investor may doubt their interpretation of financial information yet still give a forecast range for a stock's future price that is too narrow for the uncertainty they face.

The SEM compares the errors respondents expect with the errors they actually make. Understanding this comparison also requires considering why some respondents make larger errors than others. In our SOEP-IS implementation, respondents answer factual questions, so differences in accuracy may partly reflect cognitive ability and factual knowl-

[9]Let θ denote the true answer and I the respondent's information. Under squared-error loss, the optimal estimate is the posterior mean $a^* = E[\theta \mid I]$, for which $E[\theta - a^* \mid I] = 0$. This means zero expected signed error, not certainty: for any fixed estimate a and non-degenerate posterior, $E[|\theta - a| \mid I] > 0$.

[10]In our preregistered companion survey, larger subjective errors are associated with more dispersed probability distributions, supporting this interpretation of subjective error as stated uncertainty (Appendix I).

edge. Cognitive ability refers to the capacity to acquire and process information, while factual knowledge refers to what a respondent knows about a particular topic.[11] Realized errors may also contain item-specific noise. If differences in factual accuracy are not accompanied by corresponding differences in stated uncertainty, they will also affect the SEM gap. These differences may contribute to the observed behavioral associations when factual accuracy is related to the outcome of interest. Appendix J examines this concern by studying how realized and stated error are associated with the outcomes and comparing different ways to account for realized error. These analyses do not directly measure factual knowledge, but they show that accounting for realized error changes some behavioral associations more than others. We therefore interpret the SEM in this application as the gap between realized and stated subjective error, recognizing that it may reflect factual knowledge as well as miscalibration. The concern about factual knowledge is less pronounced when the SEM is applied to tasks that do not require factual recall.

As the setup makes clear, the Subjective Error Method's main advantage is its simplicity. It asks respondents to express uncertainty in the units of the question rather than through confidence intervals or probability distributions, making it easy to explain and suitable for large, diverse samples. By comparing stated and realized errors for each question, the SEM provides an individualized measure of overprecision without relying on population-level benchmarks.

3 Individual Overprecision and its Correlates

In this section, we describe the dataset used to study the relationship between overprecision and the financial and political behavior of a German sample (section 3.1), explain how we create our measure of overprecision using the Subjective Error Method (section 3.2), and report the correlation between our measure of overprecision and the sociodemographic characteristics of our sample (Section 3.3).

[11] The SOEP-IS does not measure cognitive ability directly, so we use years of education as an imperfect proxy in Table B.1 in Appendix B. The estimated relationship between stated and realized error is somewhat stronger among respondents with more education. However, unexpectedly large or small errors on individual questions can also affect these estimates. We therefore treat the comparison as descriptive and examine the role of this realization noise more closely in Appendix J.1.

3.1 Data

We included a module in the 2018 wave of the German Socio-Economic Panel Innovation Sample (SOEP-IS). The Innovation Sample is a companion panel of the larger SOEP-Core and is designed to host and test novel survey items (see, Richter and Schupp, 2015). Like the SOEP-Core, the SOEP-IS is a high-quality survey in which a professional interviewer conducts all interviews face-to-face.

Our module consists of seven contemporary history questions. In each question, we ask respondents: (a) the year in which a specific event occurred, and (b) how many years they believe their answer differs from the correct answer.[12] The questions are designed to vary in difficulty, with content ranging from the year in which the Volkswagen Beetle was introduced (1938) to the year in which the US Army captured Saddam Hussein (2003) (see Table B.2 and Table B.3 in the appendix for all questions and their correct answers in English and German, respectively). The module was administered to respondents who joined the panel in 2016 ($N = 902$). We supplement the data with additional information on personal characteristics from the survey years 2016–2018. We drop 55 respondents who did not answer any of the overprecision questions and 42 respondents with incomplete information. Our final sample consists of 805 respondents.[13]

3.2 Individual Overprecision

In Figure 2, we plot the distribution of answers $a_{i,j}$ for each question j. The red vertical line marks the correct answer. The concentration of answers around the correct answer varies across questions, indicating differences in difficulty. In Figure 3, we plot the realized error ($error_{i,j}$) on the vertical axis and subjective error ($se_{i,j}$) on the horizontal axis for each of the seven questions. Additionally, we plot a 45-degree red line; any dot above represents a respondent who is overprecise ($error_{i,j} > se_{i,j}$) in their answer, while any

[12] The questions are formulated in German. For example, when we ask about the year of Lady Diana's death, we ask: (a) *In welchem Jahr starb Lady Diana, die erste Frau von Prinz Charles?* and then (b) *Was schätzen Sie, wie viele Jahre Ihre Antwort von der richtigen Antwort entfernt ist?*

[13] To test whether our estimation sample is still representative of the German population, we compare the unweighted means of personal characteristics in our sample with the weighted means according to the sampling weights in the larger SOEP-Core, which is representative of the German population. Table B.5 in the appendix shows that our subsample remains broadly representative of the larger SOEP-Core, with only a few small but statistically significant differences. When we apply the sampling weights to our estimation sample, these differences disappear.

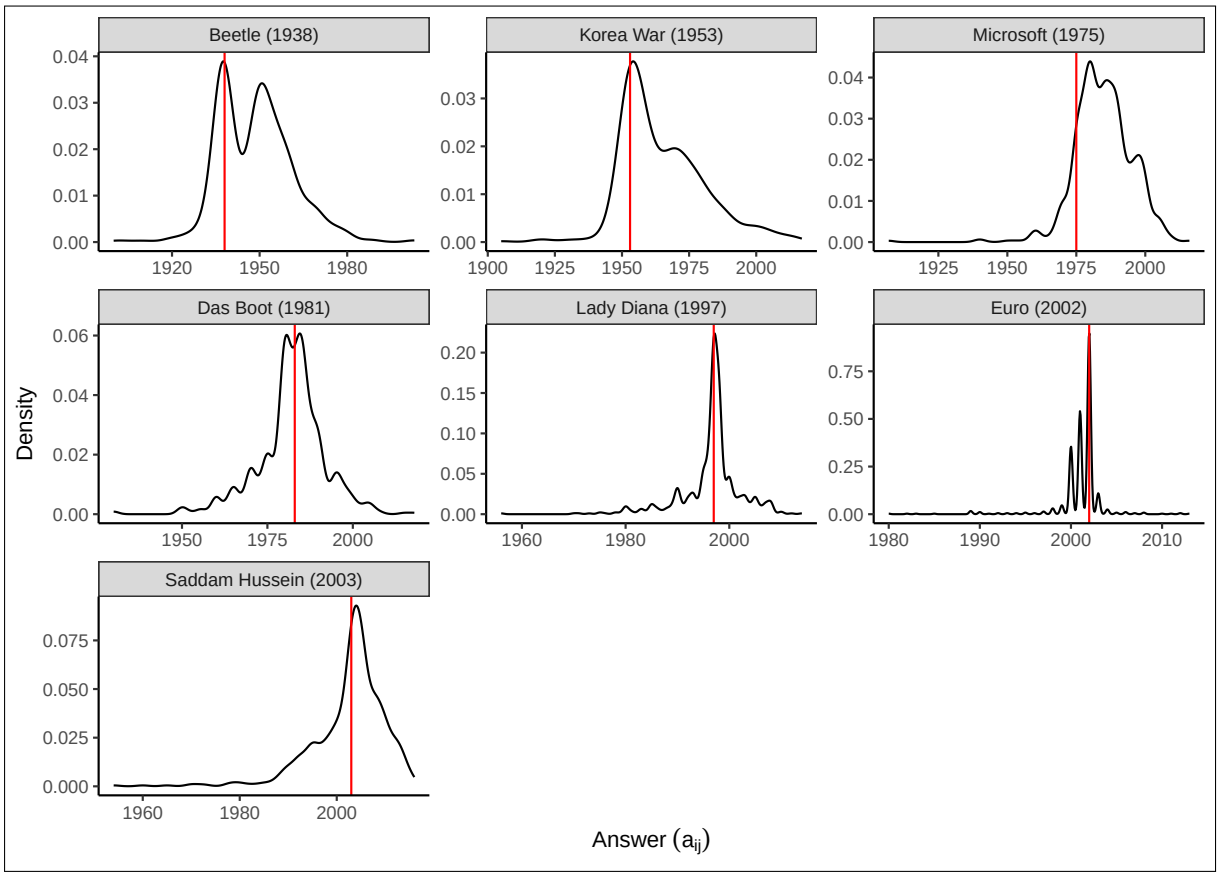


Figure 2: Density of the answers ($a_{i,j}$) for each question. The red vertical line marks the correct answer. Note that the vertical axis is different for each question.

point below corresponds to a respondent who is underprecise ($error_{i,j} < se_{i,j}$). Across all seven questions, most observations lie above the 45-degree line, indicating overprecision.

We construct a measure of overprecision for each question ($op_{i,j}$) following the outline in Section 2.2. To measure the within-respondent consistency of overprecision across the seven questions, we use congeneric reliability, commonly referred to as coefficient omega (e.g., Cho, 2016). Congeneric reliability represents the proportion of variation (variance and covariance) among a set of variables that can be explained by an unobserved factor.[14]

[14]Consider a model in which each observed outcome i of item j can be expressed as $T_{i,j} = \mu_j + \lambda_j F_i + e_{i,j}$, where $T_{i,j}$ is the ith outcome of item j, μ_j is a constant term, $e_{i,j}$ is the individual score error, and λ_j is the factor loading on the latent common factor F. To construct congeneric reliability, we estimate the factor loadings, $\hat{\lambda_j}$, for the overprecision measure of each question with respect to one common factor. This factor captures the common component across the item-level measures of overprecision. Congeneric reliability is calculated using the formula $\frac{(\sum \hat{\lambda}_j)^2}{(\sum \hat{\lambda}_j)^2 + \sum \hat{\sigma}^2_{e_j}}$, where $\hat{\sigma}^2_{e_j}$ represents the estimated variance of the error. This is a generalized version of Cronbach's alpha (Cronbach, 1951) and measures the proportion of variation among the set of items j that can be explained by the latent factor. While congeneric reliability allows for different factor loadings of the latent common factor, Cronbach's alpha

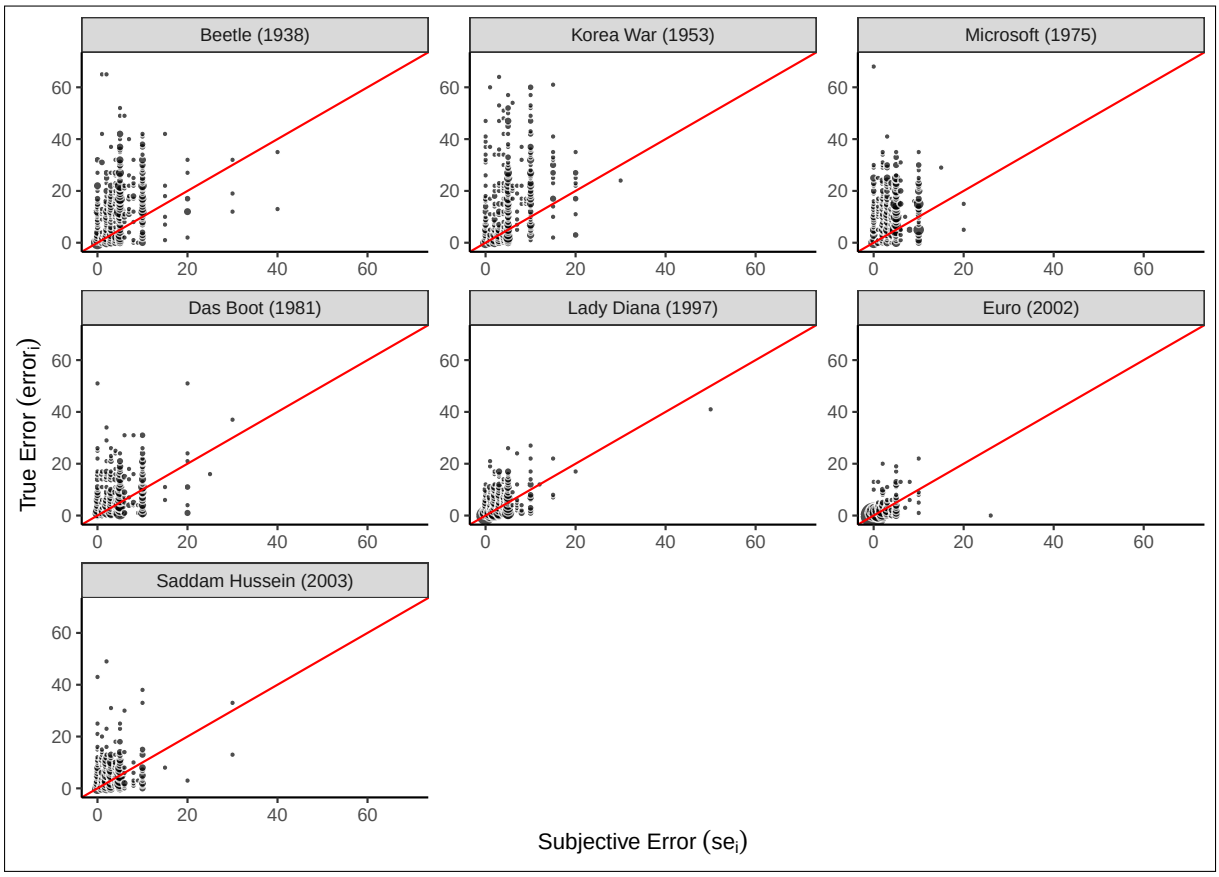


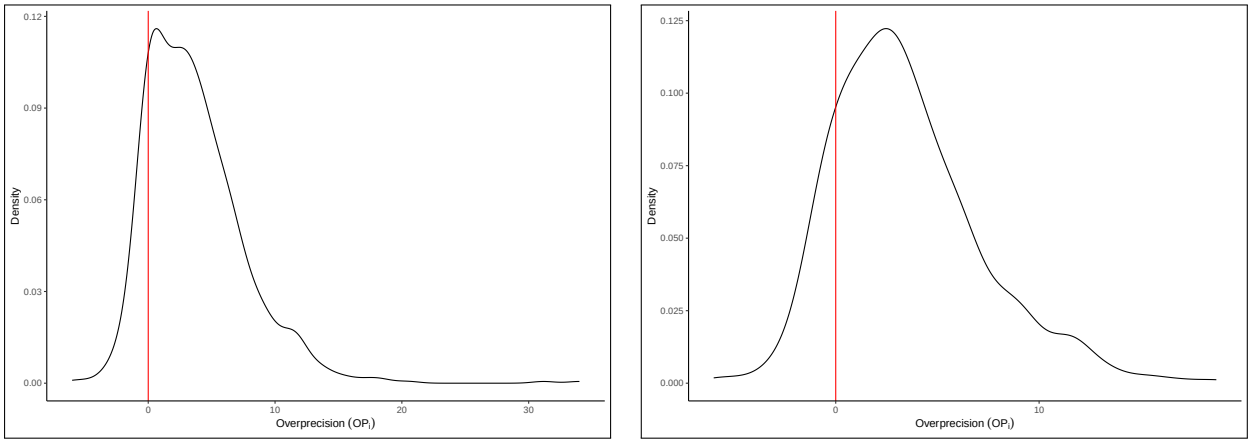
Figure 3: Relation between the realized error ($error_{i,j}$) on the vertical axis and the subjective error ($se_{i,j}$) on the horizontal axis. Any dot above (below) the 45-degree red line is an overprecise (underprecise) response.

The results indicate that 49% of the variance in the aggregate score is explained by a common factor, which we interpret as a shared component of overprecision across the seven questions.

To create a unique measure of overprecision for each respondent i (op_i), we take the average overprecision across all seven questions j.[15] Figure 4a plots the density of op_i and, in line with Figure 3, shows that the vast majority of respondents (82%) are overprecise. Interestingly, and in contrast to most literature using CIs to measure overprecision, we

assumes that the latent factor loads equally on all items and, thus, serves as a lower bound for reliability. For the case of τ-equivalence, i.e., $\lambda_j = \lambda_k \; \forall k$, all factor loadings are equal and both measures coincide.

[15] As an alternative to taking the simple average, we also construct a factor score from a one-factor measurement model of the seven item-level overprecision gaps. We estimate the model by full-information maximum likelihood, using the available responses from respondents with complete or incomplete item batteries, and predict a respondent-level factor score. This score is strongly correlated with the simple average ($\rho^{\text{Pearson}} = .88$; $\rho^{\text{Spearman}} = .84$; $N = 805$). Appendix D uses a related factor score based on a model estimated from respondents who answered all seven items, then predicts scores using each respondent's available responses.



(a) Density of op_i for all respondents (N=805) **(b)** Density of op_i for subset (N=410)

Figure 4: Density of Overprecision (op_i). In the left panel, we plot the density of op_i, which is the average overprecision for each respondent i across all questions j. In the right panel, we plot the density of op_i for respondents who answered all survey questions.

find a relatively large number of respondents who are underprecise (approximately 11%).

Moreover, 7% of respondents appear to be perfectly calibrated in the aggregate measure, as indicated by the vertical red line in Figure 4a. Of these respondents, 83% are perfectly calibrated across all the questions they answer. However, note that in the SOEP-IS, respondents can choose not to answer a question. In total, 51% of the respondents answered all questions, while 5% answered only one (see Figure A.1 in the appendix for a detailed breakdown). Among the perfectly calibrated respondents, 40% answered only one question, and just 12% answered all seven. Exact aggregate calibration is therefore more common among respondents whose measure is based on fewer questions. In Figure 4b, we plot the density function for the subset of respondents who answered all the questions. As shown, respondents are substantially less calibrated, with the mode of op_i shifting to the right, leaving only 1% of the respondents perfectly calibrated; meanwhile, there is an increase in the proportion of underprecise respondents (15%).

For ease of interpretation in the subsequent analysis, we standardize the aggregate score (op_i) to have mean zero and standard deviation one, and refer to it as sop_i. In Appendix C.1 we discuss and compare our measure of overprecision to five alternative approaches. These are: i) a *standardized* measure, which standardizes each question before aggregating, ii) a *centered* measure, which centers realized and subjective errors around their respective means before aggregation, iii) a *relative* approach, which takes into account the relative distance between the subjective error and the realized error,

iv) an *age-robust* measure, constructed using only those questions concerning events that occurred strictly after the respondent's birth year, and v) a *residual* approach following the regression methodology of Ortoleva and Snowberg (2015b).

3.3 Sociodemographic Correlates of Overprecision

In this section, we assess how overprecision is correlated with individual sociodemographic characteristics. We do so by focusing on sociodemographic characteristics that either relate to understanding the task at hand or have been examined in the literature in the context of overconfidence.[16] This exploratory analysis provides descriptive context and introduces the sociodemographic controls used in the subsequent outcome regressions.

In Table 1, we regress sop_i on a series of sociodemographic variables using five different OLS models. In all models, we control for age, gender, and years of education. In Column (2), we add the number of overprecision questions answered. In Column (3), we add the monthly gross individual income (*gross income*) measured in thousands of euros[17] as well as dummies for labor force status (e.g., employed, unemployed, maternity leave, etc.) and a dummy for those respondents who were living in East Germany in 1989. In Column (4), we add further personal characteristics: financial literacy, risk preferences, impulsivity, patience, and narcissism. Finally, in Column (5), we add federal state (*Bundesland*) and month-of-interview fixed effects.[18]

The results show a negative correlation between overprecision and age, education, and income. For example, a 2,000-euro increase in gross income is associated with a reduction in overprecision of almost one-tenth of a standard deviation, and every 2 years of education is associated with a reduction in overprecision of about one-tenth of a standard deviation. Furthermore, overprecision is negatively correlated with our measure of financial literacy and positively correlated with our measure of narcissism and the number of questions answered by respondents.[19] In all subsequent analyses, we use these variables as controls.

[16] See, e.g., Deaves et al. (2009) for gender, Campbell et al. (2004) for narcissism and risk attitudes, Visser et al. (2019) for impulsivity.

[17] Since *gross income* is only available for employed individuals, we code missing variables as 0 and include a dummy that is 1 for missing observations.

[18] Note that we only report coefficients that are statistically significant in Table 1. We report all coefficients in Table B.6 in the appendix.

[19] Item nonresponse may be related to unobserved characteristics such as cognitive ability, motivation,

Dependent Variable: *sop*	(1)	(2)	(3)	(4)	(5)
age	-0.008***	-0.008***	-0.007**	-0.005	-0.005
	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)
female	0.074	0.121	0.094	0.134*	0.111
	(0.070)	(0.074)	(0.074)	(0.074)	(0.076)
years education	-0.053***	-0.062***	-0.050***	-0.043***	-0.037***
	(0.013)	(0.013)	(0.014)	(0.013)	(0.014)
answered		0.064***	0.067***	0.086***	0.085***
		(0.023)	(0.022)	(0.023)	(0.023)
gross income			-0.049**	-0.037*	-0.039*
			(0.021)	(0.020)	(0.021)
fin. literacy				-0.481***	-0.398**
				(0.170)	(0.174)
narcissism				0.107**	0.100**
				(0.042)	(0.043)
N	805	805	805	805	805
adj. R^2	0.036	0.046	0.061	0.081	0.098
Constant Term	Yes	Yes	Yes	Yes	Yes
Employment & GDR 1989	No	No	Yes	Yes	Yes
Personal characteristics	No	No	No	Yes	Yes
Fixed Effects	No	No	No	No	Yes

Robust standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 1: Sociodemographic correlates of overprecision. In Columns (1)-(5) we run an OLS with *sop* as the dependent variable. In all Columns, we include age, gender, and education. In Column (2) we add the number of answered questions. In Column (3) we include gross income and dummies for labor force status (employed, unemployed, retired, maternity leave, not working), and whether the respondent was a citizen of the GDR before 1989. In Column (4) we include further personal characteristics: financial literacy, risk preferences, impulsivity, patience, and narcissism. In Column (5) we also include fixed effects for the federal state (*Bundesland*) where the respondents live and the time at which they responded to the questionnaire.

The results in Table 1 are broadly consistent with the existing literature. The positive correlation with narcissism accords with previous findings (e.g., Campbell et al., 2004; Hamurcu and Hamurcu, 2021), while the negative correlation with education is consistent with evidence of a negative association between cognitive ability and overprecision (e.g., Duttle, 2016). Evidence on gender is mixed. Ortoleva and Snowberg (2015b) find that women are less overprecise than men, whereas López-Pérez et al. (2021), Deaves et al.

or numerical reasoning. We retain the full sample and account for differences in response behavior by controlling for the number of questions answered.

(2009), and Wohleber and Matthews (2016) find no association. This pattern also aligns with the meta-analysis by Bandiera et al. (2022). Evidence on age is similarly mixed. Although some studies find that overprecision increases with age (Prims and Moore, 2017; Ortoleva and Snowberg, 2015b), we find a slight negative correlation.[20] This result may partly reflect the historical-question domain. When we use the age-robust measure, which uses only questions about events that occurred strictly after respondents' birth years, the correlation becomes positive (Appendix C). Evidence directly linking overprecision and financial literacy is more limited because most studies examine other forms of overconfidence. Overestimation of financial knowledge has been associated with susceptibility to investment fraud (Xiao et al., 2022), less advice seeking (Kramer, 2016), and more frequent trading (Inghelbrecht and Tedde, 2024); narcissism predicts overplacement in a financial-literacy quiz (Hamurcu and Hamurcu, 2021); and investor overestimation and overplacement persist with experience (Merkle, 2017). Together, this literature connects misjudgments of financial knowledge with financial behavior, consistent with the correlation we document between financial literacy and overprecision.

4 Overprecision and Individual Financial and Political Behavior

This section reports the main results of our paper as we examine how overprecision correlates with respondent behavior in the political and financial domains (Section 4.2) and show that the same variation in behavior that is related to overprecise behavior in the financial domain is also associated with overprecise behavior in the political domain, and *vice versa* (Section 4.3).

4.1 Methodology

To test the predictions from the theoretical literature discussed below, we use three approaches. First, we run a regression of each outcome (y_i) on our measure of overprecision

[20] Using the residual approach of Ortoleva and Snowberg (2015b) produces sociodemographic patterns closer to theirs (Table C.1). Women have lower measured overprecision than men, and the associations with age and the number of questions answered change sign relative to Table 1. The coefficients on education, financial literacy, and narcissism are no longer statistically distinguishable from zero. Appendix C discusses the interpretation of this residual construction.

and a vector of control variables of the form:

$$y_i = \alpha + \beta sop_i + \boldsymbol{\gamma'} \boldsymbol{X_i} + \epsilon_i, \tag{3}$$

where sop_i denotes the standardized overprecision measure for respondent i, $\boldsymbol{X_i}$ is a vector of control variables, and ϵ_i is the random error term. We include all available control variables from the SOEP-IS that might be correlated with the outcome we study or with overprecision, based on prior literature. These include age, gender, years of education (used as a proxy for cognitive ability), monthly gross labor income, dummy variables for the labor force status (employed, unemployed, maternity-leave, non-working, and retired), measures of impulsivity, patience, narcissism, financial literacy, and risk preferences, a dummy variable for having lived in the German Democratic Republic in 1989, the number of overprecision questions answered by each respondent, state fixed effects, and interview date (month and year) fixed effects. The latter absorbs any variation in the outcome driven by when the survey was run, e.g., in the development of asset prices. Additionally, we include a measure of political interest in the political analyses.[21] We estimate (3) using OLS and present the point estimate of the coefficient on the standardized overprecision measure sop_i from the full regression and its unadjusted p-value respectively in Columns (1) and (2) of Table 2.[22] Since we test the behavior of respondents across several dimensions, we also report the Sidak-Holm adjusted p-values to correct for multiple hypothesis testing (MHT) in Column (3).

Second, we derive the "R^2 rank" of our standardized measure of overprecision sop_i using a forward stepwise regression, similar to Cobb-Clark et al. (2022). We obtain this by running a stepwise regression in which variables are added to the model sequentially. In step 1, we regress the behavior of interest on each of the K control variables in the specification separately. Among these K regressions, we select the control variable with the highest R^2, as it explains most of the observed variation in the data. In step 2, we regress $K-1$ times the behavior of interest on the control variable selected in the first

[21] In Table B.10 in the appendix, we also include the Big Five personality traits (Rammstedt and John, 2007). These are only available from the 2017 SOEP-IS, and because not all respondents in our sample responded to them, we lose 55 observations. Yet the results remain robust to including the Big Five personality traits.

[22] Adjusting the degrees of freedom by the number of questions used to construct the measure of overprecision does not significantly affect the results.

step, plus each of the $K-1$ remaining controls. We continue until all K variables have been added to the model. The resulting R^2 rank is determined by the step at which each control variable is added to the model. Therefore, the better the "R^2 rank" of sop_i, the more the variable can explain the variation in the outcome, i.e., rank 1 delivers the highest R^2. We report the results in Column (4) of Table 2, along with the maximum number of variables included in the model, as specified above.[23]

Third, we employ a least absolute shrinkage and selection operator (LASSO) to test whether our overprecision measure has predictive power for the outcome variable in an out-of-sample prediction. LASSO is a machine learning technique frequently applied to improve the predictive power of statistical models. The objective of the LASSO approach is to choose those variables with the highest predictive power from the set of candidate variables. It does so by estimating a penalized regression by minimizing the sum of squared residuals along with a penalty term for the sum of the coefficients.[24] This is implemented via cross-validation, where the estimator partitions the data into folds for training and testing and selects the penalty term that minimizes out-of-sample prediction error on the test data.[25] If sop_i is included, the cross-validated LASSO has retained it as a predictor of the outcome. We report the results in Column (5) of Table 2, along with the number of control variables selected by LASSO and the resulting R^2 of the model in Column (6). In Column (7), we report the number of observations, which may vary due to missing data in the outcome variables.[26]

4.2 Results

Table 2 summarizes the results of our three analytical approaches. We first discuss financial behavior outcomes, then political behavior outcomes. For each outcome, we present

[23]Note that each binary representation of a possible realization of the fixed effects could be included as an independent variable. This leads to 41 and 42 as the maximum number of variables.

[24]Formally $\min_\beta \frac{1}{2N}\sum_{i=1}^{N}(y_i-\alpha-\sum_j \beta_j x_{ij})^2+\lambda\sum_j|\beta_j|$ for the linear case, where j are the coefficients which are included in the model and λ is a given tuning parameter. See Tibshirani (1996) for more details.

[25]The algorithm proceeds step-wise and estimates the model for each λ starting at the smallest λ that delivers zero non-zero coefficients and ending at a λ of 0.00005 in a grid of 100. Each step could add or remove a different number of variables from the model.

[26]A test of the means of personal characteristics for the estimation samples and the entire sample (N=805) shows no significant differences. The only exception is a slightly higher share of male respondents in the stock market regressions. Therefore, we consider the estimation samples representative of the full sample (N=805).

	(1) Point estimate	(2) Unadj. p-value	(3) MHT p-value	(4) R^2 rank	(5) LASSO included	(6) LASSO R^2	(7) N
Financial Behavior:							
DAX Error	0.083*	0.056	0.108	4/41	yes/14	0.10	548
1-year ahead	*0.529*	*0.308*		*7/41*	*yes/21*	*0.16*	*578*
2-year ahead	*3.078***	*0.016*		*4/41*	*yes/8*	*0.04*	*557*
Diversification	-0.131***	<0.001	0.002	3/41	yes/19	0.14	774
Political Behavior:							
Extremism	0.087**	0.041	0.117	6/42	yes/14	0.05	716
Left-Right	-0.008	0.854	0.854	18/42	no/14	0.08	716
Abstention	0.032**	0.011	0.041	3/42	yes/18	0.14	706

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 2: Estimation results of Section 4. The number of observations (Column (7)) varies due to missing observations in the outcome variable. The maximum number of observations is 805. Column (1) lists the point estimate of the coefficient on the standardized overprecision measure *sop* from the full regression as specified in Section 4.1 along with the unadjusted p-value (Column (2)) and the Sidak-Holm adjusted p-value for multiple hypothesis testing (MHT) (Column (3)). Column (4) displays the R^2 rank in the forward stepwise regression as specified in Section 4.1 along with the maximum possible variables to be included in the model. The regressions with political outcomes as the dependent variable also include a self-reported political interest measure. Column (5) specifies the result of the LASSO procedure as specified in Section 4.1 along with the number of control variables chosen by LASSO and the R^2 of the estimated model (Column (6)). Variable definitions are in Table B.4 in the appendix.

the literature's theoretical predictions. To get a sense of the relative effect sizes of the various regressors, the full regression results for column (1) can be found in Table B.7 of Appendix B. As robustness tests, Table B.8 of Appendix B presents the results of running a Seemingly Unrelated Regression (SURE) (Zellner, 1962),[27] and in Appendix C we report the five alternative approaches mentioned in Section 3.2. These checks clarify which associations are stable across closely related OP measures and which are sensitive to residualized or realized-error-adjusted constructions.

Additionally, in Appendix D, we conduct a specification curve analysis (Simonsohn et al., 2020) to evaluate whether the OLS associations are sensitive to modeling choices around the SEM-based overprecision measure. The grid varies controls, fixed effects, standard-error choices, outcome transformations, and two closely related aggregation

[27] We use SURE as a robustness check since some of the outcome variables are potentially correlated. Estimating them jointly via SURE increases efficiency and accounts for cross-equation error correlations. To do so, we estimate two models: One with all outcomes in Table 2 except the one- and two-year-ahead regressions (due to perfect linearity with the DAX forecast error by construction), and another that omits all DAX forecast error outcomes to minimize the loss of observations. The SURE specifications produce similar estimates in all estimable cases using the full sample.

methods—the standardized mean and an SEM latent-factor score—while keeping the target construct fixed. This exercise shows that the main associations, especially for portfolio diversification and voting abstention, are not artifacts of a narrow empirical specification. Appendix J also examines how these associations change when we account for realized error or use a population-relative benchmark.

Financial Behavior Outcomes

The first hypothesis addresses forecast errors of asset price predictions in the stock market. Benos (1998) and Odean (1998) argue that overprecise investors hold incorrect beliefs about future asset valuations because they overweight their private signals when forming expectations. Deaves et al. (2019) provide direct empirical support for the association between overprecision and forecast errors in financial markets by correlating German stock market forecasters' predictions with a measure of overprecision. Furthermore, Hilary and Menzly (2006) provide evidence supporting this association among North American analysts.[28] Therefore, we expect overprecise respondents to be less accurate in their predictions. To test this prediction, we use the absolute distance between the one- and two-year-ahead predictions of the German Stock Index (DAX) and the realized value.[29] Since individual forecast errors might be prone to random noise, we aggregate both errors using the principal component, which we standardize to be mean zero and standard deviation one (*DAX Error*).[30] Additionally, we report the results for each forecast error separately (*one-year-ahead* and *two-year-ahead*).[31]

Table 2 shows a positive association between overprecision and DAX forecast errors. An increase of one standard deviation in overprecision is associated with an increase of 0.083 standard deviations in the principal component of forecast errors. Overprecision is

[28]However, note that, unlike our method, Hilary and Menzly (2006) and Deaves et al. (2019) rely on indirect proxies to construct their measure of overprecision.

[29]We use the closing price based on the interview day to calculate exact forecast errors for each respondent.

[30]The principal component analysis shows a strong first factor with an eigenvalue of 1.65. All other factors are below the commonly used Kaiser criterion of 1.0.

[31]We include a dummy variable that indicates asset ownership as a control variable in the predictions to account for different information sets in a robustness test in Table B.11 in the appendix. This change does not affect the qualitative results, although it reduces the sample size.

also selected by the cross-validated LASSO and ranks fourth out of 41 candidate variables in the forward stepwise regression.

Next, we test the theoretical prediction by Odean (1998) that overprecision is associated with underdiversified portfolios. Overprecise investors overweight their private information, leading them to trade excessively and to concentrate on a limited number of assets. Goetzmann and Kumar (2008) provide empirical evidence supporting this prediction for traders in the US, and Merkle (2017) does so for traders in the UK. While the former study relies on an asset turnover proxy, the latter directly elicits overprecision through survey questions. While both studies rely on revealed portfolio data, we test this prediction using self-reported hypothetical allocations. Specifically, we construct a standardized diversification measure (mean zero, standard deviation one) based on participants' allocation choices across five asset categories: stocks, real estate, government bonds, savings, and gold.[32,33]

Our results are consistent with the theoretical prediction that overprecision is associated with underdiversification. The point estimate in column (1) in Table 2 shows that a 1 standard deviation increase in overprecision is associated with a 0.131 standard deviation decrease in our diversification measure. That is, the hypothetical allocations of more overprecise respondents are more concentrated in particular asset categories. Moreover, overprecision is included in the LASSO model and ranks third in the forward stepwise regression approach.

Political Views and Voting Behavior

According to Ortoleva and Snowberg (2015b), overprecision leads individuals to believe that their own experiences are more informative about politics than they really are. For

[32]The measure assumes that the "optimal" portfolio diversifies evenly across the different investment options (see Table B.4 in the appendix for details). However, one could argue that each dollar invested in stock is more diversified than its equivalent investment in gold. As a robustness check, we therefore use an alternative "optimal" portfolio benchmark in which maximum diversification is achieved by investing 35% in stocks, 35% in bonds, 10% in gold, 10% in real estate, and 10% in the savings account. Under this benchmark, the association between overprecision and diversification remains statistically significant and negative. We thank an anonymous referee for this suggestion.

[33]The measure of diversification displays an inverse-U relation with risk aversion. More risk-averse respondents skew their portfolio toward safer assets such as savings and gold. In contrast, more risk-loving respondents skew their portfolio toward riskier assets such as stocks and real estate. This lends credibility to the diversification measure. However, we do not claim an optimal degree of portfolio diversification. Rather, we test whether, conditional on the individual degree of risk aversion, overprecision is associated with a tendency towards a certain asset category.

instance, overprecise people may consult biased media outlets without fully accounting for this bias, or exchange information on social media without recognizing that much of it comes from politically like-minded peers. Against this background, the authors theoretically and empirically demonstrate that overprecision leads to ideological extremism and strengthens the identification with political parties, increasing the likelihood of voting. Yet, the literature remains inconclusive on whether these associations hold for liberals and conservatives alike. While Ortoleva and Snowberg (2015b) find that conservatives seem more susceptible to overprecision than liberals, in a different paper Ortoleva and Snowberg (2015a) show that this association holds only in election years while Moore and Swift (2011) only find a weak and non-significant correlation between overprecision and conservatism.

To test whether overprecision correlates with respondents' political preferences, we use their self-reported ideology on a scale from 0 (extreme left) to 10 (extreme right) to construct the variable *left-right*. Using the same question, we also construct *Extremism*, which measures how far respondents perceive themselves to be from the political center on a scale from 0 to 5. We standardize both variables to be mean zero and standard deviation one. Finally, to study whether overprecise respondents are more likely to vote, we use a dummy variable that equals 1 if a respondent indicated being a nonvoter in the (ex-post) opinion poll (*Sonntagsfrage*) for the 2017 federal elections to the German *Bundestag* (*Abstention*).[34]

Consistent with Ortoleva and Snowberg (2015b), the results in Table 2 suggest that overprecision is positively correlated with ideological extremism. Overprecision is among the variables chosen by the LASSO estimation and ranks high (sixth) in the forward stepwise regression approach. In line with Ortoleva and Snowberg (2015a), we find no evidence that overprecision is more strongly associated with one side of the political spectrum, as it is not correlated with political ideology and is not selected by the LASSO estimation. Furthermore, overprecision ranks quite low (18/42) in the R^2 rank approach. Finally, overprecision is associated with voting abstention in the baseline specification, as the LASSO estimation selects it and the R^2 rank approach ranks it third. As shown in Table 2, a one-standard-deviation increase in overprecision is associated with a 3 percentage

[34]The year in which the survey took place (2018) is a non-election year.

point increase in the likelihood of not voting.[35]

This finding differs from Ortoleva and Snowberg (2015b). One possible explanation is the institutional setting. Their analysis concerns a two-party system in which strongly identified partisans generally support parties with a substantial probability of winning (e.g., Miller and Conover, 2015). Germany has a multiparty system in which more extreme voters may support smaller parties with more limited electoral prospects.[36] The relationship between extremism, partisanship, and turnout may therefore differ across the two settings.

4.3 Linking Overprecise Behavior Across the Financial and Political Domains

The previous results show that overprecision correlates with respondents' financial and political behavior. However, an important question remains: does a single underlying cognitive bias relate to overprecise behavior across both domains, or do distinct aspects of overprecision independently shape financial and political decisions?

Identifying whether overprecision arises from a common bias or domain-specific mechanism has important theoretical and practical implications. Theoretically, if overprecision reflects a stable underlying domain-general bias, this would support the view that core cognitive processes, such as metacognitive monitoring and uncertainty calibration, are robust and generalizable across contexts (Koriat, 2007; Yeung and Summerfield, 2012). From a practical perspective, if a single underlying bias drives overprecision, interventions aimed at recalibrating confidence, such as training programs to better estimate uncertainty or feedback mechanisms to enhance metacognitive awareness, could enhance decision-making simultaneously in financial, political, and other domains.

To assess whether overprecise behavior across domains originates from a common cognitive bias or whether distinct facets of overprecision independently relate to behavior in each domain, we implement a two-step approach. This method disentangles domain-

[35]Both political associations are more sensitive to how we account for realized error than the diversification association. The extremism estimate changes sign across these adjustments, while the abstention estimate falls close to zero when we compare respondents with the same observed average realized error (Appendix J.3).

[36]The AfD's current electoral position should not be projected back onto the period covered by our data. In the 2017 federal election, shortly before our survey, it received 12.6% of the second vote and entered the Bundestag for the first time. By 2025, its vote share had risen to 20.8%, making it Germany's second-largest party.

specific variation from the broader latent bias of overprecision and quantifies its cross-domain influence.

In the first step, we regress the outcome Y_{ij} of individual i in domain j (i.e., financial or political behavior) on overprecision (sop_i) and a vector of control variables (X_i) as in eq. 3:

$$Y_{ij} = \alpha_j + \beta_j sop_i + \gamma_j' X_i + \epsilon_{ij}, \tag{4}$$

where α_j is a constant, β_j captures the relationship between overprecision and behavior in domain j, and ϵ_{ij} represents the error term. From this regression, we obtain the fitted values $\hat{Y}_{ij}$, which capture the variation in the respective outcomes attributable to overprecision in domain j.

In the second step, we use the first-step fitted values $\hat{Y}_{ij}$ as predictors in the models for the remaining outcomes Z_{ik} (where $j \neq k$)[37]:

$$Z_{ik} = \delta_k + \phi_{j \neq k} \hat{Y}_{ij} + \tau_k' X_i + \nu_{ik}, \tag{5}$$

where $\phi_{j \neq k}$ quantifies the extent to which overprecision-related variation in one domain's outcomes (e.g., financial) is correlated with variation in another domain's outcomes (e.g., political). The key assumption is that the first-step fitted values $\hat{Y}_{ij}$ in the second step primarily reflect the influence of overprecision in domain j. If this is the case, then ϕ provides a meaningful estimate of cross-domain correlation attributable to overprecision.

Importantly, this approach does not assume a strict causal relationship between financial and political overprecision, nor does it impose a mediation framework where financial overprecision serves as the sole channel through which overprecision influences political behavior. Instead, it isolates the component of financial behavior attributable to overprecision and examines whether this component is systematically associated with overprecise political behavior, and *vice versa*. This allows us to assess whether overprecision is a general cognitive bias that consistently manifests across decision-making contexts, or whether it operates independently within financial and political domains.[38]

[37] For example, if Y_{ij} in the first step represents Portfolio Diversification, then in the second step, Z_{ik} include *DAX Error*, *Extremism*, and Voting Abstention.

[38] See Appendix F for a more formal explanation of our statistical model and a discussion of alternative approaches.

			Dependent variable in			
			Financial Domain		Political Domain	
			DAX Error	Diversification	Extremism	Abstention
			(1)	(2)	(3)	(4)
Overprecision-related variation in	Financial Domain	DAX Error		-1.573***	1.050**	0.380**
				(0.436)	(0.512)	(0.148)
		Diversification	-0.636*		-0.667**	-0.242**
			(0.331)		(0.325)	(0.094)
	Political Domain	Extremism	0.831*	-1.407***		0.362**
			(0.501)	(0.420)		(0.141)
		Abstention	2.295*	-3.886***	2.763**	
			(1.384)	(1.159)	(1.347)	

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 3: Coefficients from the second-step regression (eq. 5), which regresses the dependent variable of the respective column separately on each of the predicted values in the respective row, along with a vector of controls. The predicted values in each row are obtained by regressing the variable in that row on overprecision and a vector of controls. The first step corresponds to the results in Table 2. Standard errors in parentheses. Variable definitions are in Table B.4 in the appendix.

Table 3 summarizes the results, presenting 12 estimates—two within the financial domain, two in the political domain, and eight across domains.[39] The coefficients show how a one standard deviation increase in overprecision-related behavior in domain j correlates with outcomes in domain k. For instance, consider the coefficient describing the relationship between overprecision in portfolio diversification and political extremism. In this case, the coefficient of -0.667 indicates that a one standard deviation increase in the overprecision-induced component of diversification is associated with a 0.667 standard deviation reduction in political extremism. The significant correlations across all tested relationships suggest that a common source is associated with overprecise behavior in multiple domains.

To our knowledge, this is the first study to provide evidence that overprecise behavior in one domain is systematically associated with overprecise behavior in another. This result matters not only because it bridges the political and financial behavior literature but also because it highlights the systematic nature of overprecision in decision-making,

[39]Given the hypothesized (and confirmed) null result for ideological inclination, we exclude this outcome from the analysis.

opening the door to interdisciplinary work on its mitigation.

5 Overprecision Across Domains

Section 4 shows that, by focusing on questions related to 20th-century history, the SEM captures a measure of overprecision associated with various aspects of respondents' political and financial behaviors, consistent with theoretical results from the literature. However, previous empirical literature suggests that overconfidence varies depending on the nature of the questions posed (e.g., Klayman et al., 1999). To examine whether the SEM produces related measures across tasks, we conduct a preregistered online survey with a representative sample of the German population (#118284 at www.aspredicted.org). In it, we measure overprecision across five distinct domains and examine whether the domain-specific measures are related using principal component analysis, partial correlations, and the leave-one-out analysis introduced in Morrison and Taubinsky (2023).[40]

5.1 Survey Design

The survey, programmed in Qualtrics, consists of five blocks of five questions each across five distinct domains (see the full set of questions in Table B.12 in the appendix). All respondents go through all domains, which are:

1. ***Neutral***: In this domain, respondents are flashed five different 20×20 matrices of black triangles and gray squares for 8 seconds each. After viewing each matrix, they estimate the number of black triangles in each of the five matrices and report their subjective error. This task, adapted from Bosch-Rosa et al. (2024), does not require recalling a previously learned factual answer. It therefore allows us to examine whether SEM responses are related across domains even when one task does not require factual recall. The matrices range from 70 to 330 black triangles.

[40]Formally, assume that through the SEM we elicit the prior precision estimates $\tilde{\sigma}_{i,n}$ of individual i across n different domains while the true level of prior precision of individual i is $\sigma_{i,n}$. If the stated prior precision of individual i is a linear transformation of the true prior precision, such that $\tilde{\sigma}_{i,n} = k_{i,n} \times \sigma_{i,n}$, then the parameter $k_{i,n}$ can be interpreted as the degree of overprecision of individual i in domain n. However, overprecision may vary across domains. In this case, the elicited degree of overprecision can be expressed as $k_{i,n} = \lambda_i + e_{i,n}$, where $e_{i,n}$ represents domain-specific characteristics, such as knowledge in that respective domain. The companion survey allows us to examine whether the domain-specific measures share a common component, represented by λ_i.

See Figure A.5 in Appendix A for an example with 120 black triangles and 280 gray squares.

2. ***Contemporary history***: This domain includes the five most frequently answered questions in the SOEP-IS survey, enabling a direct comparison between responses in our online survey and those in the SOEP-IS survey. In Appendix E, we show that respondents' answers are not substantially different across the two surveys. This domain also lets us replicate the stock price prediction findings from Section 4.2, as reported in Appendix H.2.

3. ***General knowledge***: In this domain, respondents answer five general knowledge questions and report their subjective error for each. Examples include the number of teeth in an adult polar bear, the number of keys on a concert piano, or the number of African countries in the United Nations. This domain encompasses a broad range of knowledge and covers topics commonly encountered in daily life.

4. ***Future stock prices***: In this domain, respondents predict the 28-day price forecast of five different assets (Benz, Puma, BMW, Deutsche Post, BASF) and report their subjective error for each. To do so, we used a widget from *tradingview.com* that let us present respondents with an interactive price chart for each stock over the past year. This domain lets us test overprecision in a financial context and, importantly, measure overprecision with respect to future events. This distinction matters because, unlike the other domains, the correct answer is not known at the time of questioning.

5. ***Economics***: In this domain, respondents answer five questions related to the German economy and report their subjective error for each one. To keep responses consistent, all questions in this domain are limited to percentage changes. Examples include the percentage change in the German CPI from 2011 to 2021, the percentage change in the German nominal GDP from 2006 to 2021, and the percentage change in the German DAX from 2014 to 2021. We include this domain because overprecision matters in economic decision-making and is directly relevant to the financial behavior analyzed in the SOEP-IS survey.

All respondents started with the neutral domain, since it began with three practice rounds to familiarize them with the matrices.[41] After this, we randomized the order of the remaining domains. Before each domain, respondents self-reported their knowledge of the topic on a scale from 0 (not knowledgeable at all) to 100 (very knowledgeable).

To ensure data quality, we introduced attention checks throughout the survey. Respondents who failed these checks were automatically removed from the survey. To account for the use of *Google* or other search engines, respondents were asked at the end of the survey whether they had used such methods to answer the questions. Additionally, in the domains of contemporary history, general knowledge, and economics, we included an additional question that served as a "*Google* control." These questions were presented in the same format as all others, but are difficult and unlikely to be known by respondents. Thus, any respondent answering all of them correctly is likely using external help and should be potentially excluded from the sample.[42] To our knowledge, this is the first time that such a method has been implemented, and it could prove valuable for online surveys that ask for publicly available information, particularly when the answers are monetarily incentivized (Haaland et al., 2023; Stantcheva, 2023).

Additionally, we collect sociodemographic variables including age, gender, years of education, income, and nationality. We also assess mathematical literacy by asking respondents to solve three mathematical problems and report the effort they put into answering the survey. Table B.14 in Appendix B lists all variables collected in the survey.

5.2 Data

We collected 1,000 complete responses. As preregistered, we excluded respondents who admitted to using a third party to answer our questions, respondents identified as 'Googlers'

[41] The difference between the practice rounds and the main rounds is that, in the practice rounds, respondents see the correct answer after they answer the questions, which is not the case in the main rounds.

[42] Following our pre-registration, we consider a respondent to have used a search engine if two conditions are met. In any domain: i) answering the *Google* controls correctly and stating a subjective error of zero, and ii) answering correctly and stating zero subjective error for at least three other questions in this domain. If a respondent is flagged as having used search engines, then they are excluded from the analysis. The control questions we used to determine if respondents were using *Google* included: the birth year of Joachim Sauer, Angela Merkel's husband (1949), the upper weight limit in kilograms for the female Olympic boxing Bantamweight class (54kg), and the percent change of M1 in the Euro area from 2015 to 2021 (86%). Importantly, we do not use the answers to these questions to measure respondents' overprecision.

through our control questions, and those in the bottom five percent on the self-reported effort measure. This leaves us with 839 respondents for the baseline analysis. Table B.15 in Appendix B includes summary statistics for the full and cleaned samples. The data-cleaning process does not substantially alter the sample composition with respect to the collected variables.[43]

5.3 Individual Overprecision

In Figure 5, we plot the distributions of the answers across all five domains, similar to Figure 2. For most questions across the five domains, answers cluster around the true answer, indicating that respondents were attentive and able to answer the questions. However, the dispersion of the answers and the distance of the mode from the true answer clearly indicate that some questions were easier than others. In Figure 6, we plot the realized error ($error_{i,j}$) on the vertical axis against the subjective error ($se_{i,j}$) on the horizontal axis for each of the five questions in each domain, similar to Figure 3. We add a 45-degree line, where any dot above represents an overprecise observation ($error_{i,j} > se_{i,j}$) and any point below represents an underprecise observation ($error_{i,j} < se_{i,j}$). The results show that respondents are, on average, overprecise across all questions and domains.

5.4 Overprecision Across Domains

For each domain, we construct an aggregate measure of overprecision using the procedure in Section 3.2.[44] As preregistered, to have meaningful estimates and decrease noise, we only construct the aggregate overprecision measure for each domain if the respondent answered at least four out of the five questions. For better comparability, we standardize the aggregate measure for each domain to have a mean of zero and a standard deviation of one, as we did with the SOEP-IS data.[45]

[43] In Appendix G, we use different exclusion restrictions on the sample as specified in the pre-registration and show that our results are robust to the different restrictions.

[44] Before aggregating, we divide the answers in the neutral domain by four to account for the differing scale. In our pre-registration, we specified that we would use both the simple average and the principal component across all domains for aggregation. However, because the results change only marginally, we do not report results using the principal component as an aggregation method; these results are available upon request.

[45] Standardizing the aggregate measure does not change the qualitative results.

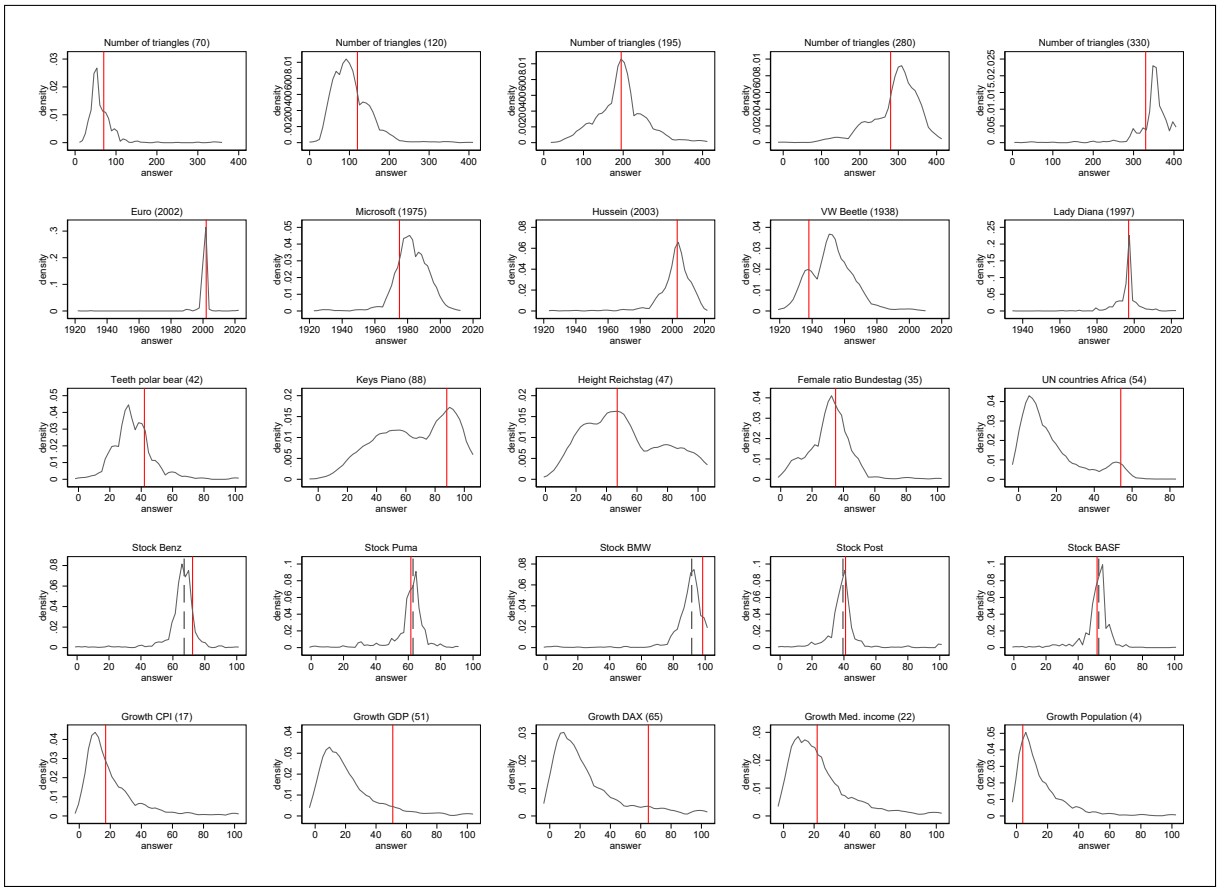


Figure 5: Density of the answers ($a_{i,j}$) for each question. The vertical red line marks the correct answer. For the stocks, the vertical red line marks the weighted average of the 28-day-ahead stock price, and the vertical black dashed line marks the weighted average of the stock price on the interview day. Note that the vertical axis scale differs across questions. The corresponding graphs for the additional questions used to detect search-engine use appear in Figure A.3 in the appendix.

We analyze the relationship among the domain-specific overprecision measures in three ways: i) through a principal component analysis, ii) through a partial correlation analysis, and iii) through a leave-one-out analysis following Morrison and Taubinsky (2023). We confirm the robustness of the results in Appendix G using different exclusion restrictions, as specified in the pre-registration.

Principal component analysis: Principal component analysis (PCA) summarizes variation across variables by constructing weighted combinations called principal components. The first component captures the largest possible share of the total variance. Our analysis includes the 552 respondents for whom we could calculate an aggregate overprecision score in each of the five domains. Only the first component meets the Kaiser criterion of an eigenvalue greater than 1.0. Its eigenvalue is 2.29, meaning that it explains

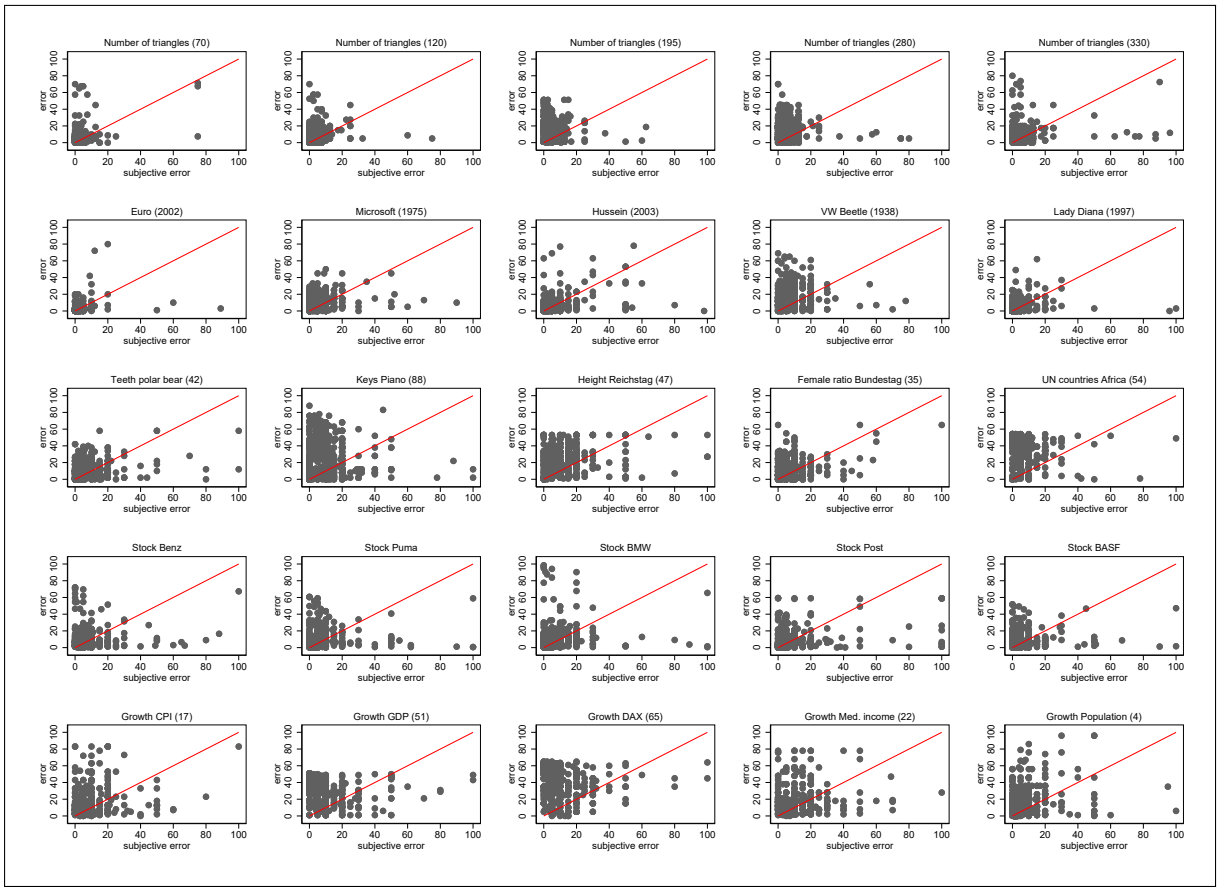


Figure 6: Relation between the realized error ($error_{i,j}$) on the vertical axis and the subjective error ($se_{i,j}$) on the horizontal axis. Any dot above the 45-degree red line is an overprecise answer by the respondent, while any dot below indicates an underprecise one. Note that we divide the results for the neutral domain by four to make them comparable to the other domains. The corresponding graphs for the additional questions used to detect the use of search engines can be found in Figure A.4 in the appendix.

45.8% of the variance across the five standardized domain measures. All five domains load positively on this component.[46] These findings are consistent with a shared component of SEM-measured overprecision across tasks, alongside task-specific variation.

Partial correlation analysis: Partial correlation analysis estimates the correlation between two domain-specific overprecision measures while holding the control variables constant. Table 4 presents the partial correlation coefficients, controlling for age, gender, education, income, nationality, Bundesland, mathematical literacy, and self-reported domain knowledge. The correlations are positive across all domain pairs and generally weaker for the triangle-count task. The strongest relationship is between contemporary

[46] The component loadings are 0.52 for the neutral task, 0.76 for contemporary history, 0.74 for general knowledge, 0.69 for stock prices, and 0.65 for economics.

Domain 1	Domain 2	Correlation	Standard error	N
Neutral	History	0.17	0.03	688
Neutral	Knowledge	0.21	0.04	626
Neutral	Stocks	0.30	0.04	676
Neutral	Economics	0.18	0.04	639
History	Knowledge	0.52	0.04	609
History	Stocks	0.35	0.04	638
History	Economics	0.37	0.05	620
Knowledge	Stocks	0.34	0.04	591
Knowledge	Economics	0.32	0.04	580
Stocks	Economics	0.28	0.04	618

Table 4: Partial correlations between domains. This table shows the partial correlations between the aggregate overprecision measures across the five domains. The control variables are age, gender, education, income, nationality, state, mathematical literacy, and a measure of self-reported knowledge on the domain topic.

history and general knowledge.[47] While these are somewhat weak correlations, they are in line with previous literature correlating measures of overprecision (e.g., Glaser and Weber, 2007) and can be visually confirmed in Figure 7, which presents a binned scatter plot of aggregate overprecision measures across domain pairs.[48]

Leave-one-out analysis: The leave-one-out analysis adopts the methodology described in Morrison and Taubinsky (2023). We first divide the overprecision sample for each domain into two groups: the top 25% (*High* group) and the bottom 75% (*Low* group) based on their overprecision scores. We then pair each domain with all others and use a t-test to check whether participants in *High* show significantly higher overprecision than those in *Low* in all other domains. The rationale is that if the SEM captures the same trait across domains, respondents' overprecision rankings should be consistent across domains, with those in the *High* group in one domain also being more overprecise than those in the *Low* group in the other domains. Table 5 shows that, across all domain pairs, respondents in the upper quartile of overprecision in one domain are significantly more overprecise in

[47]If we winsorize the data at the 99% and 1% percentiles, the average correlation drops slightly, with all values compressed toward 0.3. This is common for cognitive biases, which usually follow skewed distributions. See the winsorized correlation values in Table B.9 of Appendix B.

[48]There is no consensus in the literature for "sufficiently large" correlation values. While Evans (1996) considers a coefficient of .30 across personality traits a "weak" correlation, Cohen (1988) considers it a "medium" correlation. In fact, he notes that "Many of the correlation coefficients encountered in behavioral science are of this order of magnitude [0.3], and, indeed, this degree of relationship would be perceptible to the naked eye of a reasonably sensitive observer" (Cohen, 1988, p. 80). In our case, Figure 7 confirms this visual perception.

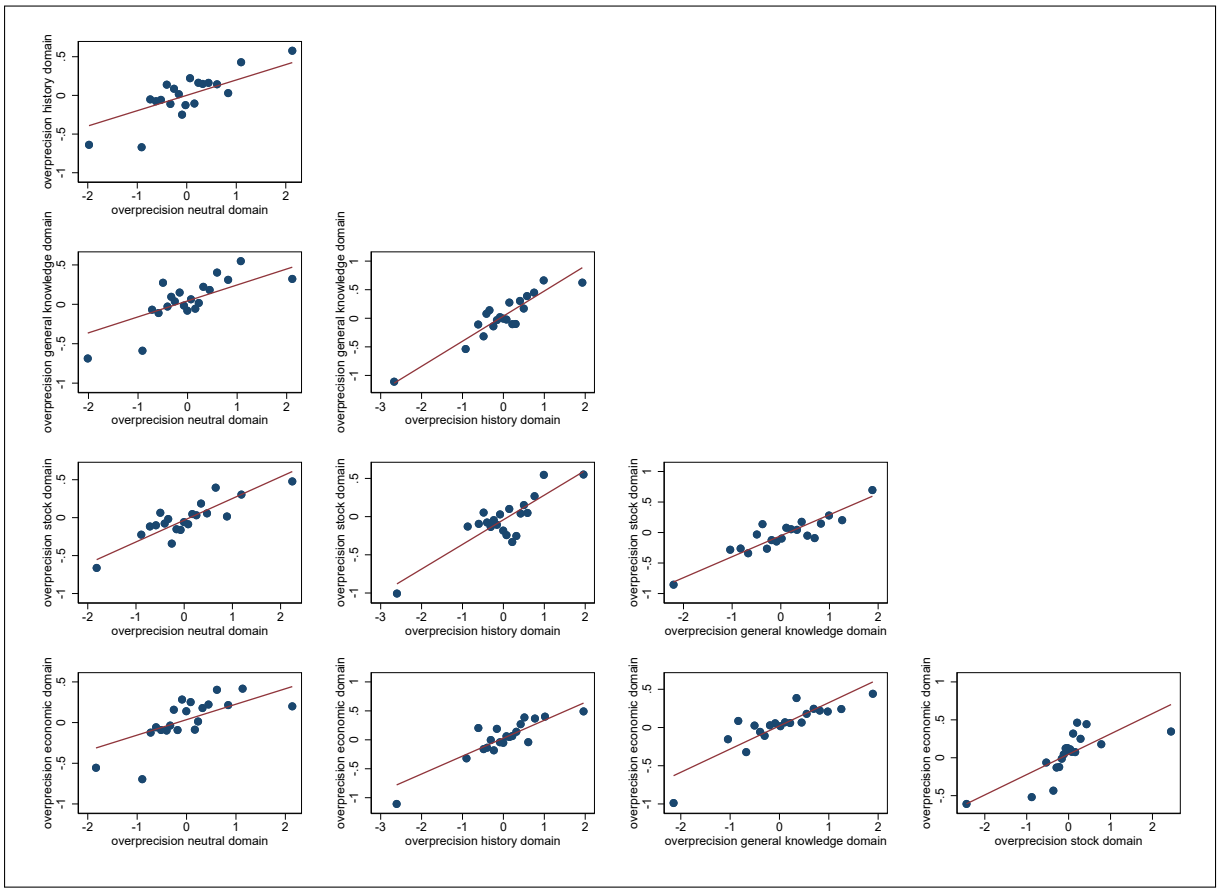


Figure 7: Correlation across domains using binned scatter plots for each pair of overprecision measures. Each plot uses 20 bins. The red line is a linear fit between the two dimensions.

the other domain than respondents in the lower quartile.[49]

Taken together, the three preregistered analyses show that the SEM measures share a meaningful common component across the five tasks, while also retaining domain-specific variation. The positive loading and correlations of the neutral triangle-count task indicate that this common pattern is not confined to factual-recall questions. The stronger relationships among the knowledge-based domains may nevertheless partly reflect their shared reliance on factual knowledge. These findings support using the SEM across different tasks, including settings that do not require factual recall, where this particular concern is less pronounced. More generally, concerns about factual knowledge are less consequential for interpreting behavioral associations when task-specific knowledge is un-

[49]While Table 5 presents the results for twenty different tests, it is clear that the t-test results are not driven by any sort of multiple hypothesis testing problem, and that any correction for it would result in qualitatively identical results.

Sample		Lower 75%			Upper 25%			Difference	
In	Out	mean	sd	N	mean	sd	N	Difference	P-value
Neutral	History	-0.08	1.09	540	0.27	0.71	163	-0.35	< 0.01
Neutral	Knowledge	-0.06	0.88	484	0.35	0.96	149	-0.41	< 0.01
Neutral	Stocks	-0.10	0.92	522	0.24	0.99	168	-0.34	< 0.01
Neutral	Economics	-0.04	0.90	495	0.29	0.69	152	-0.32	< 0.01
History	Neutral	-0.05	0.86	621	0.16	1.23	165	-0.22	0.01
History	Knowledge	-0.09	0.90	489	0.48	0.84	144	-0.57	< 0.01
History	Stocks	-0.11	0.87	536	0.32	1.09	154	-0.43	< 0.01
History	Economics	-0.05	0.90	499	0.34	0.69	148	-0.38	< 0.01
Knowledge	Neutral	-0.05	0.92	629	0.18	1.05	157	-0.23	0.01
Knowledge	History	-0.10	1.06	552	0.35	0.79	151	-0.45	< 0.01
Knowledge	Stocks	-0.09	0.88	541	0.24	1.12	149	-0.32	< 0.01
Knowledge	Economics	-0.03	0.91	499	0.27	0.67	148	-0.30	< 0.01
Stocks	Neutral	-0.10	0.88	621	0.35	1.12	165	-0.45	< 0.01
Stocks	History	-0.09	1.05	551	0.33	0.81	152	-0.42	< 0.01
Stocks	Knowledge	-0.04	0.90	499	0.33	0.90	134	-0.37	< 0.01
Stocks	Economics	-0.05	0.87	495	0.32	0.78	152	-0.37	< 0.01
Economics	Neutral	-0.07	0.93	634	0.27	0.99	152	-0.34	< 0.01
Economics	History	-0.08	1.06	561	0.31	0.77	142	-0.38	< 0.01
Economics	Knowledge	-0.02	0.89	500	0.27	0.97	133	-0.29	< 0.01
Economics	Stocks	-0.10	0.91	542	0.28	1.01	148	-0.38	< 0.01

Table 5: Results of the leave-one-out analysis. *In sample* signifies the domain in which the quartiles based on overprecision were computed. The sample is partitioned into two groups: the lower 75% and the upper 25%. *Out sample* signifies the domain in which overprecision is then measured and tested between both groups.

related to the outcome of interest.

6 Conclusion

We study how overprecision is associated with financial and political behavior using data from the 2018 wave of the German Socio-Economic Panel Innovation Sample (SOEP-IS). We measure overprecision using the Subjective Error Method (SEM), a simplified version of the MISS method (Soll et al., 2022).

Our results show that SEM-measured overprecision is associated with financial and political behavior in ways that support several theoretical predictions. We also find that the financial and political behaviors associated with the SEM score are linked for the same respondents. These findings bring together two literatures that have largely developed independently. Two preregistered companion surveys help clarify what the SEM captures. In the first, we apply the SEM in five domains and find that respondents who are more overprecise in one domain tend to be more overprecise in others. In the second, respondents who report larger subjective errors also provide more dispersed probability distributions within the same exercise, supporting the interpretation of subjective error as stated uncertainty.

Our findings have two main limitations. First, our SOEP-IS application uses factual questions, so respondents' errors may reflect differences in knowledge as well as noise in individual answers. Our additional analyses show that some associations with behavior change when we account for differences in accuracy. This concern matters less when tasks do not require factual recall or when the relevant knowledge is unrelated to the behavior being studied. Second, our main evidence relies on a single method, the SEM. Li et al. (2025) find that measures obtained through different elicitation methods are only weakly related for the same respondents. We therefore do not know whether other methods would rank respondents similarly or produce similar associations with behavior. Our second companion survey helps explain how respondents interpret the subjective-error question, but it does not establish whether different methods measure overprecision in the same way.

Overall, we show that the SEM offers a simple and scalable way to compare the errors respondents make with those they expected to make. Respondents express their uncertainty in familiar units, without having to construct confidence intervals or probability distributions. This makes the method suitable for general-purpose surveys and allows researchers to study how calibration relates to a wide range of outcomes in large, diverse

samples. Future research using multiple elicitation methods can examine whether other measures of overprecision produce similar respondent rankings and behavioral associations. Longitudinal studies could examine how calibration and behavior change over time, while experiments could help establish whether changes in calibration affect behavior.

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A Extra Figures

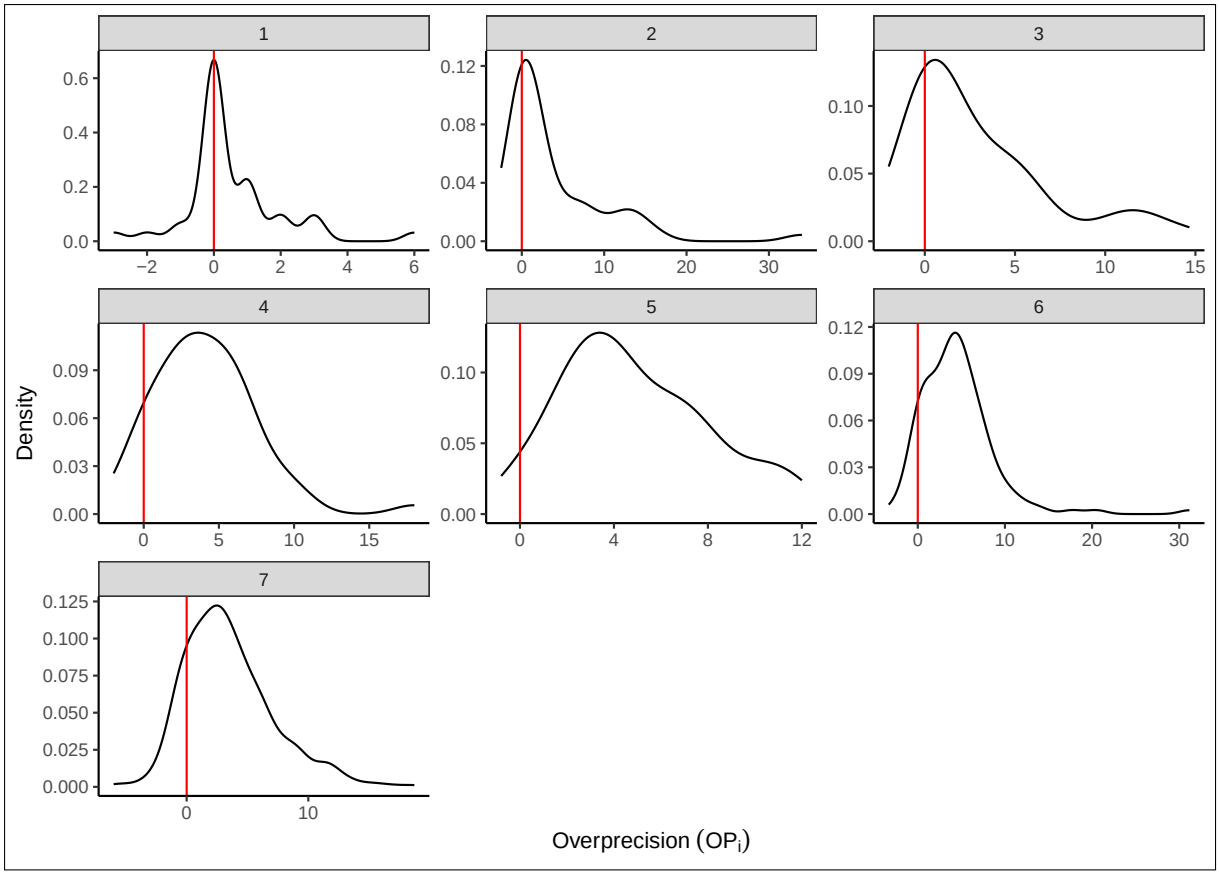


Figure A.1: Density of overprecision (op_i) for each of the subsets of questions answered by SOEP-IS respondents. We plot from left to right the densities of op_i for those respondents who answered from the minimum number of answers (1) to the maximum number of answers (7). Note that the Y-axis scale changes across panels.

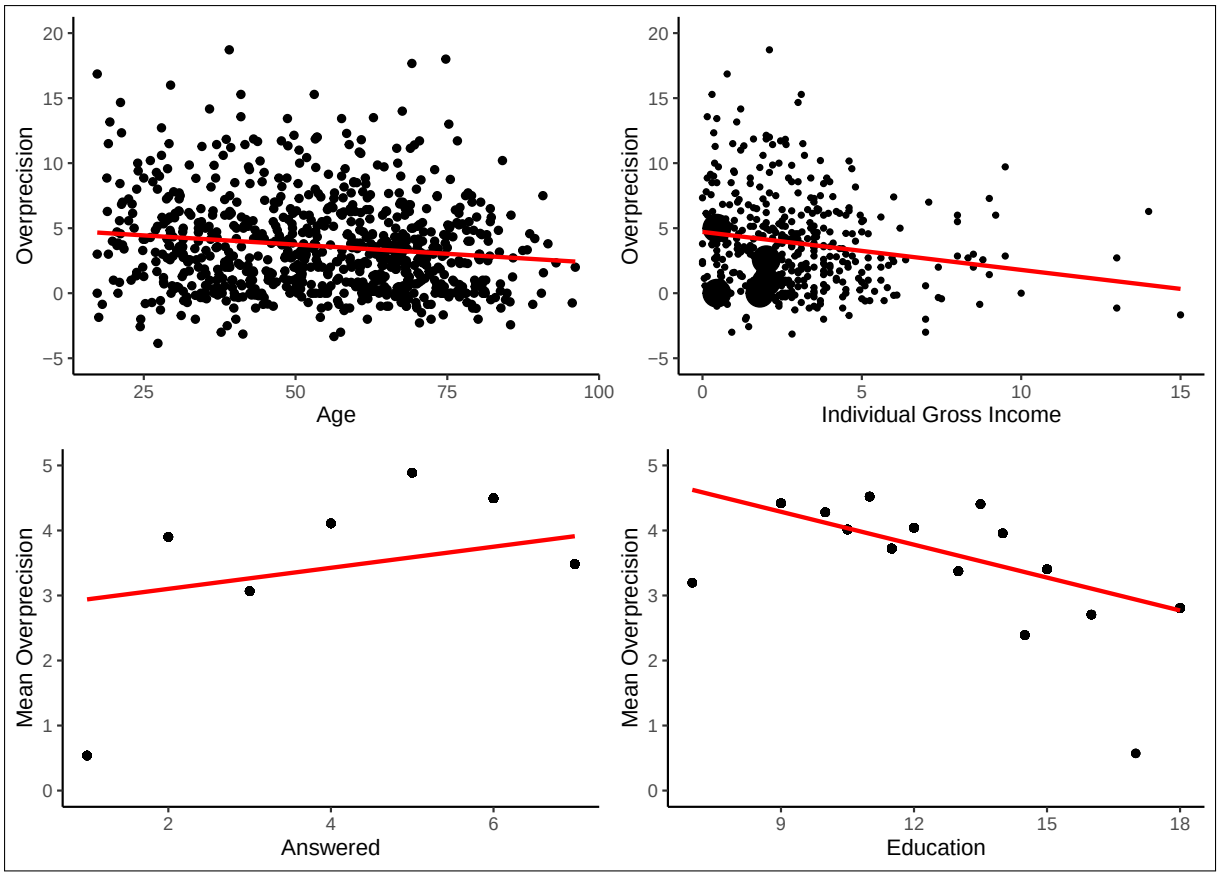


Figure A.2: Relationship between overprecision and respondent characteristics in the SOEP-IS. The upper panels plot respondent-level overprecision against age and individual gross income. The lower panels plot mean overprecision by the number of questions answered and years of education. The red lines show linear fits. We omit one respondent with an extreme value from all four panels for readability.

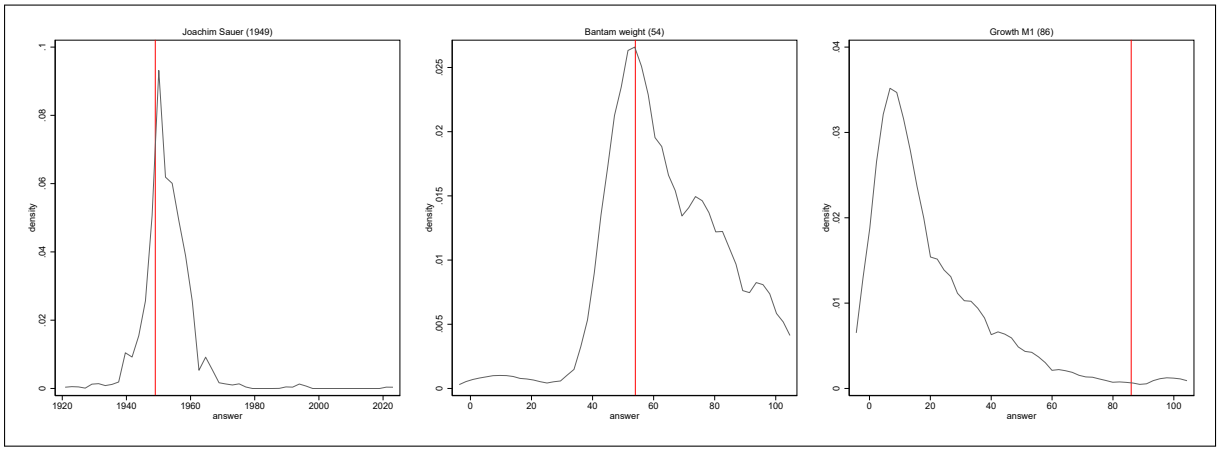


Figure A.3: Density of the answers ($a_{i,j}$) to the additional questions used to flag possible use of search engines in the companion survey. The vertical red line marks the correct answer. The vertical-axis scale differs across questions.

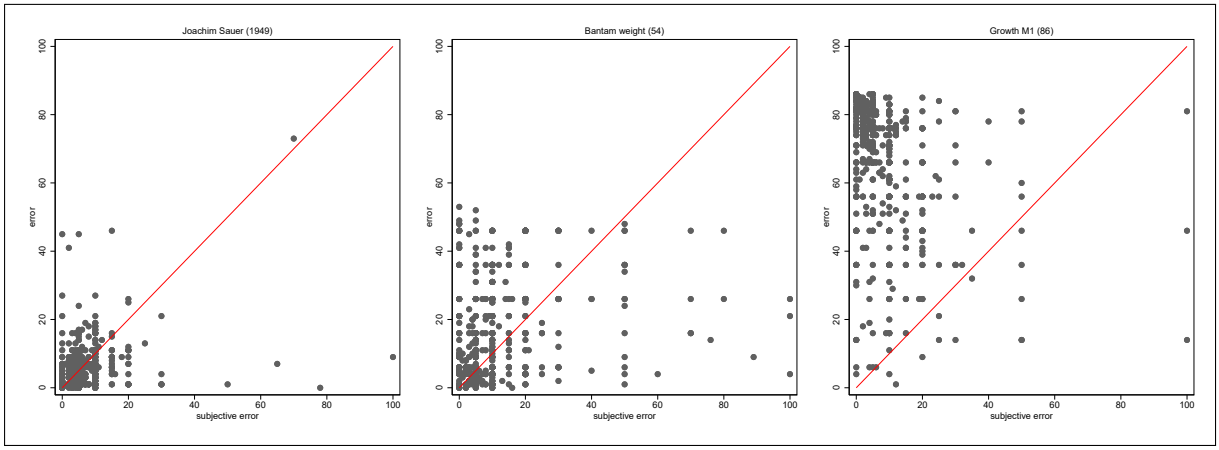


Figure A.4: Relationship between realized error ($error_{i,j}$), shown on the vertical axis, and subjective error ($se_{i,j}$), shown on the horizontal axis, for the additional questions used to flag possible use of search engines in the companion survey. Points above (below) the 45-degree red line indicate overprecise (underprecise) responses.

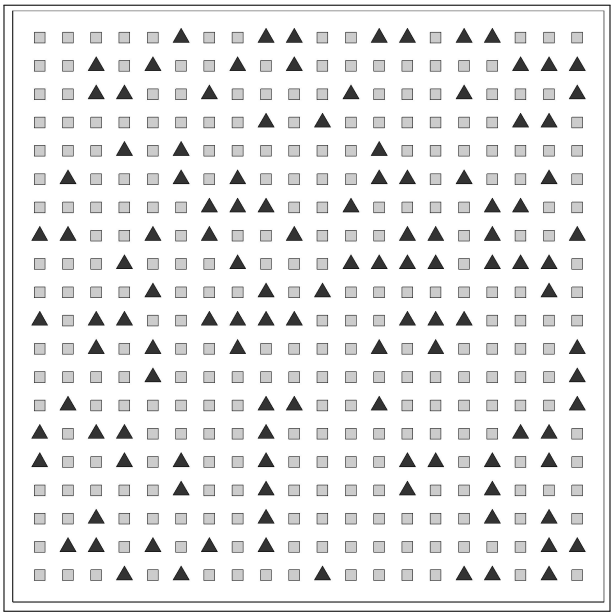


Figure A.5: Matrix example. This matrix is used in the neutral domain of the online survey. It contains 120 black triangles and 280 gray squares.

B Extra Tables

	Subjective Error				
	(1)	(2)	(3)	(4)	(5)
	OLS	Quantile	Interaction	Lower Third	Upper Third
realized error	0.154***	0.165***	0.041	0.127***	0.179***
	(0.010)	(0.005)	(0.050)	(0.017)	(0.016)
education	-0.009	-0.007	-0.073**	0.181	-0.046
	(0.030)	(0.020)	(0.031)	(0.216)	(0.059)
education×*realized error*			0.009**		
			(0.004)		
N	4562	4562	4562	1223	1950
Controls and Fixed Effects	Yes	Yes	Yes	Yes	Yes

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.1: Education, realized error, and subjective error. The dependent variable in each column is item-level subjective error. All columns include the sociodemographic controls and fixed effects used in Table 1. We report only the coefficients on education, realized error, and their interaction. Column (1) uses OLS, Column (2) median quantile regression, and Column (3) OLS with an interaction between realized error and education. Columns (4) and (5) restrict the sample to respondents in the lower and upper tertiles of education, respectively. Standard errors are clustered at the respondent level.

SOEP-IS Code	Question (a)	Answer
Q467 - IGEN02a	In which year were euro notes and coins introduced?	2002
Q470 - IGEN03a	In which year was Microsoft (Publisher of the software package Windows) founded?	1975
Q473 - IGEN04a	In which year was the movie "Das Boot" (directed by Wolfgang Peterson) first shown in German cinemas?	1981
Q476 - IGEN05a	In which year was Saddam Hussein captured by the US army?	2003
Q479 - IGEN06a	In which year was the first Volkswagen Type 1 (also known as "Volkswagen Beetle") produced?	1938
Q482 - IGEN07a	In which year did the Korean War end with a truce?	1953
Q485 - IGEN08a	In which year did Lady Diana, Prince Charles' first wife, die?	1997
	Question (b)	
	What do you think: How far is your answer from the correct answer?	

Table B.2: Original English-language history questions from the 2018 SOEP-IS

SOEP-IS Code	Questions (a)	Answer
Q467 - IGEN02a	In welchem Jahr wurden Euro-Geldscheine und -Münzen eingeführt?	2002
Q470 - IGEN03a	In welchem Jahr wurde das Unternehmen Microsoft (Herausgeber des Betriebssystems Windows) gegründet?	1975
Q473 - IGEN04a	In welchem Jahr kam der Film "Das Boot" (Regie: Wolfgang Petersen) in die deutschen Kinos?	1981
Q476 - IGEN05a	In welchem Jahr wurde Saddam Hussein von der US-Armee gefangen genommen?	2003
Q479 - IGEN06a	In welchem Jahr wurde der erste Volkswagen Typ 1(auch bekannt als "Käfer") produziert?	1938
Q482 - IGEN07a	In welchem Jahr endete der Korea-Krieg mit einem Waffenstillstand?	1953
Q485 - IGEN08a	In welchem Jahr starb Lady Diana, die erste Frau von Prinz Charles?	1997

	Question (b)
	Was schätzen Sie: wie viele Jahre liegt Ihre Antwort von der richtigen Antwort entfernt?

Table B.3: Original German-language history questions from the 2018 SOEP-IS

Variable	Definition
Financial Behavior:	
DAX forecast error	Principal component of the absolute forecast errors for respondents' one- and two-year-ahead DAX predictions. Forecast errors are computed relative to the realized index value over each horizon, using the closing price on the interview date as the baseline. Standardized to have mean 0 and standard deviation 1.
portfolio diversification	Aggregate diversification measure over five asset classes. For each asset class, a penalty score is calculated expressing the distance to an equally diversified portfolio. Diversification equals the maximum attainable penalty score less the actual penalty. The diversification measure is standardized to have mean 0 and standard deviation 1.
Political Behavior:	
extremeness	Absolute distance to the center of an ideology scale from 0 (left) to 10 (right). Standardized to have mean 0 and standard deviation 1.
left-right	Location on an ideology scale from 0 (left) to 10 (right). Standardized to have mean 0 and standard deviation 1.
non-voter	= 1 if the respondent reported not voting in the 2017 German federal election.
Controls:	
age	Difference between interview month/year and birth month/year in years.
female	= 1 if female.
GDR 1989	= 1 if living in East Germany in 1989.
years education	Years of education, including further education after primary and secondary education.

Variable	Definition
gross income	Monthly gross labor income in thousands. Missing values are coded as zero.
missing income	= 1 if gross income is missing.
fin. literacy	Share of correct answers to 6 questions related to financial knowledge.
risk aversion	Location on a risk scale from 0 (risk averse) to 10 (risk loving). Standardized to have mean 0 and standard deviation 1.
narcissism	Average narcissism measure over 6 items on a scale from 1 to 6. Standardized to have mean 0 and standard deviation 1.
impulsivity	Location on impulsivity scale from 0 (not impulsive) to 10 (fully impulsive). Standardized to have mean 0 and standard deviation 1.
patience	Location on the patience scale from 0 (not patient) to 10 (fully patient). Standardized to have mean 0 and standard deviation 1.
employed	= 1 if employed.
unemployed	= 1 if unemployed.
nonwork	= 1 if not working.
matedu	= 1 if on maternity, educational, or military leave.
retired	= 1 if retired.
answered	Number of SEM questions answered.
Additional Controls:	
assets	= 1 if the respondent owns financial assets.
pol. interest	Political interest on a scale from 1 (high) to 4 (low). Reversed and standardized to have mean 0 and standard deviation 1.

Table B.4: Definitions of the SOEP variables used in the analysis

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	SOEP-IS		SOEP Core		Difference		
	mean	SE	mean	SE	difference	p-value	N[Core]
age	53.914	(0.627)	50.535	(0.180)	-3.379	0.000	30,997
female	0.508	(0.018)	0.508	(0.005)	0.000	0.989	30,997
german	0.933	(0.009)	0.877	(0.003)	-0.056	0.000	30,997
east (current)	0.174	(0.013)	0.172	(0.003)	-0.001	0.916	30,997
GDR 1989	0.186	(0.014)	0.198	(0.004)	0.012	0.404	24,591
years education	12.704	(0.098)	12.276	(0.027)	-0.428	0.000	28,482
employed	0.534	(0.018)	0.593	(0.005)	0.058	0.001	30,967
retired	0.229	(0.015)	0.221	(0.004)	-0.007	0.627	30,967
gross income	2.943	(0.112)	2.837	(0.029)	-0.106	0.359	17,829
married	0.568	(0.017)	0.521	(0.005)	-0.047	0.009	30,896
N[SOEP-IS]	805						

Table B.5: Representativeness of the SOEP-IS subsample. The table compares selected respondent characteristics in the SOEP-IS subsample and the SOEP-Core. Statistics for the SOEP-IS in Columns (1) and (2) are unweighted, while those for the SOEP-Core in Columns (3) and (4) use the provided sampling weights. Columns (5) and (6) report the difference in means (SOEP-Core minus SOEP-IS) and its p-value. Column (7) reports the SOEP-Core sample size, which varies because of missing observations. Columns (2) and (4) report standard errors of the means.

Dependent Variable: *sop*	(1)	(2)	(3)	(4)	(5)
age	-0.008***	-0.008***	-0.007**	-0.005	-0.005
	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)
female	0.074	0.121	0.094	0.134*	0.111
	(0.070)	(0.074)	(0.074)	(0.074)	(0.076)
years education	-0.053***	-0.062***	-0.050***	-0.043***	-0.037***
	(0.013)	(0.013)	(0.014)	(0.013)	(0.014)
answered		0.064***	0.067***	0.086***	0.085***
		(0.023)	(0.022)	(0.023)	(0.023)
gross income			-0.049**	-0.037*	-0.039*
			(0.021)	(0.020)	(0.021)
employed			0.012	0.079	0.139
			(0.163)	(0.156)	(0.164)
unemployed			0.696**	0.618*	0.622*
			(0.347)	(0.334)	(0.339)
retired			-0.017	-0.008	-0.005
			(0.109)	(0.109)	(0.112)
maternity leave			0.024	0.076	0.090
			(0.217)	(0.207)	(0.209)
east 1989			0.067	0.045	-0.169
			(0.100)	(0.098)	(0.116)
fin. literacy				-0.481***	-0.398**
				(0.170)	(0.174)
risk aversion				0.056	0.048
				(0.038)	(0.038)
impulsivity				-0.019	-0.014
				(0.038)	(0.040)
patience				0.035	0.038
				(0.035)	(0.035)
narcissism				0.107**	0.100**
				(0.042)	(0.043)
N	805	805	805	805	805
adj. R^2	0.036	0.046	0.061	0.081	0.098
Constant Term	Yes	Yes	Yes	Yes	Yes
Fixed Effects	No	No	No	No	Yes

Robust standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.6: Correlates of overprecision. This is the full version of Table 1, including all coefficients. Columns (1)–(5) report OLS regressions with standardized overprecision (*sop*) as the dependent variable. All columns include age, gender, and education. Column (2) adds the number of SEM questions answered. Column (3) adds gross income, labor-force status, and whether the respondent lived in the GDR in 1989. Column (4) adds financial literacy, risk preferences, impulsivity, patience, and narcissism. Column (5) also includes federal-state and interview-month fixed effects.

	(1) DAX Error	(2) DAX 1yr	(3) DAX 2yr	(4) Diversification	(5) Extremism	(6) Left-Right	(7) Abstention
sop	0.083*	0.529	3.078**	-0.131***	0.087**	-0.008	0.032**
	(0.043)	(0.518)	(1.269)	(0.036)	(0.043)	(0.042)	(0.012)
age	-0.001	-0.036	0.020	-0.007**	-0.007**	0.007**	-0.001
	(0.004)	(0.042)	(0.104)	(0.003)	(0.003)	(0.003)	(0.001)
female	-0.008	-0.663	0.755	0.021	-0.000	-0.224***	-0.026
	(0.094)	(1.107)	(2.721)	(0.076)	(0.084)	(0.082)	(0.025)
years educ.	-0.002	-0.045	0.106	0.056***	0.029*	-0.031**	-0.013***
	(0.017)	(0.206)	(0.503)	(0.014)	(0.016)	(0.016)	(0.005)
gross income	-0.013	-0.215	-0.265	-0.009	-0.018	-0.003	0.002
	(0.026)	(0.315)	(0.756)	(0.023)	(0.027)	(0.026)	(0.008)
miss income	0.002	-0.665	0.741	-0.239	0.023	0.540**	0.058
	(0.289)	(3.408)	(8.179)	(0.207)	(0.226)	(0.222)	(0.069)
answered	-0.099***	-0.720**	-3.221***	0.064***	-0.033	-0.013	0.001
	(0.031)	(0.359)	(0.897)	(0.022)	(0.024)	(0.024)	(0.007)
employed	-0.078	-2.018	-0.212	-0.285	0.071	0.429*	0.065
	(0.295)	(3.466)	(8.338)	(0.208)	(0.226)	(0.222)	(0.069)
unemployed	0.141	3.764	-0.227	0.020	0.277	-0.087	0.149**
	(0.242)	(2.935)	(7.066)	(0.209)	(0.230)	(0.226)	(0.068)
retired	0.186	1.618	5.572	-0.003	0.104	0.068	0.040
	(0.154)	(1.819)	(4.436)	(0.121)	(0.128)	(0.126)	(0.037)
maternity	-0.043	-1.523	0.865	-0.050	-0.044	-0.344	0.028
	(0.295)	(3.504)	(8.616)	(0.231)	(0.263)	(0.258)	(0.077)
east 1989	-0.289*	-3.603*	-6.514	-0.057	0.046	-0.160	0.030
	(0.166)	(1.987)	(4.851)	(0.141)	(0.150)	(0.147)	(0.045)
fin. literacy	0.059	-3.086	9.537	0.385**	0.007	0.142	-0.053
	(0.235)	(2.698)	(6.858)	(0.162)	(0.175)	(0.171)	(0.051)
risk aversion	0.030	-0.078	2.102	0.039	0.006	-0.032	-0.011
	(0.047)	(0.557)	(1.369)	(0.037)	(0.042)	(0.041)	(0.012)
impulsivity	-0.009	0.133	-0.754	-0.081**	0.007	0.031	0.021*
	(0.047)	(0.561)	(1.362)	(0.038)	(0.042)	(0.041)	(0.012)
patience	0.083*	0.887*	2.242*	-0.027	0.005	-0.004	0.022*
	(0.045)	(0.532)	(1.297)	(0.036)	(0.040)	(0.039)	(0.011)
narcissism	0.000	-0.395	0.593	0.058	0.077*	0.050	0.010
	(0.045)	(0.539)	(1.319)	(0.038)	(0.041)	(0.041)	(0.012)
pol. interest					0.153***	-0.106**	-0.051***
					(0.044)	(0.043)	(0.013)
constant	-0.078	12.253*	12.364	-0.909**	0.231	-0.300	0.239*
	(0.528)	(6.293)	(15.214)	(0.382)	(0.431)	(0.423)	(0.125)
N	548	578	557	774	716	716	706
adj. R^2	0.076	0.125	0.035	0.112	0.024	0.062	0.111

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.7: Full estimation results for the outcome regressions in Section 4. Each column reports the corresponding full specification described in Section 4.1. Sample sizes vary because of missing outcome data; the maximum is 805 observations. Variable definitions are provided in Table B.4.

	DAX error	diversification	extremism	left-right	non-voter
overprecision	0.098*	-0.112**	0.076	-0.059	0.025*
	(0.051)	(0.048)	(0.051)	(0.050)	(0.014)
Observations	465	465	465	465	465
R^2	0.140	0.197	0.110	0.150	0.197
overprecision		-0.099**	0.098**	-0.042	0.030**
		(0.043)	(0.044)	(0.043)	(0.012)
Observations		653	653	653	653
R^2		0.158	0.080	0.123	0.165

Table B.8: Seemingly Unrelated Regression Equations (SURE). The upper panel jointly estimates equations for DAX forecast error, portfolio diversification, political extremism, left–right position, and abstention. The lower panel omits the DAX forecast-error equation to use a larger common sample. All equations include federal-state and interview-month fixed effects. Standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Domain 1	Domain 2	Correlation	Standard error	N
Neutral	History	0.19	0.03	688
Neutral	Knowledge	0.23	0.04	626
Neutral	Stocks	0.24	0.04	676
Neutral	Economics	0.20	0.04	639
History	Knowledge	0.42	0.04	609
History	Stocks	0.27	0.04	638
History	Economics	0.32	0.05	620
Knowledge	Stocks	0.29	0.04	591
Knowledge	Economics	0.30	0.04	580
Stocks	Economics	0.25	0.04	618

Table B.9: Partial correlations between aggregate overprecision measures across the five domains after winsorizing each measure at the 1st and 99th percentiles. The correlations condition on age, gender, education, income, nationality, state, mathematical literacy, and self-reported expertise in the corresponding domains.

	(1) Point estimate	(2) Unadj. p-value	(3) MHT p-value	(4) R^2 rank	(5) LASSO included	(6) R^2	(7) N
Financial Behavior:							
DAX error	0.104**	0.024	0.071	4/46	yes/13	0.09	510
1-year ahead	*0.631*	*0.251*		10/46	yes/18	0.14	*537*
2-year ahead	*3.944****	*0.004*		4/46	no/0	0.00	*519*
Diversification	-0.12***	0.002	0.009	4/46	yes/15	0.13	719
Political Behavior:							
Extremism	0.077*	0.059	0.115	8/47	yes/18	0.07	716
Left-Right	-0.019	0.643	0.643	24/47	no/17	0.10	716
Abstention	0.029**	0.015	0.060	3/47	yes/10	0.12	706

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.10: Outcome regressions including the Big Five personality traits. Column (1) reports the coefficient on standardized overprecision (*sop*) from the full specification in Section 4.1. Columns (2) and (3) report the unadjusted and Sidak–Holm adjusted p-values. Column (4) reports the ranking from the R^2 procedure and the number of candidate controls. Column (5) indicates whether LASSO selects overprecision and reports the number of selected controls; Column (6) gives the resulting R^2. Column (7) reports the sample size, which varies because of missing outcome data and has a maximum of 750. Specifications for political outcomes also include self-reported political interest.

	(1) Point estimate	(2) Unadj. p-value	(3) MHT p-value	(4) R^2 rank	(5) LASSO included	(6) LASSO R^2	(7) N
Financial Behavior:							
DAX Error	0.081*	0.065	0.125	6/42	yes/16	0.10	545
1-year ahead	*0.477*	*0.360*		8/42	yes/24	0.17	*574*
2-year ahead	*3.044***	*0.018*		5/42	no/0	0.00	*553*
Diversification	-0.131***	0.000	0.002	4/42	yes/18	0.13	763
Political Behavior:							
Extremism	0.082*	0.057	0.161	6/43	yes/14	0.06	706
Left-Right	-0.002	0.966	0.966	18/43	no/14	0.08	706
Abstention	0.031**	0.014	0.054	3/43	yes/10	0.10	694

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.11: Outcome regressions including asset ownership as a control. Column (1) reports the coefficient on standardized overprecision (*sop*) from the full specification in Section 4.1. Columns (2) and (3) report the unadjusted and Sidak–Holm adjusted p-values. Column (4) reports the ranking from the R^2 procedure and the number of candidate controls. Column (5) indicates whether LASSO selects overprecision and reports the number of selected controls; Column (6) gives the resulting R^2. Column (7) reports the sample size, which varies because of missing outcome data and has a maximum of 791. Specifications for political outcomes also include self-reported political interest.

Qualtrics Code	Question (a)	
PX110 a	How many black triangles were in the matrix?	70
PX110 b	How many black triangles were in the matrix?	120
PX110 c	How many black triangles were in the matrix?	195
PX110 d	How many black triangles were in the matrix?	280
PX110 e	How many black triangles were in the matrix?	330
PX120 a	In which year were euro notes and coins introduced?	2002
PX120 b	In which year was Microsoft (Publisher of the software package Windows) founded?	1975
PX120 c	In which year was Saddam Hussein captured by the US army?	2003
PX120 d	In which year was the first Volkswagen Type 1 (also known as "Volkswagen Beetle") produced?	1938
PX120 e	In which year did Lady Diana, King Charles' first wife, die?	1997
PX120 f	In which year was Joachim Sauer (husband of Angela Merkel) born?	1949
PX120 g	Which year is between 1993 and 1995?	1994
PX130 a	How many teeth does an adult polar bear have?	42
PX130 b	How many keys (black AND white) does a grand piano have?	88
PX130 c	How high (in meters) is the Reichstag building in Berlin?	47
PX130 d	What percentage of seats in the 20th German Bundestag (elected on September 26, 2021) are occupied by female members of the Bundestag	35
PX130 e	How many countries on the African continent are members of the United Nations?	54
PX130 f	What is the upper bantamweight limit in women's Olympic boxing weight classes?	54
PX130 g	What whole number is between 83 and 85?	84
PX140 a-e	What do you estimate: Where (in euros) will the share price be in exactly four weeks?	+ 28 days
PX140 f	In this question, the graph is empty. Please just enter the number 42 as your answer. What do you estimate: where (in euros) will the share price be in exactly four weeks?	42

PX150 a	The Consumer Price Index (CPI) is a measure of the average percentage change in the price level of certain goods and services purchased by households for consumption. The change in the consumer price index compared to the same month of the previous year or the previous year is also referred to as the rate of inflation. By how much (in percent) did the German CPI increase from the beginning of 2011 to the end of 2021?	17
PX150 b	The gross domestic product (GDP) indicates the total value of all goods and services that were produced as end products within the national borders of an economy during a year, after deduction of all intermediate consumption. By how much (in percent) is the German GDP at market prices (nominal) increased from 2006 to 2021?	51
PX150 c	The DAX (Deutscher Aktienindex) is a stock index that measures the performance of the 40 largest companies in the German stock market. By how much (in percent) did the DAX rise from the beginning of 2014 to the end of 2021?	65
PX150 d	The middle income or median income in a society or group describes the income level at which the number of households (or persons) with lower incomes is equal to the number of households with higher incomes. By how much (in percent) did median income in Germany increase from 1991 to 2018?	22
PX150 e	A census is a legally ordered survey of statistical population data. The last census in Germany took place in 2022. By how much (in percent) did the German population grow from 1991 to 2021?	4
PX150 f	M1 money supply describes the amount of cash in circulation and the amount of sight deposits (e.g. savings accounts). By how much (in percent) did the M1 money supply in the euro zone increase from the beginning of 2015 to the end of 2021?	86
	Question (b)	
	What do you think: How many [unit] is your answer away from the correct answer?	

Table B.12: Original English-language questions from the online companion survey

Qualtrics Code	Questions (a)	Answer
PX110 a	Wie viele schwarze Dreiecke waren in der Matrix?	70
PX110 b	Wie viele schwarze Dreiecke waren in der Matrix?	120
PX110 c	Wie viele schwarze Dreiecke waren in der Matrix?	195
PX110 d	Wie viele schwarze Dreiecke waren in der Matrix?	280
PX110 e	Wie viele schwarze Dreiecke waren in der Matrix?	330
PX120 a	In welchem Jahr wurden Euro-Geldscheine und -Münzen eingeführt?	2002
PX120 b	In welchem Jahr wurde das Unternehmen Microsoft (Herausgeber des Betriebssystems Windows) gegründet?	1975
PX120 c	In welchem Jahr wurde Saddam Hussein von der US-Armee gefangen genommen?	2003
PX120 d	In welchem Jahr wurde der erste Volkswagen Typ 1 (auch bekannt als "Käfer") produziert?	1938
PX120 e	In welchem Jahr starb Lady Diana, die erste Frau von König Charles III.?	1997
PX120 f	In welchem Jahr wurde Joachim Sauer (Ehemann von Angela Merkel) geboren?	1949
PX120 g	Welches Jahr liegt zwischen 1993 und 1995?	1994
PX130 a	Wie viele Zähne hat ein ausgewachsener Eisbär?	42
PX130 b	Wie viele Tasten (schwarz UND weiß) hat ein Konzertflüegel?	88
PX130 c	Wie hoch (in Metern) ist das Berliner Reichstagsgebäude?	47
PX130 d	Wie viel Prozent der Sitze im 20. Deutschen Bundestag (gewählt am 26. September 2021) sind durch weibliche Bundestagsabgeordnete besetzt?	35
PX130 e	Wie viele Staaten auf dem Afrikanischen Kontinent sind Mitglied der Vereinten Nationen?	54
PX130 f	Bei wie viel Kilogramm liegt die Obergrenze des Bantamgewichts in den Gewichtsklassen der Frauen beim Olympischen Boxen?	54
PX130 g	Welche ganze Zahl liegt zwischen 83 und 85?	84
PX140 a-e	Was schätzen Sie: wo (in Euro) liegt der Aktienkurs in genau vier Wochen?	+28 Tage

PX140 f	In dieser Frage ist die Grafik leer. Bitte geben Sie einfach die Zahl 42 als Antwort ein. Was schätzen Sie: wo (in Euro) liegt der Aktienkurs in genau vier Wochen?	42
PX150 a	Der Verbraucherpreisindex (VPI) ist ein ist ein Maß der durchschnittlichen prozentualen Veränderung des Preisniveaus bestimmter Waren und Dienstleistungen, die von privaten Haushalten für Konsumzwecke gekauft werden. Die Veränderung des Verbraucherpreisindex zum Vorjahresmonat bzw. zum Vorjahr wird auch als Teuerungsrate oder als Inflationsrate bezeichnet. Um wie viel (in Prozent) ist der deutsche VPI von Anfang 2011 bis Ende 2021 gestiegen?	17
PX150 b	Das Bruttoinlandsprodukt (BIP) gibt den Gesamtwert aller Waren und Dienstleistungen an, die während eines Jahres innerhalb der Landesgrenzen einer Volkswirtschaft als Endprodukte hergestellt wurden, nach Abzug aller Vorleistungen. Um wie viel (in Prozent) ist das deutsche BIP zu Marktpreisen (nominal) von 2006 bis 2021 gestiegen?	51
PX150 c	Der DAX (Deutscher Aktienindex) ist ein Aktienindex, der die Wertentwicklung der 40 größ ten Unternehmen des deutschen Aktienmarkts misst. Um wie viel (in Prozent) ist der DAX von Anfang 2014 bis Ende 2021 gestiegen?	65
PX150 d	Das mittlere Einkommen oder Medianeinkommen in einer Gesellschaft oder Gruppe bezeichnet die Einkommenshöhe, von der aus die Anzahl der Haushalte (bzw. Personen) mit niedrigeren Einkommen gleich groß ist wie die der Haushalte mit höheren Einkommen. Um wie viel (in Prozent) ist das Medianeinkommen in Deutschland von 1991 bis 2018 gestiegen?	22
PX150 e	Eine Volkszählung oder auch Zensus ist eine gesetzlich angeordnete Erhebung statistischer Bevölkerungsdaten. Der letzte Zensus in Deutschland fand 2022 statt. Um wie viel (in Prozent) ist die deutsche Bevölkerung von 1991 bis 2021 gewachsen?	4

PX150 f	Die Geldmenge M1 bezeichnet die Menge an Bargeld im Umlauf sowie die Höhe an Sichteinlagen (bspw. Sparkonten). Um wie viel (in Prozent) ist die Geldmenge M1 in der Eurozone von Anfang 2015 bis Ende 2021 gestiegen?	86
	Question (b)	
	Was schätzen Sie: wie viele [Einheit] liegt Ihre Antwort von der richtigen Antwort entfernt?	

Table B.13: Original German-language questions from the online companion survey

Variable	Definition
Controls:	
age	Reported age.
gender	= 1 if female.
education	Years of primary and secondary education.
income	Monthly gross labor income in thousands, reported in bins.
nationality	Reported primary nationality.
mathematical literacy	Share of correct answers to three statistical problems.
expertise	Self-reported expertise in the corresponding domain on a scale from 0 to 100.

Table B.14: Definitions of the online companion-survey variables used in the analysis

	count	mean	sd	p25	p50	p75
Full sample						
age	1000	44.950	14.315	33.000	46.000	58.000
gender	998	0.489	0.500	0.000	0.000	1.000
german	1000	0.962	0.191	1.000	1.000	1.000
east (current)	1000	0.200	0.400	0.000	0.000	0.000
education	988	10.685	1.871	10.000	10.000	12.000
gross income	924	2.456	1.856	1.500	2.500	4.000
N	1000					
Subsample						
age	839	44.897	14.213	33.000	46.000	57.000
gender	837	0.483	0.500	0.000	0.000	1.000
german	839	0.969	0.173	1.000	1.000	1.000
east (current)	839	0.203	0.402	0.000	0.000	0.000
education	831	10.693	1.863	10.000	10.000	12.000
gross income	778	2.485	1.898	1.500	2.500	4.000
N	839					

Table B.15: Summary statistics for the main variables in the online companion survey

C Alternative Measures of Overprecision

This appendix considers five alternative constructions of the overprecision measure. Section C.1 defines the measures, Section C.2 compares their sociodemographic correlates, and Section C.3 reports their associations with the financial and political outcomes.

C.1 Alternative Measures

Standardized measure (op_i^{std}): Since the overprecision measure of Ortoleva and Snowberg (2015b) standardizes the measure with respect to the entire population, we further construct a *standardized* measure op_i^{std} of overprecision where we standardize the absolute measure op_i of the respective question to be mean zero and standard deviation one before aggregation to avoid the aggregated measure to be biased by a specific question and to relate the level to the entire population. We then take the mean across the seven questions.

Centered measure (op_i^{cent}): We construct a centered measure by subtracting the question-specific sample means from realized and subjective error before taking their difference. We then average these centered gaps across each respondent's available paired items. This construction provides a check on sensitivity to question-specific differences in average realized and subjective error.

Relative measure (op_i^{rel}): we compute a *relative* measure op_i^{rel} by dividing the absolute measure op_i in a specific question with the respective subjective error. Accounting for relative distance makes the measure more comparable across respondents while preserving the relative distance between the subjective error and the realized error (see Figure C.1).

Assume that, as in Figure 1, the true error is normally distributed with mean 0 and variance σ^2 (solid curve). Moreover, the perceived distribution may not coincide with the true distribution. If the perceived variance $\hat{\sigma}^2$ is smaller, i.e., the precision $\rho = 1/\hat{\sigma}^2$ is larger, then we call this respondent overprecise (dashed curve). As long as respondents have the same idea in mind when asked for the error they expect to make, the absolute overprecision measure is comparable across subjects. However, when respondents differ substantially, e.g., by having different confidence intervals in mind, the ranking computed with the absolute measure may no longer be consistent, even if the sign of the deviation remains correct. Taking the example in Figure C.1, where the respondents have the

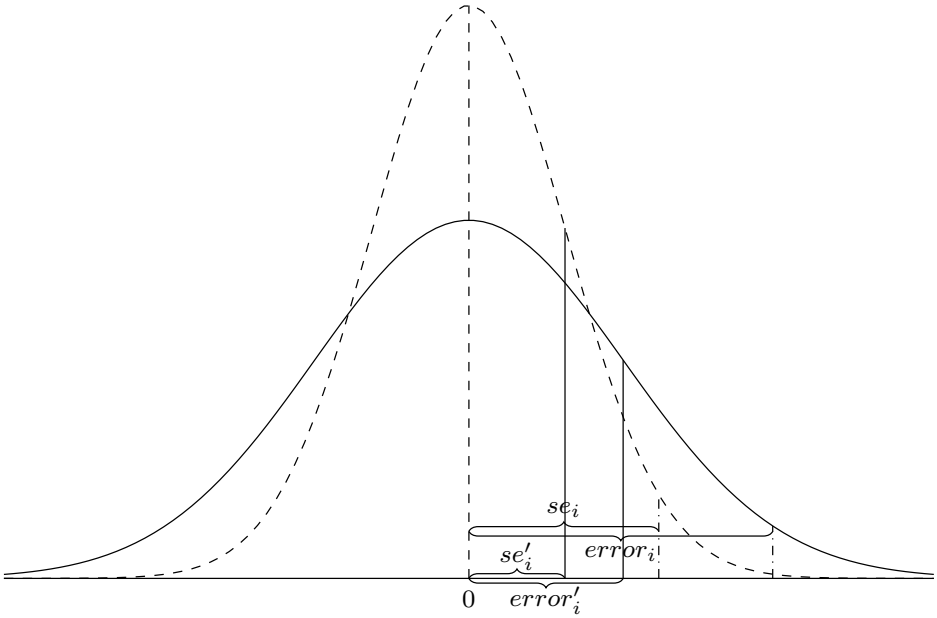


Figure C.1: Two distributions of the (subjective) error. The solid curve shows the true distribution of the error with a standard deviation of 2 (precision of .25). The dashed red curve shows the perceived distribution by an overprecise respondent with a standard deviation of 1.25 (precision of .64). The solid and dashed vertical lines indicate the subjective errors (se) and the true errors ($error$) resulting from respondents with two different ideas about the nature of the subjective error asked in the second question.

same degree of overprecision since the perceived precision of .64 deviates from the true precision of .25, for a respondent with having 95% confidence in mind (se and $error$) the absolute overprecision measure would yield 1.47 whereas for the respondent with having 68% confidence in mind (se' and $error'$) it would yield .75. Thus, the second respondent would incorrectly be classified as less overprecise.

The relative measure corrects this inconsistency by scaling the absolute overprecision measure by the subjective absolute error, making it comparable across subjects. In the above example, the relative measure yields .6 in both cases, which equals the relative difference between the standard deviations of the respective distributions.

Turning to the SOEP data, the correlation between the absolute and relative measures across the seven questions ranges from $\rho^{Spearman} = .91$ to $\rho^{Spearman} = .96$, consistent with respondents interpreting the subjective error question in the same way.[50] Given the high correlation between both approaches, using the absolute measure is preferable because it does not make item-level ratios undefined when stated subjective error is zero and therefore retains more respondents in the aggregate measure.

[50]Note that the relationship between the absolute and the relative measure is non-linear. Therefore, we report only the Spearman correlation coefficient.

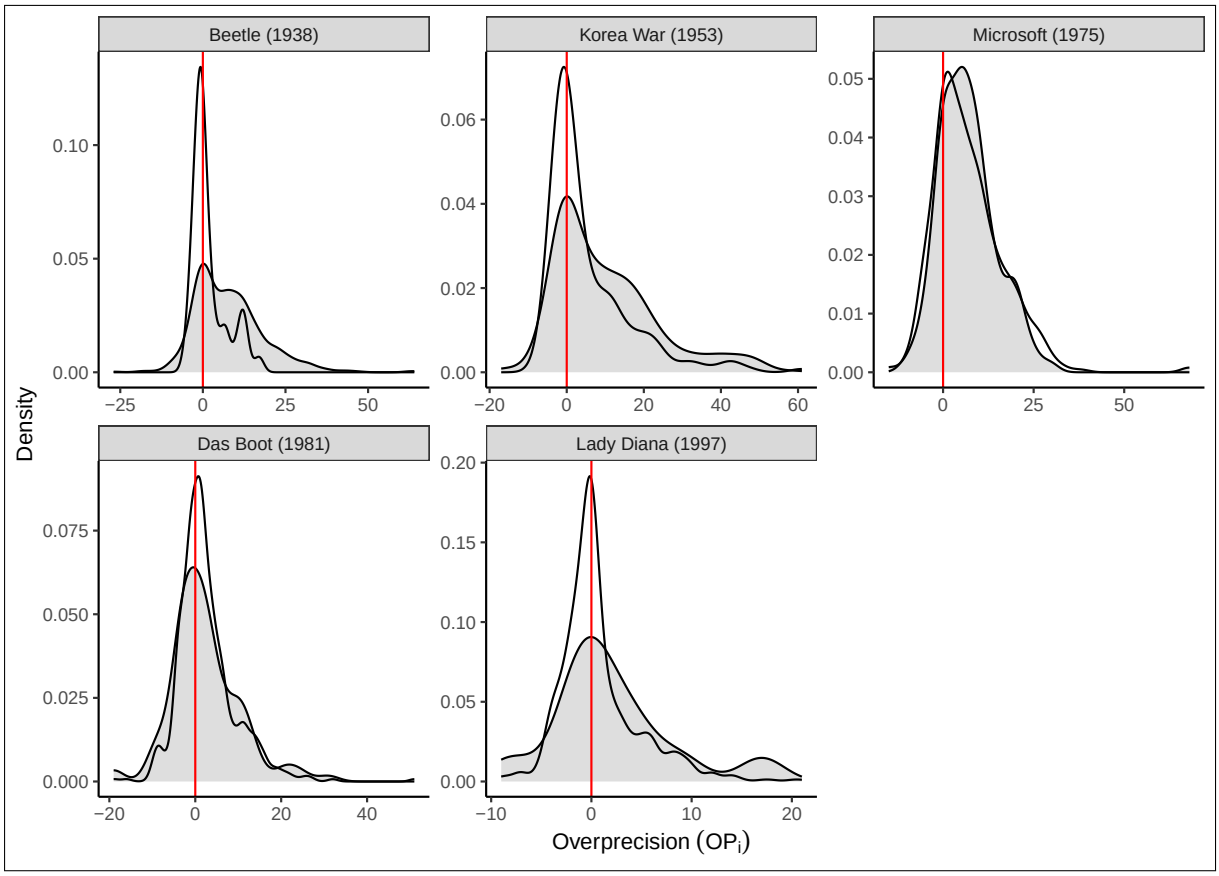
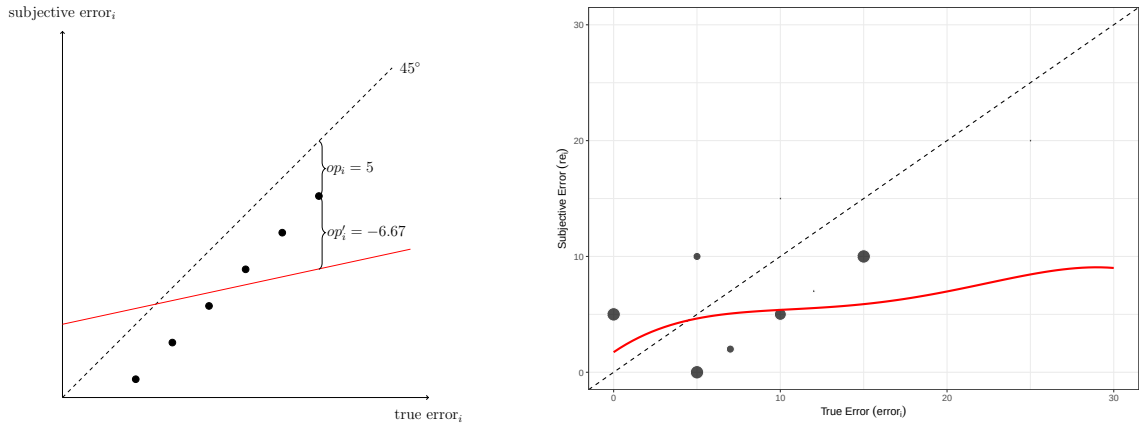


Figure C.2: Distribution of item-level overprecision ($op_{i,j}$) according to whether respondents were born before or after the event in question. The panels are ordered from the earliest to the most recent event. Gray densities represent respondents born after the event; unshaded densities represent respondents born in the year of the event or earlier. The vertical red line marks perfect calibration. Vertical-axis scales differ across panels. We omit questions about events after 2000 because no respondents were born after those events.

Age-robust measure (op_i^{age}): Because our main implementation uses historical questions, the negative correlation between age and overprecision may partly reflect whether respondents were alive when the events occurred. Figure C.2 shows that respondents born before an event tend to be better calibrated on that item. We therefore construct an *age-robust* measure as in Section 2.1, using only questions about events that occurred strictly after the respondent's birth year. This restriction addresses one potential source of factual-knowledge heterogeneity but shortens the item battery, which may reduce precision and places greater weight on more recent events.

Residual measure (op_i^{resid}): Following Ortoleva and Snowberg (2015b), we construct a residual measure based on the relationship between confidence and realized error in the population. Ortoleva and Snowberg (2015b) elicit respondents' assessments of inflation and unemployment, along with confidence on a six-point scale. They regress confidence



(a) Illustration **(b)** Data

Figure C.3: Absolute and population-relative calibration benchmarks. Panel (a) provides a stylized illustration, and Panel (b) uses responses to one of the SEM questions. The 45° line represents the baseline absolute-calibration benchmark. Observations below it report less uncertainty than their realized error. The red fitted line represents the population-relative benchmark used by the residual measure. An observation below the 45° line can therefore be classified as underprecise by the residual measure if it lies above the fitted line. Panel (a) uses a first-order fit for illustration, while Panel (b) uses the fourth-order polynomial employed in the residual construction.

on a fourth-order polynomial of accuracy and use the principal component of the residuals as their measure of overconfidence. To implement their approach as closely as possible, we multiply reported subjective error by -1, so that higher values indicate greater implied confidence. For each question, we divide the resulting measure into quintiles and regress the quintile scores on a fourth-order polynomial of realized error. A one-factor measurement model combines the residuals across the seven questions into a respondent-level score. Estimation by full-information maximum likelihood allows respondents with incomplete item batteries to contribute their available responses. The predicted factor score defines op_i^{resid}. We therefore follow their population-relative residualization approach while using a factor model to accommodate incomplete item batteries.

The residual index therefore compares a respondent's confidence with that of other respondents who have similar realized errors. The baseline SEM instead compares the respondent's stated uncertainty with their own realized error. We treat the residual measure as a useful complement rather than a direct replacement for the baseline SEM. Appendix J reports a direct residualization of subjective error that does not require inversion

or division into quintiles; see Table J.4.

C.2 Robustness of Descriptive Results

Table C.1 compares the sociodemographic correlates of the baseline measure in Column (1) with those of the five alternative measures in Columns (2)–(6). The standardized and centered measures retain several features of the baseline pattern, with the centered measure producing the most similar results. Under the standardized measure, the age and education coefficients are closer to zero than in the baseline. The coefficient on the number of questions answered changes from positive to negative. None of these three coefficients is statistically distinguishable from zero.

The relative measure in Column (4) is available for a smaller sample because item-level ratios are undefined when stated subjective error is zero; the respondent-level aggregate remains defined whenever at least one usable item ratio is available. The negative association with financial literacy and the positive association with narcissism remain, although several estimates are less precise. Under the age-robust measure, the age coefficient becomes positive, while the associations with education and narcissism attenuate. These changes may reflect both the age-based restriction and the shorter, differently composed item battery. The residual measure gives a different picture. Older respondents have higher scores, while women and respondents who answered more questions have lower scores. The associations with education, income, financial literacy, and narcissism are no longer statistically distinguishable from zero.

Taken together, the alternative constructions preserve some, but not all, features of the baseline pattern. We retain the baseline as the primary measure because it corresponds directly to the comparison elicited from each respondent, uses all available items, and does not require a population-level adjustment. The alternative measures show how the results change with standardization, scaling, item selection, and the calibration benchmark. The baseline remains strongly correlated with the standardized measure ($\rho^{Pearson} = .85$; $\rho^{Spearman} = .86$; $N = 805$), the centered measure ($\rho^{Pearson} = .96$; $\rho^{Spearman} = .93$; $N = 805$), and the relative measure ($\rho^{Pearson} = .68$; $\rho^{Spearman} = .82$; $N = 702$). Section C.3 compares their associations with the financial and political outcomes.

	Baseline	Standardized	Centered	Relative	Age robust	Residual
	(1)	(2)	(3)	(4)	(5)	(6)
age	-0.005	-0.000	-0.005*	0.003	0.018***	0.006**
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
female	0.111	0.087	0.177**	0.079	-0.013	-0.199***
	(0.076)	(0.074)	(0.076)	(0.086)	(0.073)	(0.073)
years education	-0.037***	-0.014	-0.037***	-0.003	-0.008	0.002
	(0.014)	(0.014)	(0.014)	(0.020)	(0.013)	(0.014)
answered	0.085***	-0.031	-0.010	0.033	0.075***	-0.104***
	(0.023)	(0.026)	(0.023)	(0.029)	(0.023)	(0.019)
gross income	-0.039*	-0.040**	-0.033	-0.019	-0.032	0.010
	(0.021)	(0.020)	(0.021)	(0.027)	(0.020)	(0.025)
fin. literacy	-0.398**	-0.306*	-0.364**	-0.469*	-0.347**	-0.149
	(0.174)	(0.172)	(0.177)	(0.257)	(0.161)	(0.158)
risk aversion	0.048	0.037	0.036	-0.016	0.017	0.007
	(0.038)	(0.042)	(0.039)	(0.046)	(0.040)	(0.039)
impulsivity	-0.014	-0.005	-0.018	-0.006	0.020	0.034
	(0.040)	(0.040)	(0.040)	(0.048)	(0.037)	(0.037)
patience	0.038	0.042	0.038	0.058*	0.033	0.044
	(0.035)	(0.034)	(0.035)	(0.035)	(0.033)	(0.034)
narcissism	0.100**	0.079*	0.098**	0.118**	-0.003	0.042
	(0.043)	(0.040)	(0.041)	(0.047)	(0.036)	(0.037)
N	805	805	805	702	800	805
adj. R^2	0.098	0.066	0.100	0.045	0.121	0.117
Constant Term	Yes	Yes	Yes	Yes	Yes	Yes
Employment & GDR 1989	Yes	Yes	Yes	Yes	Yes	Yes
Personal characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C.1: Correlates of overprecision using alternative measures. Column (1) uses the baseline measure. Columns (2)–(6) use the standardized, centered, relative, age-robust, and residual measures, respectively. All specifications include indicators for labor-force status, whether the respondent lived in the GDR in 1989, and the personal characteristics reported in Table 1. They also include federal-state and interview-month fixed effects.

C.3 Outcome Associations Using Alternative Overprecision Measures

We next compare the outcome associations obtained using the standardized, centered, relative, age-robust, and residual measures. Tables C.2–C.6 use the baseline specifications from Section 4.2.

Table C.2 reports results for the standardized measure. All associations retain their baseline directions. The aggregate DAX association is estimated more precisely, while the associations with diversification, political extremism, and abstention remain statistically distinguishable from zero. The association with left–right placement remains imprecise.

Table C.3 reports results for the centered measure. Consistent with its correlation of .96 with the baseline measure, the results follow a similar pattern. The association with diversification remains negative and statistically distinguishable from zero. The estimated coefficients for DAX forecast errors, political extremism, and abstention retain their baseline signs but are somewhat smaller in magnitude.

Table C.4 reports results for the relative measure. Because item-level ratios are undefined when stated subjective error is zero, the aggregate measure cannot be constructed for respondents with no valid ratios. The resulting estimation samples are therefore smaller. The associations with DAX forecast errors, diversification, and political extremism retain their baseline directions, with the clearest evidence emerging for political extremism. The association with abstention attenuates to approximately zero.

Table C.5 reports results for the age-robust measure, which uses only historical questions about events that occurred strictly after respondents' birth years. The associations with diversification and political extremism retain their baseline directions but are smaller and estimated imprecisely. The association with abstention remains positive and statistically significant before, but not after, adjustment for multiple testing. Because this restriction changes both the number and type of questions included, the results show that the findings are sensitive to question selection.

Table C.6 reports results for the residual measure. All point estimates are substantially smaller, and none is statistically distinguishable from zero. As discussed in Section C.1, this measure evaluates respondents relative to others with similar realized errors. The baseline measure instead compares each respondent's stated uncertainty with their own realized error. The two measures therefore use different benchmarks and answer different questions.

	(1) Point estimate	(2) Unadj. p-value	(3) MHT p-value	(4) R^2 rank	(5) LASSO included	(6) R^2	(7) N
Financial Behavior:							
DAX error	0.11***	0.010	0.039	3/41	yes/15	0.10	548
1-year ahead	*0.382*	*0.450*		*19/41*	*no/15*	*0.14*	*578*
2-year ahead	*4.776****	*0.000*		*1/41*	*no/0*	*0.00*	*557*
Diversification	-0.104***	0.004	0.019	3/41	yes/18	0.13	774
Political Behavior:							
Extremism	0.09**	0.025	0.072	3/42	yes/13	0.06	716
Left-Right	-0.017	0.662	0.662	18/42	no/14	0.08	716
Abstention	0.026**	0.025	0.050	4/42	yes/19	0.13	706

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C.2: Outcome associations using the standardized overprecision measure. Column (1) reports the coefficient on the measure from the full specification in Section 4.1. Columns (2) and (3) report the unadjusted and Sidak–Holm adjusted p-values. Column (4) reports the ranking from the R^2 procedure and the number of candidate controls. Column (5) indicates whether LASSO selects overprecision and reports the number of selected controls; Column (6) gives the resulting R^2. Column (7) reports the sample size. Specifications for political outcomes also include self-reported political interest.

	(1) Point estimate	(2) Unadj. p-value	(3) MHT p-value	(4) R^2 rank	(5) LASSO included	(6) R^2	(7) N
Financial Behavior:							
DAX error	0.08*	0.066	0.127	4/41	yes/14	0.10	548
1-year ahead	*0.387*	*0.456*		*12/41*	*yes/21*	*0.16*	*578*
2-year ahead	*3.202***	*0.012*		*4/41*	*no/1*	*0.00*	*557*
Diversification	-0.125***	0.001	0.003	3/41	yes/19	0.13	774
Political Behavior:							
Extremism	0.081*	0.059	0.166	6/42	yes/14	0.05	716
Left-Right	0.003	0.944	0.944	18/42	no/14	0.08	716
Abstention	0.026**	0.034	0.128	4/42	yes/18	0.13	706

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C.3: Outcome associations using the centered overprecision measure. Column (1) reports the coefficient on the measure from the full specification in Section 4.1. Columns (2) and (3) report the unadjusted and Sidak–Holm adjusted p-values. Column (4) reports the ranking from the R^2 procedure and the number of candidate controls. Column (5) indicates whether LASSO selects overprecision and reports the number of selected controls; Column (6) gives the resulting R^2. Column (7) reports the sample size. Specifications for political outcomes also include self-reported political interest.

	(1) Point estimate	(2) Unadj. p-value	(3) MHT p-value	(4) R^2 rank	(5) LASSO included	(6) R^2	(7) N
Financial Behavior:							
DAX error	0.062	0.166	0.421	3/41	yes/16	0.11	501
1-year ahead	*0.19*	*0.721*		*9/41*	*no/20*	*0.16*	*530*
2-year ahead	*2.822***	*0.033*		*2/41*	*no/1*	*0.01*	*510*
Diversification	-0.068*	0.080	0.284	18/41	yes/14	0.11	681
Political Behavior:							
Extremism	0.113***	0.007	0.033	2/42	yes/15	0.06	624
Left-Right	-0.03	0.465	0.713	18/42	no/15	0.08	624
Abstention	0.003	0.800	0.800	3/42	no/4	0.09	616

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C.4: Outcome associations using the relative overprecision measure. Column (1) reports the coefficient on the measure from the full specification in Section 4.1. Columns (2) and (3) report the unadjusted and Sidak–Holm adjusted p-values. Column (4) reports the ranking from the R^2 procedure and the number of candidate controls. Column (5) indicates whether LASSO selects overprecision and reports the number of selected controls; Column (6) gives the resulting R^2. Column (7) reports the sample size. Specifications for political outcomes also include self-reported political interest.

	(1) Point estimate	(2) Unadj. p-value	(3) MHT p-value	(4) R^2 rank	(5) LASSO included	(6) R^2	(7) N
Financial Behavior:							
DAX error	0.036	0.399	0.399	8/41	no/8	0.07	546
1-year ahead	*-0.442*	*0.379*		*4/41*	*no/16*	*0.15*	*576*
2-year ahead	*2.868***	*0.021*		*5/41*	*no/0*	*0.00*	*555*
Diversification	-0.048	0.200	0.359	6/41	yes/16	0.12	769
Political Behavior:							
Extremism	0.065	0.102	0.350	5/42	yes/13	0.05	712
Left-Right	-0.061	0.119	0.315	6/42	no/11	0.07	712
Abstention	0.026**	0.022	0.106	3/42	yes/19	0.14	701

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C.5: Outcome associations using the age-robust overprecision measure. Column (1) reports the coefficient on the measure from the full specification in Section 4.1. Columns (2) and (3) report the unadjusted and Sidak–Holm adjusted p-values. Column (4) reports the ranking from the R^2 procedure and the number of candidate controls. Column (5) indicates whether LASSO selects overprecision and reports the number of selected controls; Column (6) gives the resulting R^2. Column (7) reports the sample size. Specifications for political outcomes also include self-reported political interest.

	(1) Point estimate	(2) Unadj. p-value	(3) MHT p-value	(4) R^2 rank	(5) LASSO included	(6) R^2	(7) N
Financial Behavior:							
DAX error	-0.021	0.626	0.948	14/41	no/12	0.09	548
1-year ahead	*-0.269*	*0.606*		*9/41*	*no/20*	*0.16*	*578*
2-year ahead	*-0.267*	*0.835*		*12/41*	*no/7*	*0.04*	*557*
Diversification	-0.029	0.440	0.902	13/41	no/19	0.12	774
Political Behavior:							
Extremism	-0.046	0.256	0.773	8/42	yes/13	0.05	716
Left-Right	-0.008	0.831	0.831	17/42	no/14	0.08	716
Abstention	-0.003	0.798	0.959	18/42	no/17	0.13	706

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C.6: Outcome associations using the residual measure adapted from Ortoleva and Snowberg (2015b). Column (1) reports the coefficient on the measure from the full specification in Section 4.1. Columns (2) and (3) report the unadjusted and Sidak–Holm adjusted p-values. Column (4) reports the ranking from the R^2 procedure and the number of candidate controls. Column (5) indicates whether LASSO selects overprecision and reports the number of selected controls; Column (6) gives the resulting R^2. Column (7) reports the sample size. Specifications for political outcomes also include self-reported political interest.

D Specification Curve Analysis

D.1 Specification Sensitivity Across Modeling Choices

To assess the robustness of our results concerning overprecision, we conduct a comprehensive specification curve analysis, following the methodology of Simonsohn et al. (2020). This method systematically evaluates the stability of estimated coefficients across a large set of plausible model specifications, addressing concerns about researcher degrees of freedom and the sensitivity of empirical findings to modeling choices.

We conduct the analysis separately for four outcomes: *DAX Error*, *Diversification*, *Extremism*, and *Abstention*. Each linear specification takes the form

$$y_i = \alpha + \beta \text{op}_i + \mathbf{X}_i'\boldsymbol{\gamma} + \varepsilon_i, \tag{6}$$

where y_i denotes the outcome of interest, op_i is the overprecision measure, $\mathbf{X}_i$ is a vector of control variables, and ε_i is the error term. The coefficient of interest is β.

We estimate the model over a grid that varies the set of controls, the inclusion of state and interview-month fixed effects, the aggregation of the item-level SEM gaps (the standardized SEM mean or a standardized factor score), and the outcome construction, including standardization and winsorization.[51] For each model, we calculate both conventional and heteroskedasticity-robust standard errors. For abstention, we also vary the estimator between a linear probability model and logit. For diversification, we use two alternative portfolio benchmarks. Table D.1 lists all choices included in the specification curves.

For each model–inference combination, we store the estimated coefficient $\hat{\beta}$ and its associated p-value. Figures D.1–D.4 present the resulting specification curves. In each figure, the upper panel plots 500 selected coefficient estimates, ordered from smallest to largest, while the lower panel identifies the corresponding analytical choices, including the

[51] The alternative aggregation is a factor score from a one-factor measurement model of the seven item-level SEM gaps. We estimate the model by maximum likelihood using the 410 respondents who answered all seven items and then predict scores from each respondent's available item responses, yielding scores for 805 respondents. In contrast to the full-information construction described in Section 3.2, respondents with incomplete item batteries contribute only at the score-prediction stage. Among respondents with complete item responses, the factor score is highly correlated with the first principal component of the seven gaps ($\rho^{\text{Pearson}} = .95$; $\rho^{\text{Spearman}} = .93$). This approach allows us to retain respondents who would be excluded by the complete-case PCA implementation.

Category	Variable	DAX	Divers.	Extrem.	Abst.
Demographics	Age	✓	✓	✓	✓
	Female	✓	✓	✓	✓
	Education (years)	✓	✓	✓	✓
	Gross income	✓	✓	✓	✓
	Missing income	✓	✓	✓	✓
	Employed	✓	✓	✓	✓
	Unemployed	✓	✓	✓	✓
	Retired	✓	✓	✓	✓
	Maternity leave	✓	✓	✓	✓
	GDR 1989 dummy	✓	✓	✓	✓
	Financial literacy	✓	✓	✓	✓
	Married	✓	✓	✓	✓
	German nationality	✓	✓	✓	✓
Psych. Traits	Risk preferences	✓	✓	✓	✓
	Impulsivity	✓	✓	✓	✓
	Patience	✓	✓	✓	✓
	Narcissism	✓	✓	✓	✓
	Self control	✓	✓	✓	✓
	Life satisfaction	✓	✓	✓	✓
	Risk preferences and financial lit	✓	✓	✓	✓
	Big Five	✓	✓	✓	✓
Other Controls	# Questions answered	✓	✓	✓	✓
	Political interest	–	–	✓	✓
	Asset ownership	✓	✓	–	–
Fixed Effects	State fixed effects	✓	✓	✓	✓
	Interview month FE	✓	✓	✓	✓
Spec. Variants	SE type (conventional/robust)	✓	✓	✓	✓
	Logit model	–	–	–	✓
	Standardized outcome	✓	✓	✓	–
	Winsorized outcome	✓	✓	✓	–
	OP measure variants	✓	✓	✓	✓
Diversification Options	Equal weights	–	✓	–	–
	35/10/35/10/10	–	✓	–	–

Table D.1: Variables used in specification curves. The columns indicate the variables that are included in the specification curve permutations, along with the type of outcome analyzed in each. A checkmark indicates that a dimension is varied, not that the variable is included in every specification. Married, German nationality, self-control, and life satisfaction enter one at a time in the full grids but are held at the no-additional-control category in the displayed 500-point curves. The diversification percentages list stocks, real estate, government bonds, savings, and gold, in that order.

controls, fixed effects, SEM aggregation, outcome construction, and standard-error type. The curves therefore show the range of estimates and how their direction changes across the displayed combinations. Because the complete grids contain too many combinations to display clearly, each figure includes:

- the smallest and largest coefficient estimates;
- random samples of nine estimates from the smallest 1,000 and nine estimates from the largest 1,000, excluding the two extrema;
- a random sample of 479 estimates from the remaining interior estimates; and
- the estimate from the specification reported in the main text.

This procedure always includes the extrema and overrepresents both tails, so the displayed points are not a representative sample of the complete grid. The apparent distribution of the coefficients and the fraction of estimates with $p < .10$ should therefore not be interpreted as their frequencies in the complete grid.

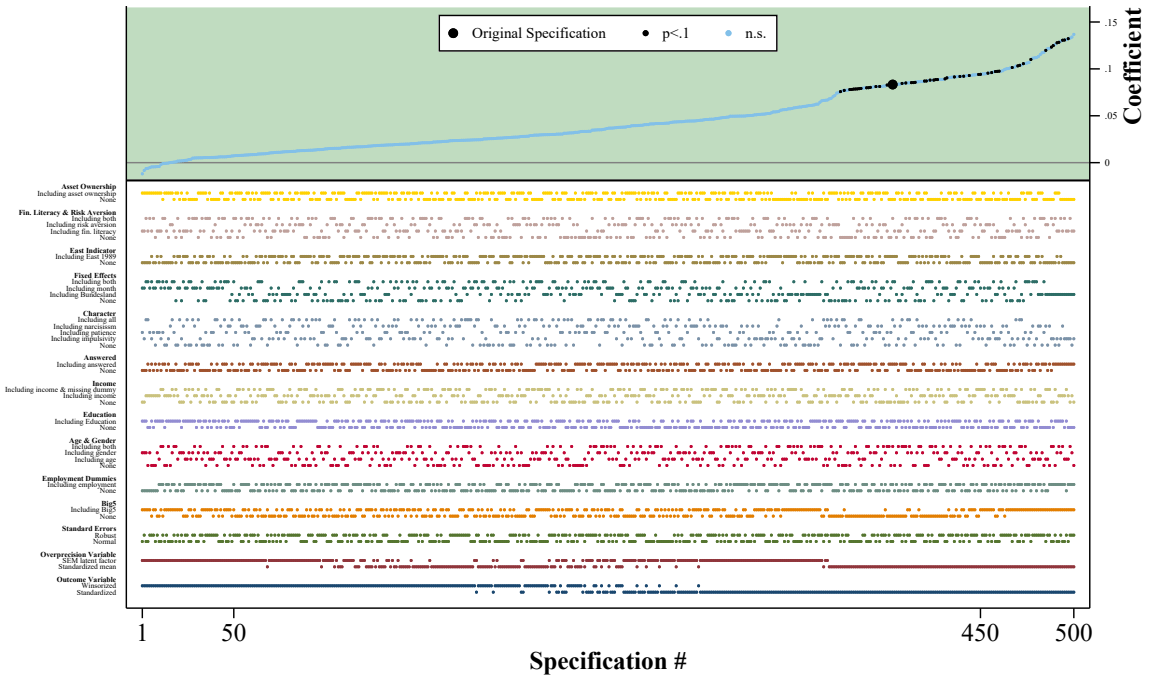


Figure D.1: Specification curve for DAX forecast error. Each point represents one of the 500 displayed specifications. Black points indicate coefficients with $p < .10$. The upper panel orders the coefficient estimates from smallest to largest, while the lower panel identifies the analytical choices used in each specification. The large dark point represents the specification reported in Table 2.

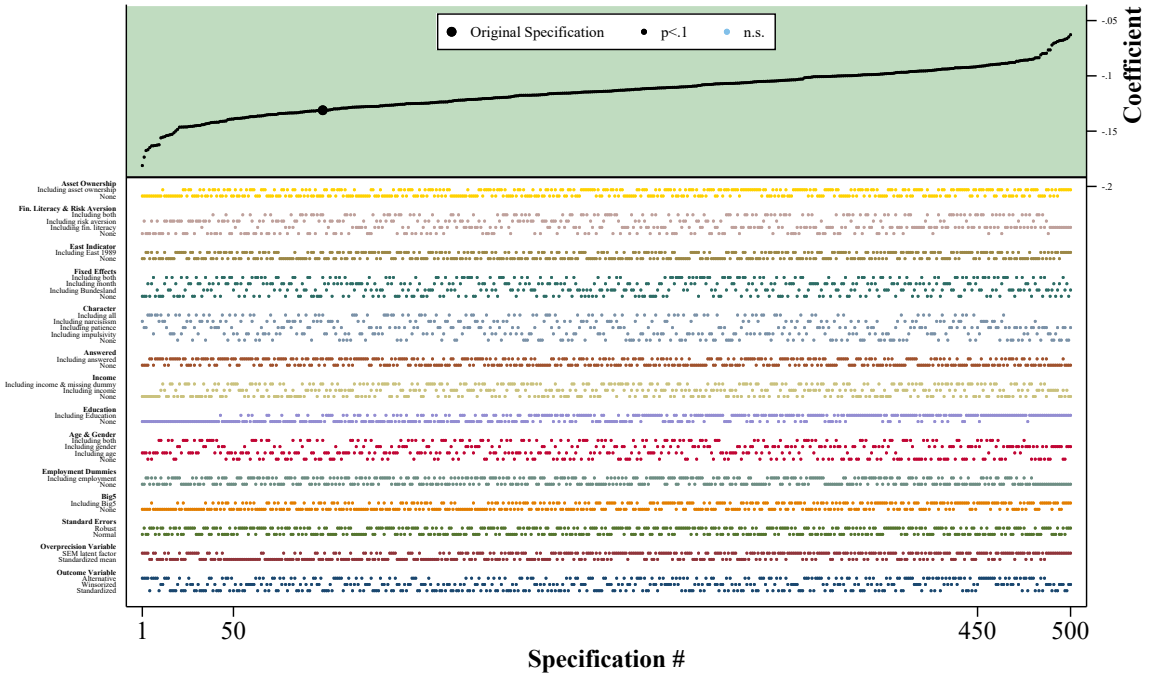


Figure D.2: Specification curve for portfolio diversification. Each point represents one of the 500 displayed specifications. Black points indicate coefficients with $p < .10$. The upper panel orders the coefficient estimates from smallest to largest, while the lower panel identifies the analytical choices used in each specification. The large dark point represents the specification reported in Table 2.

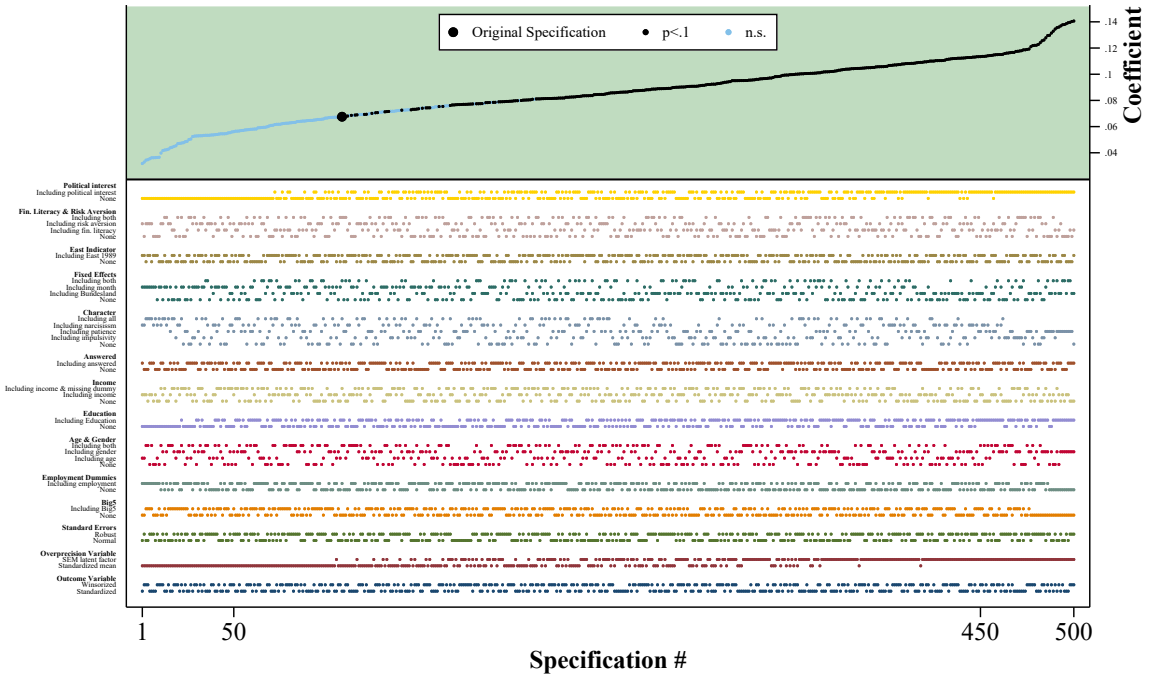


Figure D.3: Specification curve for political extremism. Each point represents one of the 500 displayed specifications. Black points indicate coefficients with $p < .10$. The upper panel orders the coefficient estimates from smallest to largest, while the lower panel identifies the analytical choices used in each specification. The large dark point represents the specification reported in Table 2.

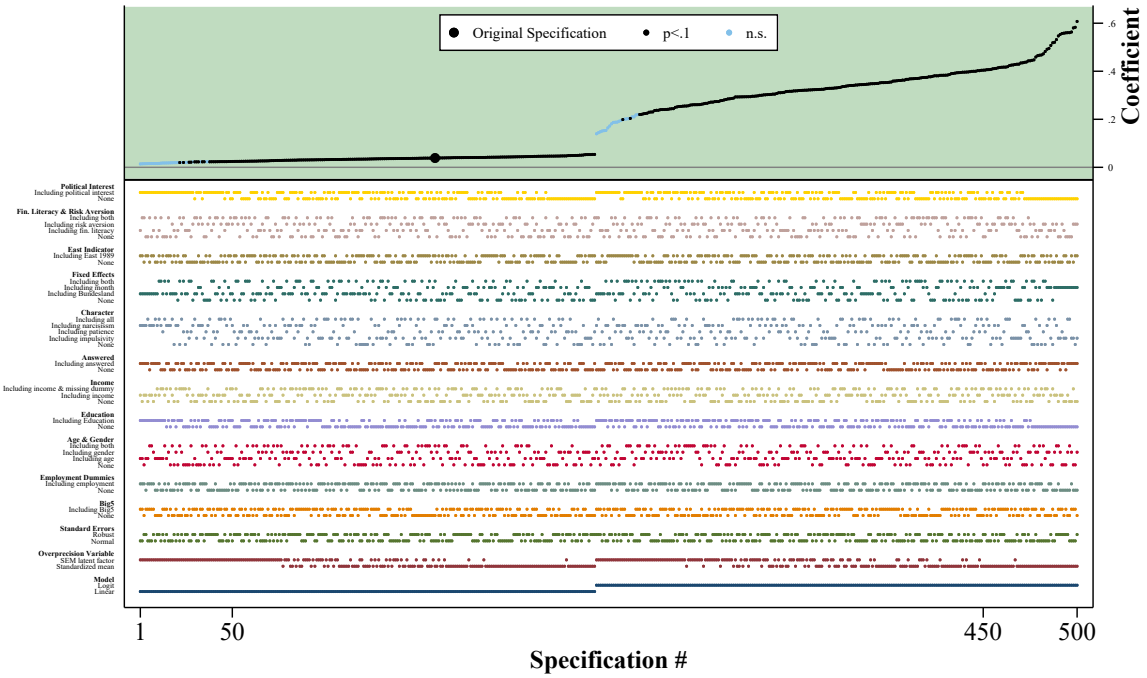


Figure D.4: Specification curve for abstention. Each point represents one of the 500 displayed specifications. Black points indicate coefficients with $p < .10$. The upper panel orders the coefficient estimates from smallest to largest, while the lower panel identifies the analytical choices used in each specification,including the standardized SEM mean and the standardized SEM latent-factor score. The left and right coefficient bands correspond to linear and logit specifications, respectively, so their magnitudes are not directly comparable. The large dark point represents the specification reported in Table 2.

The four specification curves summarize how the estimates change across modeling choices. Across the displayed combinations, the estimates generally retain their baseline directions: positive for *DAX Error*, *Extremism*, and *Abstention*, and negative for *Diversification*. The clearest and most stable pattern is for portfolio diversification. The associations with DAX forecast errors and political extremism are more sensitive. Winsorizing the DAX outcome reduces the DAX estimates, while omitting political interest reduces the extremism estimates, particularly when the standardized SEM mean is used.

The displayed estimates span relatively narrow ranges for diversification and political extremism, whereas the range is broader for *DAX Error*. For abstention, the two distinct bands of coefficients reflect the change from a linear probability model to logit. Because the two estimators use different scales, their coefficient magnitudes are not directly comparable. Beyond the role of political interest noted above, the lower panels reveal no clear clustering associated with individual controls.

D.2 Construct Sensitivity Across Alternative Measures

The preceding specification curves assess sensitivity to modeling choices within the SEM approach. We now examine whether the results change under alternative constructions of overprecision. Table D.2 compares the baseline SEM mean with five alternatives: the factor score described in the preceding subsection; the age-robust, relative, and Ortoleva–Snowberg residual measures introduced in Appendix C.1; and the direct SE-on-TE residual measure described in Appendix J.4.[52]

For this comparison, we use a smaller grid than in the preceding subsection and apply the same grid to each overprecision measure. We vary four dimensions: the control set, the fixed effects, the estimation sample, and the inclusion of an outcome-specific control.[53] The four fixed-effect specifications include neither state nor interview-month effects, state effects only, interview-month effects only, or both. For the estimation sample, we use either all available observations or only respondents who answered all seven questions. Finally, we vary whether the model includes asset ownership for the financial outcomes or political interest for the political outcomes. We calculate both conventional and heteroskedasticity-robust standard errors for each regression.

For the four continuous outcomes—DAX forecast errors, diversification, political extremism, and left–right placement—we consider the standardized baseline outcome and a version winsorized at the 1st and 99th percentiles. For diversification, we additionally use a measure based on the alternative 35/10/35/10/10 portfolio benchmark described in Table D.1. Because abstention is binary, we retain its original coding and vary the estimator between a linear probability model and logit.

The four common dimensions yield $4 \times 4 \times 2 \times 2 = 64$ model specifications for each overprecision-measure–outcome pair. For outcomes with two variants, this produces 128 model specifications and, after applying both standard-error choices, 256 model–inference

[52]The direct residual measure compares each respondent's stated error with the value predicted by the population relationship between stated and realized error for that question. For each question, we regress stated error on a fourth-order polynomial in realized error, reverse the sign of each residual, and average the residuals across the respondent's available questions before standardizing. A higher value indicates less stated uncertainty than predicted.

[53]The four control sets are: (i) no controls; (ii) age and gender; (iii) age, gender, education, gross labor income, an indicator for missing income, and the number of questions answered; and (iv) all controls in (iii), together with employment-status indicators, residence in East Germany in 1989, financial literacy, risk preferences, impulsivity, patience, and narcissism.

combinations. Diversification instead produces 192 model specifications and 384 model–inference combinations because it has three outcome variants. Across all six overprecision measures and five outcomes, the exercise therefore includes 4,224 model specifications and 8,448 model–inference combinations.

	DAX error	**Diversif.**	**Extremism**	**Left–right**	**Abstention**
Reference sign	+	−	+	n/a	+
SEM mean (sop_i)	14	100	28	0	50
SEM latent factor	0	98	82	5	46
Age-robust SEM (op_i^{age})	0	34	7	18	59
Direct residual SE-on-TE ($\widetilde{G}_i$)	0	5	0	0	0
Ortoleva–Snowberg residual (op_i^{resid})	0	14	0	0	0
Relative SEM (op_i^{rel})	0	59	50	0	2

Table D.2: Construct sensitivity across alternative overprecision measures. Cells report the percentage of model–inference combinations with $p < .10$ in the direction of the corresponding baseline estimate in Table 2. Percentages are rounded to the nearest whole number, so a displayed zero can represent a positive share below 0.5%. Left–right placement is evaluated irrespective of sign. For the other outcomes, zero does not rule out significant estimates in the opposite direction. The positive reference direction for abstention is empirical and runs counter to Ortoleva and Snowberg (2015b). Each cell contains 256 combinations, except diversification, which contains 384; all p-values are two-sided. Samples vary with the outcome, controls, and OP construction. The relative measure omits item ratios when stated subjective error is zero but averages any remaining usable ratios; it is available for 702 of 805 respondents.

The percentages in Table D.2 report the share of model–inference combinations that produce a coefficient with the same sign as the baseline estimate and a two-sided p-value below .10. For diversification, for example, the value of 100 under the SEM mean indicates that every combination produces a negative coefficient with $p < .10$. Because left–right placement has no reference direction, we count estimates with $p < .10$ in either direction. These shares describe sensitivity across the choices included in the grid. They should not be interpreted as independent replications because the combinations use overlapping observations and related modeling choices. Estimation samples can also differ across measures because the measures have different data requirements.

Diversification shows the clearest stability across measure constructions. Support remains very high under the latent-factor score and is present in a majority of combinations under the relative SEM, but falls substantially under the age-robust and residual measures. Positive abstention estimates are also common under the SEM mean, latent-factor score, and age-robust SEM, although this direction runs counter to Ortoleva and Snowberg (2015b). The DAX and extremism results are more sensitive, and the residual constructions sometimes produce statistically significant estimates in the direction oppo-

site to the baseline association. Evidence for left–right placement appears mainly under the age-robust measure. Overall, the exercise shows that stability across modeling choices within one measure does not necessarily imply stability across alternative constructions of overprecision.

E Comparing the Companion Online Survey with SOEP-IS

We compare responses to the SEM history questions in the companion online survey with those in SOEP-IS. To limit the influence of outliers, all variables are trimmed separately within each dataset at the 1st and 99th percentiles.

Figure E.1 compares the distributions of answers to the history questions in SOEP-IS and the companion online survey. Except for the VW Beetle question, the distributions are broadly similar across the two samples. Table E.1 reports tests of equality of the means and standard deviations. The differences in mean answers are generally small in years, although they are statistically significant for four of the five questions. For four questions, the SOEP-IS mean is closer to the correct answer. The dispersions are also broadly similar for most questions, with larger differences for the Hussein and VW Beetle questions. Overall, the answer distributions are broadly similar, although not identical, across the two samples.

Figure E.2 compares the distributions of the overprecision measure for the five history questions in SOEP-IS and the companion online survey. The distributions overlap considerably across the two samples. Table E.2 shows that mean overprecision is higher in SOEP-IS for three of the five questions, while the dispersion differs across all five questions. Taken together, the empirical distributions are broadly similar but not identical across the two samples, with differences in magnitude that vary by question.

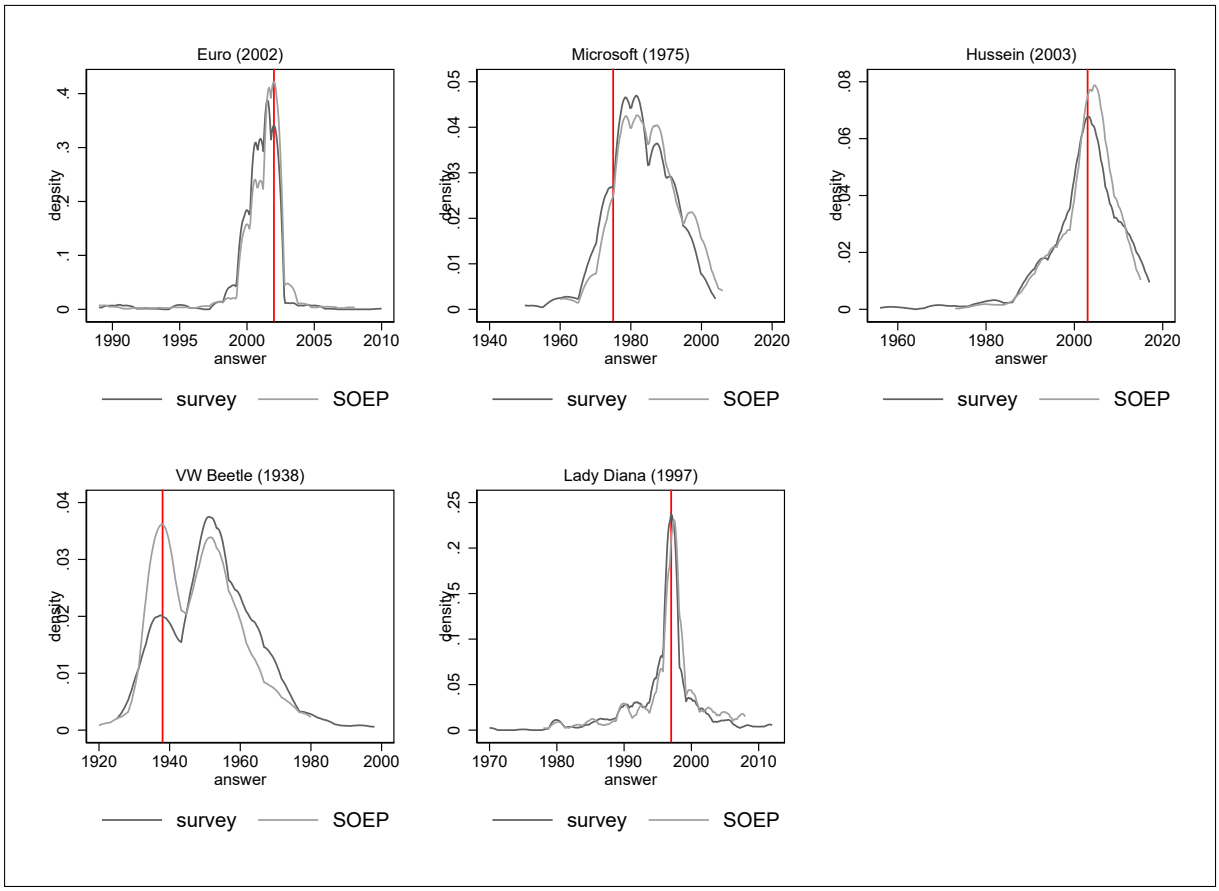


Figure E.1: Density of answers ($a_{i,j}$) to the history questions in SOEP-IS and the companion online survey. The vertical line marks the correct answer. The vertical-axis scale differs across questions.

Question	Mean (online)	Mean (SOEP-IS)	Δ	p-value	SD (online)	SD (SOEP-IS)	Δ	p-value
Euro (2002)	2000.835	2001.124	-0.289	0.004	1.980	2.055	-0.075	0.294
Microsoft (1975)	1982.691	1984.929	-2.237	< 0.001	8.974	9.108	-0.134	0.697
Hussein (2003)	2002.212	2002.955	-0.742	0.077	8.718	6.568	2.150	< 0.001
VW Beetle (1938)	1952.268	1948.535	3.734	< 0.001	12.561	11.502	1.059	0.016
Lady Diana (1997)	1995.975	1996.680	-0.704	0.010	5.367	5.141	0.226	0.240

Table E.1: Comparison of answers to the history questions in the companion online survey and SOEP-IS. Means, standard deviations, and differences are measured in years; Δ is the online-survey value minus the SOEP-IS value. The first p-value tests equality of means using a two-sample t-test, and the second tests equality of standard deviations. Variables are trimmed separately within each sample at the 1st and 99th percentiles. Sample sizes vary across questions because of item nonresponse and trimming.

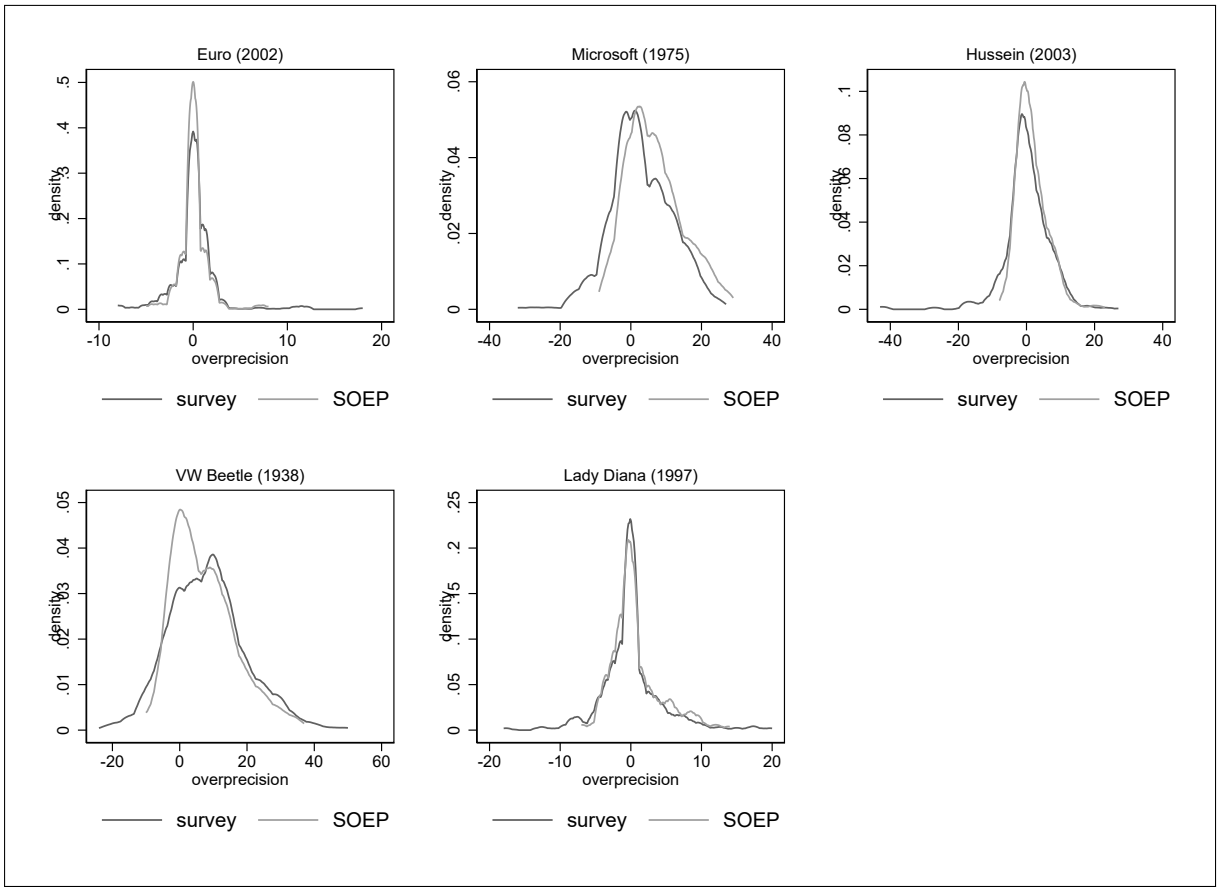


Figure E.2: Density of the overprecision measure ($op_{i,j}$) for the history questions in SOEP-IS and the companion online survey. The vertical-axis scale differs across questions.

Question	Mean (online)	Mean (SOEP-IS)	Δ	p-value	SD (online)	SD (SOEP-IS)	Δ	p-value
Euro (2002)	0.143	0.106	0.037	0.728	2.474	1.562	0.913	< 0.001
Microsoft (1975)	2.953	6.662	-3.708	< 0.001	8.687	7.979	0.708	0.031
Hussein (2003)	0.293	1.562	-1.269	< 0.001	6.707	4.554	2.153	< 0.001
VW Beetle (1938)	8.222	7.238	0.984	0.076	11.352	9.290	2.062	< 0.001
Lady Diana (1997)	0.096	0.663	-0.567	0.008	4.412	3.622	0.790	< 0.001

Table E.2: Comparison of the overprecision measure across the companion online survey and SOEP-IS. Means, standard deviations, and differences are measured in years; Δ is the online-survey value minus the SOEP-IS value. The first p-value tests equality of means using a two-sample t-test, and the second tests equality of standard deviations. Variables are trimmed separately within each sample at the 1st and 99th percentiles. Sample sizes vary across questions because of item nonresponse and trimming.

F Linking Overprecise Behavior Across the Financial and Political Domains

As shown in section 4, overprecision correlates with respondents' financial and political behaviors. However, it remains possible that *distinct* aspects of overprecision are related to behavior in each domain separately. To formalize this possibility, assume that our measured overprecision (op) comprises two orthogonal factors,[54] F_1 and F_2 plus an idiosyncratic error term such that:

$$op = F_1 + F_2 + \epsilon_{op}. \tag{7}$$

Consider two outcome variables, Y_1 (say, a financial behavior) and Y_2 (a political behavior) and suppose that domain 1 depends only on F_1 while domain 2 depends only on F_2. Mathematically, this would imply:

$$Y_1 = \beta_1 op + \eta_1 = \beta_1(1 \cdot F_1 + 0 \cdot F_2 + \epsilon_{op}) + \eta_1 = \beta_1 F_1 + \epsilon_1, \tag{8}$$

$$Y_2 = \beta_2 op + \eta_2 = \beta_2(0 \cdot F_1 + 1 \cdot F_2 + \epsilon_{op}) + \eta_2 = \beta_2 F_2 + \epsilon_2. \tag{9}$$

When estimating the two relationships separately, the estimated coefficients $\hat{\beta}_1$ and $\hat{\beta}_2$ would capture the effect of distinct sources of variation, or factors, in the overprecision measure. Instead, we propose a two-step procedure to determine whether a common source of variation in the overprecision measure correlates with both outcomes. This predicted-component, two-step design follows an approach used elsewhere in the empirical literature to assess whether predictable components of two variables share a common source (Hughes et al., 2008).

Let us take domain 1 as an example. In the first step, we regress the outcome Y_1 on the measure of overprecision:

$$Y_1 = \alpha_1 + \beta_1 op + \epsilon_1. \tag{10}$$

[54] The assumption of orthogonality is for ease of exposition; non-orthogonal factors introduce additional covariances but do not alter the fundamental logic of our two-step procedure.

From it, we obtain the fitted values,

$$\hat{Y}_1 = \hat{\alpha}_1 + \hat{\beta}_1 op, \tag{11}$$

where the fitted values $\hat{Y}_1$ capture the component of *op* variation that predicts Y_1.

In the second step, we regress the outcome in the other domain (Y_2) on the fitted values from the first step such that:

$$\begin{aligned} Y_2 &= \alpha_2 + \delta_2 \hat{Y}_1 + \mu_2 = \alpha_2 + \delta_2(\hat{\alpha}_1 + \hat{\beta}_1 op) + \mu_2, \\ &= (\alpha_2 + \delta_2 \hat{\alpha}_1) + \delta_2(\hat{\beta}_1 op) + \mu_2. \end{aligned} \tag{12}$$

If the estimated coefficient $\hat{\delta}_2$ is statistically significant, this suggests that overprecision correlates with outcomes in both domains.

In practice, true behavior is likely more complex than the simplified factorization in Equations (8) and (9). We treat this approach as a diagnostic tool for detecting a *common* source of overprecision rather than a test of causality. Additionally, this two-step approach does not fully account for latent variables that may jointly influence overprecision across domains. A structural equation model would complement this approach by explicitly modeling direct and indirect pathways between financial and political behavior. Unlike our two-step regression framework, structural models allow for the simultaneous estimation of multiple relationships and incorporate latent constructs that account for unobserved factors affecting both domains.

However, structural equation models also require strong distributional assumptions. They also require predefined theoretical pathways that specify whether financial overprecision directly influences political overprecision or whether another factor mediates the relationship. By contrast, our two-step approach is assumption-light, empirically testing whether overprecision-related behavior in one domain is associated with overprecision-related behavior in another while letting the data determine the nature of these relationships. This approach is ideal for exploratory analysis and opens the door to future research that can establish *a priori* links, to further disentangle causal mechanisms and uncover unbiased correlations between overprecision in financial, political, and other domains.

G Robustness Checks for the Companion Online Survey

In the following, we test the robustness of the companion survey analyses in Section 5 using the alternative exclusion rules specified in the preregistration. In the baseline analysis, we exclude respondents who report using external help, respondents identified as 'Googlers' using our control questions, and respondents in the bottom 5% of the self-reported effort distribution. A respondent is identified as a 'Googler' if, in any of the three factual domains, the respondent answers that domain's hard control question correctly and reports zero subjective error, while displaying the same pattern on at least three of the five main questions in that domain.

We implement seven robustness checks:

(1) We flag respondents who answer any hard control question correctly and report zero subjective error, without requiring the same pattern on three other questions.

(2) We exclude respondents who answer at least four of five questions correctly in any of the three factual domains.

(3) We exclude respondents who fall in the bottom response-time decile on at least 20% of the 25 main questions.

(4) We exclude respondents who fall in the top response-time decile on at least 20% of the 25 main questions.

(5) We exclude the bottom 10% of the self-reported effort distribution.

(6) We exclude the bottom 20% of the self-reported effort distribution.

(7) We exclude respondents who give the same numerical answer to at least three questions within any domain.

Table G.1 reports the principal-component results, Table G.2 the partial correlations, and Table G.3 the leave-one-out comparisons. The first-component eigenvalue remains between 2.23 and 2.34, and all component loadings remain positive, ranging from 0.48 to 0.78. The partial correlations also remain positive, ranging from 0.16 to 0.55. Every leave-one-out comparison retains the expected direction. All but two remain statistically significant at the 5% level; the remaining two are significant at the 10% level under

the stricter exclusion rules. Thus, the main patterns are stable across the preregistered alternatives, although the smaller samples make a few estimates less precise.

		Robustness						
	Baseline	1	2	3	4	5	6	7
First-component eigenvalue	2.29	2.30	2.28	2.34	2.23	2.27	2.32	2.25
Neutral loading	0.52	0.52	0.52	0.53	0.60	0.51	0.49	0.48
History loading	0.76	0.76	0.76	0.76	0.72	0.76	0.78	0.77
Knowledge loading	0.74	0.75	0.74	0.75	0.72	0.73	0.75	0.74
Stocks loading	0.69	0.69	0.68	0.71	0.68	0.68	0.67	0.68
Economics loading	0.65	0.65	0.65	0.65	0.61	0.66	0.67	0.65
N	552	544	547	487	440	535	476	500

Table G.1: Robustness of the principal-component analysis. The Baseline column applies the preregistered baseline exclusions, while Columns (1)–(7) apply the alternative exclusion rules defined in the text. N is the number of respondents for whom an aggregate score can be constructed in all five domains.

				Robustness													
		Baseline		1		2		3		4		5		6		7	
Domain 1	Domain 2	ρ	N	ρ	N	ρ	N	ρ	N	ρ	N	ρ	N	ρ	N	ρ	N
Neutral	History	0.17	688	0.17	680	0.17	679	0.18	588	0.19	560	0.16	658	0.18	578	0.16	632
Neutral	Knowledge	0.21	626	0.22	618	0.20	617	0.23	539	0.22	503	0.19	603	0.18	538	0.18	570
Neutral	Stocks	0.30	676	0.30	668	0.29	670	0.32	577	0.33	546	0.28	648	0.24	575	0.25	619
Neutral	Economics	0.18	639	0.18	632	0.18	633	0.17	552	0.20	518	0.17	613	0.19	542	0.17	580
History	Knowledge	0.52	609	0.52	600	0.52	600	0.52	531	0.49	488	0.52	586	0.55	522	0.53	554
History	Stocks	0.35	638	0.35	629	0.35	632	0.37	557	0.29	514	0.35	614	0.37	544	0.35	584
History	Economics	0.37	620	0.37	612	0.37	614	0.37	542	0.31	499	0.37	596	0.38	526	0.37	565
Knowledge	Stocks	0.34	591	0.34	582	0.33	585	0.37	516	0.32	471	0.33	573	0.34	509	0.32	537
Knowledge	Economics	0.32	580	0.33	572	0.32	574	0.33	507	0.28	461	0.32	562	0.35	502	0.30	525
Stocks	Economics	0.28	618	0.28	610	0.27	613	0.30	540	0.24	500	0.29	594	0.29	529	0.28	561

Table G.2: Robustness of the partial-correlation analysis. Each domain pair reports the partial correlation ρ and the corresponding number of observations. The Baseline columns apply the preregistered baseline exclusions, while Columns (1)–(7) apply the alternative exclusion rules defined in the text.

				Robustness													
		Baseline		1		2		3		4		5		6		7	
Grouping domain	Comparison domain	Δ	p	Δ	p	Δ	p	Δ	p	Δ	p	Δ	p	Δ	p	Δ	p
Neutral	History	-0.348	< 0.01	-0.335	< 0.01	-0.363	< 0.01	-0.394	< 0.01	-0.335	< 0.01	-0.346	< 0.01	-0.352	< 0.01	-0.335	< 0.01
Neutral	Knowledge	-0.406	< 0.01	-0.422	< 0.01	-0.373	< 0.01	-0.519	< 0.01	-0.368	< 0.01	-0.397	< 0.01	-0.409	< 0.01	-0.396	< 0.01
Neutral	Stocks	-0.339	< 0.01	-0.343	< 0.01	-0.341	< 0.01	-0.354	< 0.01	-0.338	< 0.01	-0.301	< 0.01	-0.273	< 0.01	-0.328	< 0.01
Neutral	Economics	-0.323	< 0.01	-0.322	< 0.01	-0.340	< 0.01	-0.326	< 0.01	-0.338	< 0.01	-0.311	< 0.01	-0.364	< 0.01	-0.311	< 0.01
History	Neutral	-0.218	0.01	-0.184	0.03	-0.221	0.01	-0.243	0.01	-0.288	< 0.01	-0.207	0.02	-0.273	< 0.01	-0.163	0.07
History	Knowledge	-0.566	< 0.01	-0.572	< 0.01	-0.547	< 0.01	-0.605	< 0.01	-0.547	< 0.01	-0.549	< 0.01	-0.534	< 0.01	-0.556	< 0.01
History	Stocks	-0.432	< 0.01	-0.436	< 0.01	-0.453	< 0.01	-0.439	< 0.01	-0.477	< 0.01	-0.421	< 0.01	-0.368	< 0.01	-0.387	< 0.01
History	Economics	-0.385	< 0.01	-0.393	< 0.01	-0.383	< 0.01	-0.366	< 0.01	-0.439	< 0.01	-0.370	< 0.01	-0.405	< 0.01	-0.390	< 0.01
Knowledge	Neutral	-0.230	0.01	-0.244	< 0.01	-0.215	0.01	-0.247	0.01	-0.186	0.05	-0.206	0.02	-0.150	0.10	-0.213	0.02
Knowledge	History	-0.451	< 0.01	-0.480	< 0.01	-0.463	< 0.01	-0.381	< 0.01	-0.454	< 0.01	-0.468	< 0.01	-0.419	< 0.01	-0.488	< 0.01
Knowledge	Stocks	-0.323	< 0.01	-0.336	< 0.01	-0.314	< 0.01	-0.349	< 0.01	-0.341	< 0.01	-0.315	< 0.01	-0.271	< 0.01	-0.325	< 0.01
Knowledge	Economics	-0.301	< 0.01	-0.307	< 0.01	-0.291	< 0.01	-0.301	< 0.01	-0.308	< 0.01	-0.281	< 0.01	-0.320	< 0.01	-0.279	< 0.01
Stocks	Neutral	-0.449	< 0.01	-0.418	< 0.01	-0.436	< 0.01	-0.451	< 0.01	-0.467	< 0.01	-0.421	< 0.01	-0.406	< 0.01	-0.407	< 0.01
Stocks	History	-0.421	< 0.01	-0.408	< 0.01	-0.421	< 0.01	-0.416	< 0.01	-0.467	< 0.01	-0.416	< 0.01	-0.411	< 0.01	-0.361	< 0.01
Stocks	Knowledge	-0.368	< 0.01	-0.408	< 0.01	-0.360	< 0.01	-0.432	< 0.01	-0.381	< 0.01	-0.364	< 0.01	-0.352	< 0.01	-0.325	< 0.01
Stocks	Economics	-0.371	< 0.01	-0.395	< 0.01	-0.369	< 0.01	-0.437	< 0.01	-0.394	< 0.01	-0.397	< 0.01	-0.459	< 0.01	-0.348	< 0.01
Economics	Neutral	-0.338	< 0.01	-0.313	< 0.01	-0.358	< 0.01	-0.344	< 0.01	-0.346	< 0.01	-0.323	< 0.01	-0.302	< 0.01	-0.319	< 0.01
Economics	History	-0.384	< 0.01	-0.367	< 0.01	-0.386	< 0.01	-0.336	< 0.01	-0.445	< 0.01	-0.376	< 0.01	-0.359	< 0.01	-0.341	< 0.01
Economics	Knowledge	-0.292	< 0.01	-0.306	< 0.01	-0.277	< 0.01	-0.273	0.01	-0.320	< 0.01	-0.281	< 0.01	-0.301	< 0.01	-0.262	< 0.01
Economics	Stocks	-0.381	< 0.01	-0.386	< 0.01	-0.330	< 0.01	-0.364	< 0.01	-0.360	< 0.01	-0.363	< 0.01	-0.359	< 0.01	-0.340	< 0.01

Table G.3: Robustness of the leave-one-out analysis. The grouping domain defines the upper-quartile and lower-75% groups; we then compare overprecision in the comparison domain. We define Δ as the mean for the lower 75% minus the mean for the upper quartile, so a negative value indicates greater overprecision among respondents in the upper quartile of the grouping domain. The Baseline columns apply the preregistered baseline exclusions, while Columns (1)–(7) apply the alternative exclusion rules defined in the text.

H Further Exploratory Analyses in the Companion Survey

We report two additional exploratory analyses using data from the companion online survey in Section 5. We did not include these analyses in the preregistration.

H.1 Overprecision and Self-Reported Expertise

The literature suggests that greater expertise can improve accuracy without necessarily reducing overprecision. Önkal et al. (2003) find that professional forecasters make more accurate point forecasts of exchange rates than amateurs, but show similar overconfidence when providing 90% prediction intervals. McKenzie et al. (2008) examine the accuracy and width of intervals together. For questions asking respondents to estimate percentages, experts provide intervals whose midpoints are closer to the true answer, but which are also narrower than those of novices. These two effects offset one another, leaving experts and novices with similar levels of overprecision.

Next, we examine this relationship in our data using a self-reported measure of expertise. To do so, we regress the mean realized error, mean subjective error, and the aggregate measure of overprecision in each domain on self-reported expertise in that domain. Because self-reported expertise may itself reflect a respondent's overprecision, in Columns (2), (4), and (6) we additionally control for the first principal component of the standardized aggregate overprecision measures in the other four domains. This component summarizes the comovement in those four measures without using the measure from the domain being analyzed.

Table H.1 shows that self-reported expertise in a domain is associated with decreases in both mean realized error and mean subjective error. The association between self-reported expertise and overprecision in Columns (5) and (6) is small and imprecise in the history, general-knowledge, and economics domains. In the stock domain, it is positive and statistically significant in Column (5) without the cross-domain component, but smaller and imprecise once we include the component in Column (6). Overall, self-reported expertise is associated with lower realized and subjective errors, but not systematically with the difference between them.

These findings complement the analysis in Appendix J.1, which examines how stated and realized error vary together. Here, we examine how each component relates to self-reported expertise. Expertise can have similar associations with both components even

	mean realized error		mean subjective error		overprecision measure	
	(1)	(2)	(3)	(4)	(5)	(6)
History:						
expertise	-0.036***	-0.039***	-0.044***	-0.041***	0.001	0.000
	(0.008)	(0.008)	(0.013)	(0.014)	(0.002)	(0.002)
cross-domain first component		0.389*		-3.437***		0.581***
		(0.208)		(0.856)		(0.129)
N	703	552	703	552	703	552
adj. R^2	0.032	0.042	0.025	0.312	-0.001	0.291
Knowledge:						
expertise	-0.030*	-0.030*	-0.054***	-0.056***	0.002	0.002
	(0.015)	(0.017)	(0.017)	(0.017)	(0.002)	(0.002)
cross-domain first component		1.287***		-4.277***		0.484***
		(0.414)		(0.596)		(0.055)
N	633	552	633	552	633	552
adj. R^2	0.005	0.028	0.023	0.370	0.001	0.272
Stocks:						
expertise	-0.026**	-0.028**	-0.058***	-0.046***	0.003**	0.002
	(0.011)	(0.012)	(0.012)	(0.011)	(0.001)	(0.001)
cross-domain first component		0.983*		-3.655***		0.443***
		(0.555)		(0.735)		(0.075)
N	690	552	690	552	690	552
adj. R^2	0.005	0.020	0.024	0.214	0.004	0.219
Economics:						
expertise	-0.039***	-0.043***	-0.066***	-0.050***	0.002	0.001
	(0.011)	(0.011)	(0.015)	(0.014)	(0.002)	(0.001)
cross-domain first component		0.429*		-3.839***		0.376***
		(0.250)		(0.747)		(0.067)
N	647	552	647	552	647	552
adj. R^2	0.019	0.025	0.030	0.230	0.002	0.188

Robust standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table H.1: Self-reported expertise and the components of overprecision. This table reports results from regressing the mean realized error, mean subjective error, and the overprecision measure in each domain on self-reported expertise in the same domain. In Columns (2), (4), and (6), we additionally control for the first principal component of the standardized aggregate overprecision measures in the other four domains. Self-reported expertise is measured on a scale from 0 to 100.

when other sources of variation in realized error are accompanied by smaller differences in stated uncertainty. The results therefore need not imply a one-for-one relationship between stated and realized error. Because expertise is self-reported, this analysis describes perceived knowledge and does not independently measure actual knowledge.

H.2 Overprecision and One-Month-Ahead Stock-Price Forecast Errors

This section examines whether the association between overprecision and stock price forecast errors reported in Section 4.2 also appears over a different time horizon. We calculate the absolute one-month-ahead forecast error for each stock and regress the resulting forecast error on a measure of overprecision with and without control variables (Columns (1)–(10) in Table H.2). We also compute the first principal component of the five absolute forecast errors and regress this component on a measure of overprecision and control variables in Column (11).

The first panel applies the analysis in Section 4.2 using the standardized aggregate overprecision measure in the contemporary history domain. The coefficient on the first principal component of forecast errors is positive but imprecise. Among the stock-specific estimates, the coefficients for Benz and BASF are positive and statistically significant without controls; only the Benz estimate remains marginally significant with controls. Thus, the history-domain results are consistent in direction with the positive SOEP-IS estimate but provide less precise evidence at the shorter horizon. In the second panel, we use the standardized aggregate overprecision measure from the stock forecast domain instead of the history domain. All five stock-specific coefficients and the coefficient on the first principal component of forecast errors are positive and statistically significant. This relationship is partly mechanical because the realized forecast errors used as outcomes also enter the stock-domain overprecision measure.

To reduce this overlap, in the third and fourth panels, we use the first principal component of the standardized aggregate overprecision measures across all five domains.[55] Because this component still includes the stock-forecast domain, it reduces rather than eliminates the direct overlap. Requiring an overprecision measure in every domain reduces the number of observations, but the resulting measure summarizes information across all five domains and may be less affected by item-specific noise.[56] The estimates are positive for most individual stocks and for the first principal component of forecast errors. In the fourth panel, we winsorize the forecast errors at the 5th and 95th percentiles

[55]The first component has an eigenvalue of 2.29 and positive loadings across the five domains (0.52, 0.76, 0.74, 0.69, and 0.65).

[56]The principal component calculated across all questions, instead of across all domain-specific aggregate overprecision measures, exhibits a Spearman correlation coefficient of 0.96 with the principal component calculated across all domain-specific aggregate overprecision measures.

to avoid outliers influencing the results. Winsorization generally reduces the coefficient magnitudes, but the aggregate association remains positive and statistically significant; several stock-specific estimates are less precise.

In the fifth panel, we exclude the stock forecast domain from the aggregate overprecision measure, thereby removing the direct overlap with the forecast-error outcomes. The estimates are smaller and less precise: four of the five controlled stock-specific coefficients remain positive, and the coefficient on the first principal component of forecast errors is positive but imprecise. The Benz and BASF estimates remain statistically significant at conventional levels.

Forecast error in:	Benz		Puma		BMW		Post		BASF		First PC
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Overprecision in history domain:											
overprecision measure	0.924**	0.779*	0.303	0.167	0.066	-0.074	-0.431	-0.577	0.581*	0.441	0.018
	(0.466)	(0.446)	(0.432)	(0.428)	(0.498)	(0.501)	(0.557)	(0.568)	(0.297)	(0.297)	(0.041)
N	665	663	669	667	665	663	657	655	664	662	641
adj. R^2	0.008	0.036	-0.000	0.005	-0.001	0.031	0.000	0.028	0.005	0.001	0.028
controls	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	Yes
Overprecision in stock forecast domain:											
overprecision measure	5.400***	5.141***	4.981***	4.831***	6.460***	6.025***	2.718***	2.467***	3.631***	3.463***	0.566***
	(1.000)	(0.963)	(0.831)	(0.845)	(1.503)	(1.462)	(0.794)	(0.819)	(0.660)	(0.666)	(0.106)
N	687	683	687	683	686	682	683	679	687	683	666
adj. R^2	0.272	0.276	0.231	0.225	0.224	0.225	0.063	0.071	0.198	0.175	0.314
controls	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	Yes
First principal component of overprecision:											
overprecision measure	2.769***	2.638***	1.970**	1.862**	2.990**	2.740**	0.845	0.555	1.864***	1.736***	0.243**
	(0.903)	(0.885)	(0.806)	(0.806)	(1.241)	(1.217)	(0.781)	(0.788)	(0.577)	(0.570)	(0.098)
N	550	548	551	549	549	547	548	546	551	549	539
adj. R^2	0.083	0.094	0.043	0.043	0.052	0.070	0.005	0.021	0.057	0.041	0.075
controls	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	Yes
First principal component of overprecision and winsorized outcomes:											
overprecision measure	1.011***	0.880***	1.328***	1.215**	0.521	0.421	0.969*	0.767	1.134***	1.016***	0.094**
	(0.323)	(0.313)	(0.510)	(0.505)	(0.332)	(0.331)	(0.516)	(0.513)	(0.321)	(0.313)	(0.043)
N	550	548	551	549	549	547	548	546	551	549	539
adj. R^2	0.029	0.045	0.032	0.042	0.005	0.015	0.016	0.040	0.040	0.025	0.036
controls	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	Yes
First principal component of overprecision excluding the stock-forecast domain:											
overprecision measure	1.515**	1.339**	0.705	0.508	1.405*	1.029	0.143	-0.188	0.971**	0.788*	0.086
	(0.636)	(0.618)	(0.648)	(0.649)	(0.842)	(0.835)	(0.693)	(0.709)	(0.441)	(0.438)	(0.071)
N	556	554	557	555	556	554	552	550	557	555	542
adj. R^2	0.024	0.039	0.004	0.008	0.010	0.033	-0.002	0.018	0.014	0.001	0.023
controls	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	Yes

Robust standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table H.2: Forecast errors and the overprecision measure. This table reports the results of regressing the absolute one-month-ahead forecast errors for each of the stocks (Columns (1)–(10)) and the first principal component of the five forecast errors (Column (11)) on overprecision. Odd-numbered columns omit control variables, even-numbered columns include them, and Column (11) includes controls.

I Calibration Analysis

Individuals may interpret the second question of the Subjective Error Method differently. In particular, a reported subjective error may correspond to different percentiles of respondents' perceived absolute-error distributions. Consider the example in Figure I.1, which illustrates a hypothetical subjective distribution of possible signed errors for two individuals. Both individuals hold the same beliefs about the correct answer and make the same realized error (vertical red line). Since the Subjective Error Method asks for the absolute error, we transform the dotted signed-error distribution to a half-normal distribution and its cumulative distribution (both represented by continuous lines).[57] As shown in the graph, individual i, who reports the 95th percentile of the perceived absolute-error distribution, reports a larger subjective error than individual j, who reports its 68th percentile. Consequently, two people with the same expected error distribution and realized error could be classified as over- or underprecise, depending on how they interpret the Subjective Error Method's second question.

To investigate where the reported subjective error lies within respondents' elicited absolute-error distributions, we run a second online companion survey using a representative sample of the German population. We archived the preregistration and analysis plan at AsPredicted (#123146), available at https://aspredicted.org/2QD_7R1.

I.1 Survey Design

As in the first companion survey, respondents estimated the year in which each event covered by the five most frequently answered SOEP questions occurred and reported their subjective error. In this second survey, they also reported a binned probability distribution over possible correct years using the click-and-drag interface developed by Crosetto and De Haan (2022).[58] Expressing the possible years relative to each respondent's point estimate yielded an implied distribution of signed errors, with errors of 20 years or more

[57] Note that this assumes that the distribution is symmetric around zero.

[58] For each item, respondents saw a graph covering 20 years on either side of their point estimate. They assigned probability mass by clicking and dragging points on the graph, and the interface normalized the assigned weights to sum to 100%.

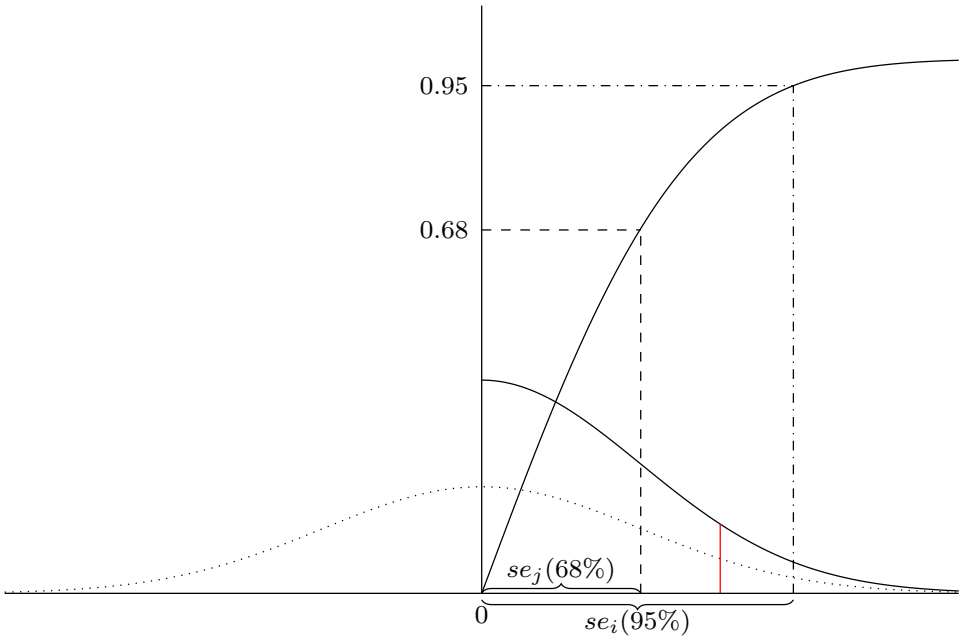


Figure I.1: Hypothetical distribution of the subjective error. The dotted normal density with a mean of zero and a standard deviation of two represents the subjective distribution of signed errors for respondents i and j. The solid curves show the corresponding density and cumulative distribution of the absolute error. The vertical red line marks the realized error for respondents i and j. The dot-dashed line marks the 95th percentile reported by respondent i, while the dashed line marks the 68th percentile reported by respondent j.

in either direction pooled in the two outer bins.[59] Half of the respondents completed the distribution elicitation before reporting their subjective error, while the other half completed the two tasks in the reverse order.

To support data quality, we included attention checks and screened out respondents who failed them. At the end of the survey, we also asked whether respondents had used Google or another search engine to answer the questions. In addition, we included a deliberately difficult "Google-control" item asking for the year in which the Treaty of Luxembourg—the first budgetary treaty of the European Communities—was signed (1970).[60] As preregistered, we retained the Google-control item as a sixth item in the analysis to preserve statistical power.

[59]Suppose, for example, that a respondent gives 1995 as the year of Princess Diana's death and reports a subjective error of one year. If the respondent assigns 60% probability to 1995, 10% each to 1994 and 1996, 5% each to 1993 and 1997, 2.5% each to 1992 and 1998, and the remaining 5% to years outside this range, then 80% of the probability mass lies within one year of the point estimate. The reported subjective error therefore lies at the 80th percentile of the implied absolute-error distribution. Respondents were not required to provide symmetric distributions.

[60]Following our preregistration, we flagged a respondent as a likely search-engine user if they answered the Google-control item correctly with a subjective error of zero and displayed the same pattern on at least three of the five main questions.

At the end of the survey, we collected respondents' age, gender, years of education, income, and nationality. We measured mathematical literacy using three problems. Respondents also rated their knowledge of contemporary history on a scale from 0 (not knowledgeable at all) to 100 (very knowledgeable) and reported the effort they had devoted to the survey.

I.2 Data

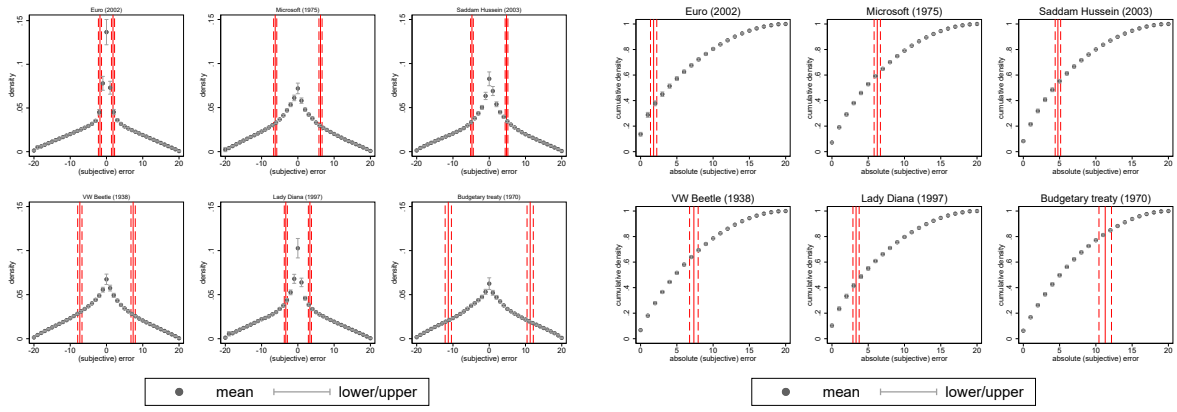
We collected 1,000 completed surveys. As preregistered, we excluded respondents who reported receiving outside assistance, those flagged as likely search-engine users based on the control item described in Section I.1, and those in the bottom 5% of the self-reported effort distribution. These restrictions left 818 eligible respondents. For analyses using the elicited distributions, we further required respondents to provide at least four of the six distributions, reducing the sample to 650. Finally, we excluded six respondents who, in at least one distribution, placed 100% of the probability mass in one of the two outermost bins at either end of the elicited range. This yielded a final analysis sample of 644.

I.3 Results

Figure I.2a presents the probability density functions of the possible signed errors, averaged across respondents for each question. The gray dots represent the mean probabilities, the gray bars their 90% confidence intervals, and the red lines the mean subjective errors and their 90% confidence intervals. Both the elicited distributions and the reported subjective errors vary across questions. For easier questions, such as the Euro question, respondents provide narrower distributions and report smaller subjective errors.

Figure I.2b presents the corresponding cumulative distribution functions of the absolute errors implied by the elicited distributions. The position of the average subjective error within these distributions also varies across questions. It lies at approximately the 40th percentile for the Euro question but at approximately the 80th percentile for the difficult Google-control item. These aggregate patterns suggest that, for more difficult questions, respondents not only report greater uncertainty but also place their subjective error at a higher percentile of their elicited absolute-error distribution.

The raw patterns do not account for differences in respondents' realized errors or effort in completing the distribution questions. To examine the first issue, Figure I.3a divides



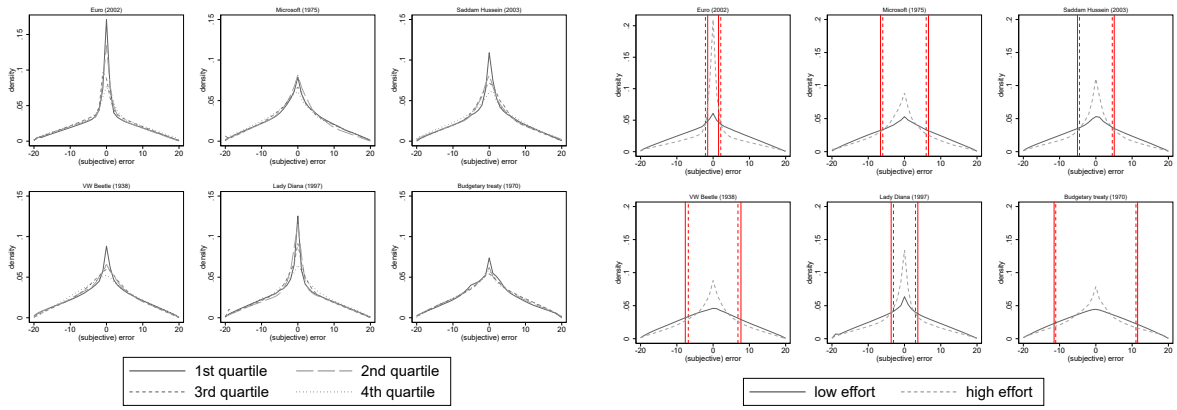
(a) PDF **(b)** CDF

Figure I.2: Subjective error and the elicited distributions of possible errors in the raw data. Panel (a) shows the probability density functions and Panel (b) the corresponding cumulative distribution functions. Gray points show the mean probability assigned to each possible error, and gray bars show its 90% confidence interval. The solid vertical red line denotes the average subjective error, while the dashed vertical red lines mark its 90% confidence interval.

respondents into four quartiles of realized error for each question and plots the average elicited distribution within each quartile. Realized error may reflect factual knowledge as well as question-specific noise. Respondents with smaller realized errors provide narrower distributions on average.

The click-and-drag interface automatically generates a triangular distribution after the first click, so respondents who interact little with the interface may submit a distribution close to this initial shape. Figure I.3b examines this possibility by dividing responses according to whether the time spent constructing the distribution is below or above the median. Distributions in the shorter-time group tend to be closer to triangular, whereas those in the longer-time group are narrower. Because response time may reflect both effort and uncertainty, we treat it as an imperfect proxy for effort. Reported subjective errors, however, differ little between the two groups.

As an additional exploratory check, we classify each elicited distribution by shape, as shown in Figure I.4. We label perfectly triangular distributions "low effort" and all other distributions "high effort." For each question and group, we locate the group's mean reported subjective error within its average elicited absolute-error distribution. For



(a) Realized error **(b)** Effort

Figure I.3: Elicited error distributions grouped by realized error and response time. Panel (a) groups responses by quartiles of item-level absolute realized error. Panel (b) compares responses with time spent constructing the distribution at or below the median ("low effort") and above the median ("high effort"). In Panel (b), solid and dashed vertical red lines mark the mean reported subjective error for the low- and high-effort groups, respectively. Each mean is marked at both its positive and negative value.

triangular distributions (solid curves), this location ranges from approximately 15% to 85% across the six questions. For non-triangular distributions (dashed curves), it remains around 60% for the five main questions, but is higher for the Google-control item. Across the five main questions, the percentile locations therefore vary less for non-triangular distributions. Because distribution shape may capture aspects of interface use other than effort, we treat this comparison as descriptive. In the analyses below, we report both the raw results and results adjusted for realized error and the two effort proxies.

To assess how consistently respondents interpret the second SEM question, we calculate the within-respondent standard deviation of the percentile location of their reported subjective error within the elicited absolute-error distribution. We calculate this measure across the available questions for respondents who provided at least four of the six distributions. Larger values indicate greater variation in percentile locations across questions.

Alongside the raw measure, we construct adjusted percentiles by subtracting the estimated contributions of realized error and the two effort proxies.[61] Figure I.5a plots the

[61]For each question, we regress the observed percentile location on realized error, time spent on the distribution question, and an indicator for a perfectly triangular distribution. The regressions also control

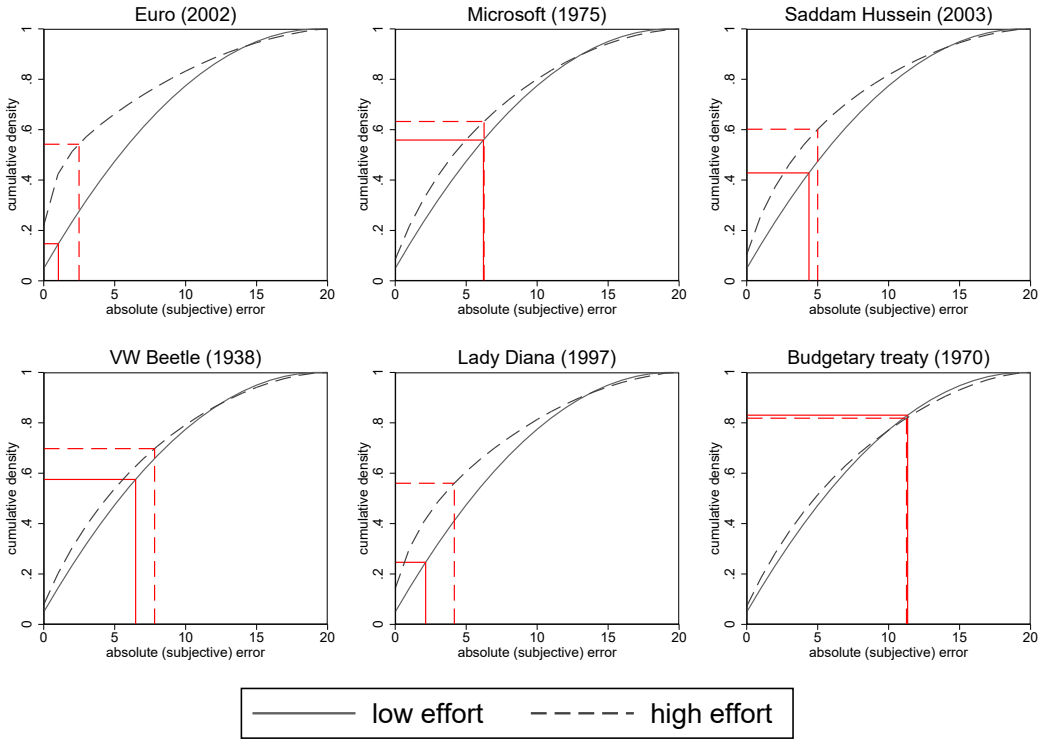
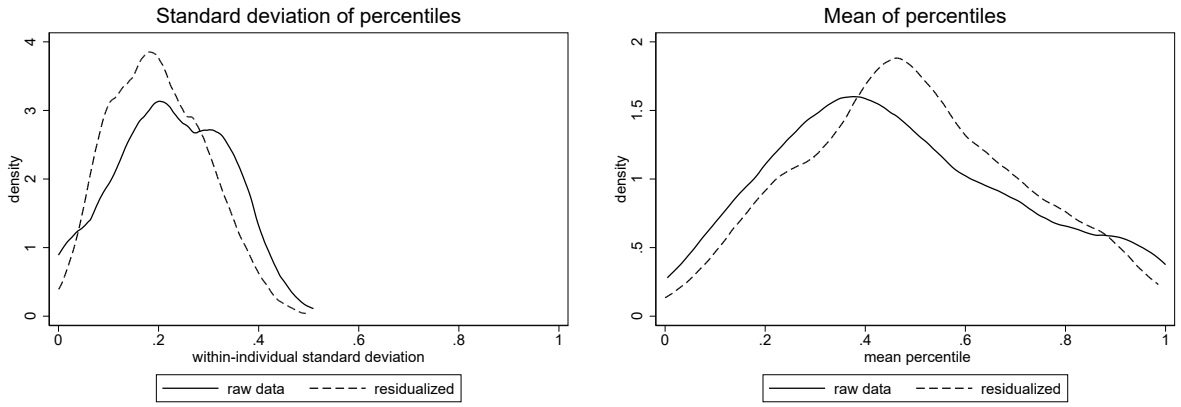


Figure I.4: Cumulative distributions of absolute errors by distribution shape. Solid curves represent perfectly triangular distributions ("low effort"), while dashed curves represent all other distributions ("high effort"). The corresponding solid and dashed vertical red lines mark each group's mean reported subjective error. Horizontal red lines show the percentile location of that mean within the group's average elicited absolute-error distribution. We use distribution shape only as a proxy for effort.

distributions of the raw and adjusted within-respondent standard deviations. The figure shows lower dispersion in percentile locations after adjustment. The median adjusted standard deviation is 0.19, corresponding to 19 percentile points.

We also calculate each respondent's mean percentile location across the available questions, using up to six distributions. This provides a summary of where their reported subjective error typically falls within their elicited absolute-error distributions. We calculate these respondent means using both the raw and adjusted percentiles described above. Figure I.5b plots both measures across respondents. The adjusted means are less dispersed than the raw means. Their median is 0.48, corresponding to the 48th percentile, and the middle half lies between 0.33 and 0.64. These results suggest that respondents typically report a subjective error near the middle of their elicited absolute-error distribution,

for age, gender, years of education, income categories, German citizenship, self-reported expertise in contemporary history, mathematical literacy, the number of distributions provided, and state fixed effects. We subtract the estimated contributions of realized error and the two effort proxies, and set adjusted values below zero or above one to zero or one, respectively. This adjustment was not preregistered.



(a) Standard deviation **(b)** Mean location

Figure I.5: Densities of the within-individual standard deviation of the location of the subjective error and the individual-specific mean percentile. "Raw data" uses the observed percentiles, while "residualized" adjusts them for realized error and the two effort proxies.

although its location varies across respondents.

Finally, we ask whether reported subjective errors are associated with the dispersion of the probability distributions elicited in the same exercise. At the respondent–question level, the analysis relates subjective error to the variance of the elicited distribution and includes the "Google-control" item. Because variance is measured in squared years and subjective error in years, the coefficient cannot be interpreted as a one-for-one calibration benchmark. Instead, the analysis tests whether the two reports vary together as expected.

The specifications use levels or logarithms, include question fixed effects and either respondent characteristics or respondent fixed effects, and cluster standard errors by respondent. Columns (1)–(4) of Table I.1 report the level specifications, while Columns (5)–(8) use the logarithms of both subjective error and variance.[62] The association is positive and statistically significant in every specification: respondents who report larger subjective errors also provide more dispersed probability distributions. This result supports interpreting subjective error as stated uncertainty within the exercise. Because both responses come from the same task, however, it does not validate the measure against an

[62]The logarithmic specifications contain fewer observations because they exclude observations for which variance or subjective error equals zero.

	Subjective Error							
	Levels				Logarithmic			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
variance of elicited distribution	0.082***	0.030***	0.126***	0.130***	0.179***	0.098***	0.307***	0.301***
	(0.012)	(0.008)	(0.014)	(0.015)	(0.016)	(0.015)	(0.017)	(0.017)
age		-0.065***	-0.012	-0.014		-0.009***	-0.001	-0.001
		(0.017)	(0.012)	(0.012)		(0.002)	(0.001)	(0.001)
female		-0.320	-0.049	0.105		-0.011	0.042	0.019
		(0.548)	(0.379)	(0.358)		(0.051)	(0.029)	(0.029)
years education		0.057	-0.122	-0.119		-0.005	-0.019**	-0.018**
		(0.174)	(0.104)	(0.101)		(0.018)	(0.008)	(0.008)
number of distributions completed		0.372	-0.905***	-0.940**		0.074	-0.057*	-0.074**
		(0.523)	(0.324)	(0.364)		(0.057)	(0.032)	(0.033)
gross income		-0.179	-0.091	-0.080		-0.012	-0.003	-0.004
		(0.109)	(0.070)	(0.068)		(0.011)	(0.006)	(0.006)
income missing		-1.910***	-0.419	-0.341		-0.172**	-0.022	-0.038
		(0.556)	(0.441)	(0.458)		(0.080)	(0.055)	(0.053)
expertise		-0.034***	-0.002	-0.004		-0.005***	-0.002**	-0.001**
		(0.011)	(0.009)	(0.008)		(0.001)	(0.001)	(0.001)
math. literacy		0.230	-1.467**	-1.414**		0.100	-0.081*	-0.092**
		(0.753)	(0.670)	(0.645)		(0.074)	(0.043)	(0.042)
mean percentile location			19.894***	20.196***			2.329***	2.354***
			(2.057)	(2.138)			(0.110)	(0.114)
SD of percentile location				-4.346				0.617***
				(2.676)				(0.161)
N	3442	3437	3437	3433	2670	2679	2679	2677
adj. R^2	0.586	0.174	0.404	0.408	0.595	0.317	0.553	0.557
Question Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual Fixed Effects	Yes	No	No	No	Yes	No	No	No

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table I.1: Relationship between subjective error and the dispersion of elicited distributions. Columns (1)–(4) use subjective error and variance in levels, and Columns (5)–(8) use their logarithms. All specifications include question fixed effects. Columns (1) and (5) include respondent fixed effects; the remaining columns include respondent characteristics. Columns (3), (4), (7), and (8) also include the respondent's mean percentile location, while Columns (4) and (8) add its within-respondent standard deviation. Standard errors are clustered at the respondent level.

independent elicitation method or establish that other methods would produce similar respondent rankings.

J Overprecision, Realized Error, and Knowledge Sensitivity

For each answered question, the SEM subtracts the error a respondent expected to make (SE) from the error they actually made (TE) and then averages these differences across questions. Because such a measure uses differences in levels, the resulting gap depends partly on the size of the errors. Thus, even when two respondents underestimate their errors by the same percentage, the respondent with the larger realized error will have the larger SEM gap.

This scale dependence matters because realized errors may reflect factual knowledge as well as question-specific noise. Our baseline regressions already control for education and financial literacy, but these variables may not fully capture respondents' knowledge of the factual questions. If knowledge remains related to both realized error and an outcome after accounting for these controls, it may contribute to the observed association between the SEM measure and that outcome. This appendix examines how stated and realized errors vary together and how the behavioral associations change when we account for realized error in different ways.

The analyses proceed in four steps. First, we examine how closely stated and realized error move together and assess whether item-specific noise weakens their estimated relationship. Second, we retain the baseline gap and add the combined error level to assess whether the gap remains associated with behavior after this adjustment. Third, we retain the gap but adjust directly for respondent-level mean realized error to examine whether the associations persist among respondents with similar observed accuracy. Fourth, we construct an alternative measure that compares stated uncertainty with the uncertainty predicted for similar realized errors on the same item, using the fitted sample relationship as the benchmark.

The results show that respondents who make larger errors also tend to report greater uncertainty. Even so, differences in accuracy can still influence the SEM gap. This matters for interpreting the behavioral findings, since the estimated associations change when we account for accuracy or construct the SEM measure differently. We discuss these findings and their implications in the final subsection of this appendix.

J.1 The relationship between stated and realized error

We begin by examining how stated uncertainty varies with realized error and what this means for the SEM gap. Respondents can be overprecise even if stated and realized error increase one-for-one, provided that stated error remains consistently lower. A slope below one indicates that the gap tends to grow as realized error increases. Yet, because realized error combines a predictable component with item-specific realization noise, the raw relationship may understate how strongly stated uncertainty responds to predictable error. To illustrate how realization noise can weaken the estimated relationship, consider the stylized model

$$TE_{i,j} = m_{i,j} + u_{i,j}, \qquad SE_{i,j} = \alpha + \lambda m_{i,j} + e_{i,j},$$

where $m_{i,j}$ is the predictable component of realized error, $u_{i,j}$ is item-specific realization noise, and $e_{i,j}$ captures the remaining variation in subjective error. Assume that, within each item, u is uncorrelated with m and e is uncorrelated with TE. The population slope from regressing SE on TE with item fixed effects is then

$$\beta_{\text{OLS}} = \lambda r_1, \qquad r_1 = \frac{\text{Var}(m)}{\text{Var}(m) + \text{Var}(u)},$$

where both variances are calculated after subtracting item-specific means. The ratio r_1 captures the share of variation in realized error attributable to its predictable component. If realization noise has positive variance, then $r_1 < 1$. The estimated slope can therefore be below one even when $\lambda = 1$, meaning that stated uncertainty responds one-for-one to predictable error. On its own, the raw slope cannot tell us how much of a low estimate reflects realization noise and how much reflects a weaker response of stated uncertainty.

The three specifications in Table J.1 assess whether item-specific realization noise weakens the estimated relationship between stated and realized error. Column (1) provides the item-level benchmark. Column (2) uses respondent-level averages across questions to reduce the influence of single-item noise. Column (3) instruments realized error on a given question with the respondent's errors on the remaining questions, drawing on persistent differences in accuracy across questions. Larger coefficients in Columns (2) and (3) would be consistent with attenuation in Column (1), provided that these ap-

proaches reduce the influence of realization noise.

	(1) Item-FE OLS	(2) Respondent means	(3) Proxy IV
Realized error (TE)	0.125***	0.257***	0.383***
	(0.012)	(0.028)	(0.059)
N	4562	805	4523
KP rk Wald F			136.45

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table J.1: Complementary diagnostics of the SE–TE relationship. Column (1) reports item-level OLS with item fixed effects. Column (2) reports OLS using respondent-level means. Column (3) reports item-level 2SLS, instrumenting focal-item TE with the leave-one-out mean of the respondent's realized errors on the remaining items after removing item effects. Standard errors are clustered by respondent in Columns (1) and (3) and heteroskedasticity-robust in Column (2). N denotes respondent–item observations in Columns (1) and (3) and respondents in Column (2). Column (3) includes 766 respondents and excludes 39 for whom the leave-one-out instrument cannot be constructed.

Columns (1) and (2) of Table J.1 show a positive relationship between SE and TE. The item-level slope in Column (1) is well below one, but increases when we average both measures across questions in Column (2). This increase is consistent with averaging reducing the influence of item-specific realization noise. At the same time, averaging changes the variation used in the regression, and respondents' averages can cover different sets of answered questions. Column (2) is therefore consistent with noise weakening the item-level relationship, but it does not tell us how large that effect is.

Column (3) of Table J.1 reports an IV slope larger than the item-level OLS slope in Column (1). The leave-one-out instrument strongly predicts focal-item TE, and the OLS slope is essentially unchanged in the IV sample, suggesting that sample differences do not explain the increase. Interpreting the IV slope as the relationship between SE and predictable error requires the instrument to be unrelated to focal-item realization noise and to subjective-error variation not explained by predictable error. Persistent respondent characteristics could violate these assumptions if they affect both other-question performance and reported uncertainty beyond their relationship with predictable error. Under these assumptions, the IV estimate indicates that stated uncertainty increases less than one-for-one with predictable error, and we reject a unit slope.[63]

[63]The IV estimate is similar when the leave-one-out instrument is constructed without first removing item effects.

J.2 Paired gap and component decomposition

We now examine whether the behavioral associations depend only on the SEM gap or also on the combined level of realized and stated error. Two respondents can have the same gap even though one makes small errors and reports little uncertainty, while the other makes larger errors and reports greater uncertainty. A regression using only the gap treats these cases alike, holding the other controls constant.

The baseline regressions in Table 2 use a standardized version of the gap, $\overline{TE}_i - \overline{SE}_i$. Using only this difference imposes equal and opposite coefficients on realized and stated error. Adding their sum, $\overline{TE}_i + \overline{SE}_i$, relaxes this restriction. This specification is algebraically equivalent to replacing the gap with respondent-level mean realized and stated error, entered jointly. To preserve this equivalence, we construct both component means using the same questions. For respondent i, let J_i denote the number of questions with both realized and stated error observed. Averaging over these questions gives

$$\overline{TE}_i = \frac{1}{J_i}\sum_{j=1}^{J_i} TE_{ij}, \qquad \overline{SE}_i = \frac{1}{J_i}\sum_{j=1}^{J_i} SE_{ij}.$$

The mean item gap is therefore exactly

$$\overline{OP}_i = \overline{TE}_i - \overline{SE}_i.$$

We use the same scale for both components so that their difference equals the standardized SEM gap. Specifically, we center both component means at their sample means and divide them by the gap's standard deviation. Let μ_{TE} and μ_{SE} denote the sample means of $\overline{TE}_i$ and $\overline{SE}_i$, respectively. The unstandardized gap, $\overline{OP}_i$, has sample mean $\mu_{OP} = \mu_{TE} - \mu_{SE}$ and sample standard deviation σ_{OP}.[64] We then define

$$TE_i^* = \frac{\overline{TE}_i - \mu_{TE}}{\sigma_{OP}}, \qquad SE_i^* = \frac{\overline{SE}_i - \mu_{SE}}{\sigma_{OP}}.$$

The standardized gap G_i and the combined level L_i are therefore

$$G_i = TE_i^* - SE_i^* = \frac{\overline{OP}_i - \mu_{OP}}{\sigma_{OP}}, \qquad L_i = TE_i^* + SE_i^*.$$

[64] These means and σ_{OP} are calculated in the full paired-item sample used to standardize the baseline SEM measure and held fixed across the estimation samples for the different outcomes.

By construction, G_i is exactly the standardized gap used in the baseline outcome regressions. For each outcome, we estimate

$$Y_i = \alpha + \delta_G G_i + \delta_L L_i + X_i'\gamma + \varepsilon_i,$$

where X_i contains the controls and fixed effects from the corresponding baseline regression in Table 2. The coefficient δ_G captures the association between the outcome and a larger gap G_i, holding the combined error level L_i fixed. At a fixed L_i, a larger gap means higher TE_i^* and lower SE_i^*. Realized error itself is therefore not held fixed. The coefficient δ_L captures the association with a higher combined error level while holding the gap fixed. In this case, realized and stated error increase together. Panel A of Table J.2 reproduces the corresponding baseline estimates from Table 2. Panel B adds L_i, relaxing the baseline restriction $\delta_L = 0$. We test this restriction to assess whether the combined error level is also associated with the outcome after accounting for the gap.

	DAX error	Diversification	Extremism	Left–right	Abstention
Panel A: Paired-gap benchmark					
Paired gap	0.083*	-0.131***	0.087**	-0.008	0.032**
	(0.043)	(0.036)	(0.043)	(0.042)	(0.012)
Panel B: Common-scale components and conditional gap					
Component representation					
Realized error, TE^*	0.095**	-0.134***	0.097**	-0.010	0.032***
	(0.044)	(0.036)	(0.043)	(0.042)	(0.012)
Stated error, SE^*	0.071	0.069	0.030	-0.021	-0.024
	(0.088)	(0.074)	(0.078)	(0.077)	(0.023)
Equivalent gap-and-sum representation					
Gap | $TE + SE$, δ_G	0.012	-0.102**	0.034	0.005	0.028*
	(0.056)	(0.047)	(0.052)	(0.051)	(0.015)
Equal-and-opposite test (p)	0.043	0.336	0.074	0.660	0.693
N	548	774	716	716	706

Table J.2: Paired gap and component decomposition on a common scale. All measures use the same paired items. Panel A reports the baseline coefficient on the standardized paired gap $G_i = (\overline{OP}_i - \mu_{OP})/\sigma_{OP}$. Panel B presents the same model in two equivalent forms. The component rows report β_{TE} and β_{SE} from a regression containing TE_i^* and SE_i^*. The gap row reports δ_G from the equivalent regression containing the gap G_i and the sum $L_i = TE_i^* + SE_i^*$. Both respondent-level component means are centered using the full paired-item sample and divided by σ_{OP} rather than by their own standard deviations. The conditional-gap coefficient is $\delta_G = (\beta_{TE} - \beta_{SE})/2$. The reported p-value tests $\delta_L = (\beta_{TE} + \beta_{SE})/2 = 0$, which is equivalent to $\beta_{TE} = -\beta_{SE}$. All specifications use the controls and fixed effects from the corresponding regression in Table 2. Political outcomes also control for political interest. Conventional OLS standard errors are reported in parentheses. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table J.2 shows that accounting for the combined error level generally yields smaller

and less precise gap estimates. Some evidence of a behavioral association remains after this adjustment. These comparisons hold the combined level of realized and stated error fixed while allowing realized error itself to vary. They therefore do not establish whether the associations persist among respondents with similar observed accuracy.

The restriction tests provide evidence against the equal-and-opposite restriction for some outcomes, indicating that the combined error level is associated with behavior at a fixed gap. Because realized error may partly reflect factual knowledge, the next subsection adjusts directly for respondent-level mean realized error.

J.3 Adjustment and stratification by realized error

This subsection examines whether respondents with similar observed accuracy but different stated uncertainty also differ in their behavior. We retain the standardized gap G_i and account for mean realized error, $\overline{TE}_i$, in four ways. We begin by adding mean realized error as a regressor. The gap coefficient then captures the association with lower stated uncertainty while holding mean realized error fixed. In the previous subsection, the gap coefficient instead held the combined level of realized and stated error fixed.[65]

The other three specifications allow more flexible adjustments. A quartic polynomial allows the relationship between mean realized error and the outcome to be nonlinear. Quintile fixed effects compare respondents within broad ranges of mean realized error, allowing some accuracy differences within each group. Exact-value fixed effects compare respondents with identical observed mean realized errors, so the gap association comes from differences in their stated uncertainty. This last approach makes the accuracy comparison tighter, although limited variation within groups may reduce precision.

[65] With the same sample and controls, the linear specification yields the same fitted values as the preceding gap-plus-level model because $TE_i^* = (G_i + L_i)/2$. Its coefficient on G_i equals $-\beta_{SE}$. It therefore provides another interpretation of the same fitted model rather than an independent test.

	DAX error	Diversification	Extremism	Left–right	Abstention
Baseline paired gap	0.083*	-0.131***	0.087**	-0.008	0.032**
	(0.043)	(0.036)	(0.043)	(0.042)	(0.012)
Gap + linear TE	-0.071	-0.069	-0.030	0.021	0.024
	(0.088)	(0.074)	(0.078)	(0.077)	(0.023)
Gap + quartic in TE	-0.050	-0.077	-0.016	0.039	0.028
	(0.090)	(0.076)	(0.080)	(0.078)	(0.023)
Gap + TE-quintile FE	0.022	-0.107*	0.060	-0.007	0.028
	(0.066)	(0.057)	(0.067)	(0.065)	(0.019)
Gap + exact-value TE FE	0.135	-0.151	-0.083	0.030	-0.002
	(0.120)	(0.098)	(0.104)	(0.105)	(0.030)
N	548	774	716	716	706

Table J.3: Adjustment and stratification by respondent-level realized error. Entries report coefficients on the standardized paired SEM gap. Mean realized error is computed over the same item set. Exact-value fixed effects compare respondents with the same observed mean realized error where that value is repeated. All specifications use the controls and fixed effects from the corresponding baseline outcome regression. Political outcomes also control for political interest. Across the five outcome samples, the exact-value specifications contain 213–241 mean-TE strata; 124–151 repeated strata contain 459–684 observations (median repeated-stratum size: 3). Conventional OLS standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The estimates in Table J.3 are sensitive to how observed accuracy is accounted for. They are generally less precise than the baseline estimates and provide limited evidence that the baseline associations persist among respondents with similar mean realized errors.

These comparisons do not isolate the contribution of factual knowledge. Respondents with identical mean realized errors may still differ in their underlying knowledge: their averages may cover different sets of questions and contain item-specific realization noise. Changes in the estimates therefore cannot be attributed to factual knowledge alone.

J.4 Outcome regressions using a population-relative residual measure

As a final exercise, we replace the standardized gap G_i with a population-relative measure constructed from item-specific residuals. For each item j, we regress stated error on a quartic polynomial in realized error:

$$SE_{ij} = \alpha_j + \sum_{k=1}^{4} \gamma_{jk} \, TE_{ij}^{k} + r_{ij}. \tag{13}$$

The fitted value provides a sample benchmark for stated uncertainty at the respondent's observed realized error on that question. The residual $\widehat{r}_{ij}$ compares reported uncertainty

with this benchmark. We reverse its sign, defining $\widetilde{OP}_{ij} = -\widehat{r}_{ij}$, so that positive values indicate less stated uncertainty than predicted. We then average these values over each respondent's paired items and standardize the respondent mean:

$$\widetilde{G}_i = z\left(\frac{1}{J_i}\sum_{j=1}^{J_i}\widetilde{OP}_{ij}\right).$$

Higher values of $\widetilde{G}_i$ indicate greater overprecision relative to the fitted sample benchmark. We repeat the baseline outcome regressions using $\widetilde{G}_i$ in place of G_i.

Here, we account for realized error when constructing the measure, removing the fitted relationship between stated and realized error separately for each item before averaging across questions. We do not additionally hold respondent-level mean realized error, $\overline{TE}_i$, fixed in the outcome regressions.

	DAX error	Diversification	Extremism	Left–right	Abstention
Baseline standardized gap (G_i)					
G_i	0.083*	-0.131***	0.087**	-0.008	0.032**
	(0.043)	(0.036)	(0.043)	(0.042)	(0.012)
Population-relative residual index ($\widetilde{G}_i$)					
$\widetilde{G}_i$	-0.022	-0.016	-0.026	-0.004	0.008
	(0.045)	(0.037)	(0.040)	(0.039)	(0.012)
N	548	774	716	716	706
Adjusted R^2, baseline G_i	0.076	0.112	0.024	0.062	0.111
Adjusted R^2, residual index $\widetilde{G}_i$	0.070	0.097	0.018	0.062	0.103

Table J.4: Outcome regressions using the baseline gap and population-relative residual index. Each row uses the controls and fixed effects from the corresponding Table 2 outcome regression; political outcomes additionally include political interest. Conventional OLS standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table J.4 shows that the estimated associations are smaller in absolute magnitude when we replace G_i with $\widetilde{G}_i$, and none is statistically distinguishable from zero. However, the two measures ask different questions. The baseline directly assesses whether respondents underestimate their own errors, while the alternative asks whether they report less uncertainty than is typical among respondents making similar errors on the same question. Someone can underestimate their error while still reporting more uncertainty than others, so the two measures can give different answers. We therefore treat the alternative as a useful complement to the baseline.

J.5 Discussion

The additional analyses show that accounting for realized error changes some behavioral associations more than others. How useful this adjustment is depends on what we want to learn from the regression and whether accuracy may have a separate relationship with the outcome. For our SOEP-IS analysis, the concern is that factual knowledge may affect both accuracy on the SEM questions and behavior in ways that the baseline controls do not fully capture.

Table J.1 shows that respondents who make larger realized errors generally report greater uncertainty. The estimated slope increases when we average across questions or use errors on other questions as a proxy for persistent differences in accuracy. This pattern is consistent with noise on individual questions weakening the raw relationship. Under the proxy-IV assumptions, however, stated uncertainty still increases less than one-for-one with predictable error. Differences in accuracy may therefore contribute to the SEM gap even when respondents making larger errors also report greater uncertainty.

The SEM gap remains associated with some outcomes after we account for the combined level of realized and stated error (Section J.2). Since this comparison still allows accuracy to vary, Section J.3 examines the associations among respondents with similar observed accuracy. Section J.4 takes a different approach by comparing stated uncertainty with the uncertainty predicted for similar errors on the same question. With this population-relative measure, the associations are smaller, and none is statistically distinguishable from zero. These exercises help us interpret the baseline findings by showing how they change under different comparisons and definitions of overprecision.

Since realized error is part of the SEM gap, controlling for it directly can leave less variation with which to estimate the association and may reduce precision. We use these specifications as complements to the baseline analysis. Changes in the estimates do not tell us how much factual knowledge contributes to the baseline associations, because realized error may also contain item-specific noise or reflect other respondent characteristics. The population-relative residual measure also changes which differences between respondents are captured by the score. Therefore, in the SOEP-IS analysis, we interpret the SEM gap as a mismatch between realized and stated error, recognizing that differences in both accuracy and reported uncertainty may contribute to its behavioral associations.

Evidence elsewhere in the paper complements the analyses in this appendix. Appendix C examines the behavioral associations using relative and age-robust SEM measures (Tables C.4 and C.5). The relative measure divides item-level gaps by stated error, while the age-robust measure uses only historical questions about events that occurred strictly after the respondent's birth year. For the five main outcomes, the estimated associations retain their baseline directions under both constructions, although their magnitudes and statistical support vary. This consistency further supports the baseline patterns across changes in how the gap is scaled and which questions are included.

The companion survey in Section 5 supports the SEM's broader scope. The triangle-count task does not require factual recall, yet its SEM score is positively related to scores in the other domains and loads positively on the first principal component. Although these relationships are weaker than those among the knowledge-based domains, they suggest that the shared pattern in SEM responses extends beyond factual recall. Because the survey uses a separate sample, it does not identify the source of the SOEP-IS behavioral associations. For future applications, everyday tasks such as estimating an object's weight or predicting the average time it takes to read a text offer a way to use the SEM without asking respondents to recall a previously known answer.